

TYPE II REGULARITY FOR NAVIER–STOKES

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ABSTRACT. This document consolidates the Type II regularity argument into one working draft. The source material has been placed in a single LaTeX file, with duplicated scaffolding removed and shared notation, references, and bibliography handled by one preamble.

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1. LOCAL SINGLE-CORE CRITERION

Abstract. We prove a local compact single-core exclusion criterion for Type II Navier–Stokes concentration. The theorem is formulated on fixed compact parabolic cylinders with explicit Caffarelli–Kohn–Nirenberg bounds, positive core concentration, localized scale-translation representation, local pressure reconstruction, and vanishing localized dissipation on selected windows. Compactness and pressure stability then force the selected limit to be spatially constant. A zero constant contradicts the positive CKN core; a nonzero constant gives a bounded selected-window limit, which is excluded by the definition of the local Type II regime.

1.1. Introduction. This section isolates a local compactness mechanism for Type II concentration in the three-dimensional Navier–Stokes equations. The setting is a sequence of renormalized solutions on fixed parabolic cylinders, centered at a selected concentration core and written in scale-translation variables. The result excludes the case in which one compact core carries a fixed positive Caffarelli–Kohn–Nirenberg density while the localized dissipation along selected windows vanishes. The theorem is conditional on the local representation and pressure decomposition/stability inputs stated below; the compact-cylinder A, E, C, D bounds are part of the branch hypothesis.

The argument is local. Uniform Caffarelli–Kohn–Nirenberg bounds and bounded modulation coefficients give compactness on smaller cylinders. Vanishing localized dissipation forces the compactness limit to be spatially constant. If the constant is zero, strong velocity convergence and pressure stability contradict the positive core density. If the constant is nonzero, the sequence has a nonzero bounded selected-window limit, which is incompatible with the local Type II condition used here.

1.2. Setup and definitions. For $R > 0$, write

$$Q_R = B_R \times (-R^2, 0).$$

For a suitable pair (v, q) on Q_R , define

$$A_v(R) = R^{-1} \operatorname{ess\,sup}_{-R^2 < s < 0} \int_{B_R} |v(y, s)|^2 dy,$$

$$E_v(R) = R^{-1} \int_{Q_R} |\nabla v|^2 dy ds,$$

$$C_v(R) = R^{-2} \int_{Q_R} |v|^3 dy ds,$$

$$D_q(R) = R^{-2} \int_{-R^2}^0 \int_{B_R} |q(y, s) - (q)_{B_R}(s)|^{3/2} dy ds.$$

All pressure quantities are understood modulo functions of time.

Definition 1.1 (Bounded selected-window limit). Let (V_n, P_n) be a sequence of represented local rescalings on a compact cylinder Q_r . A selected-window subsequence has a bounded selected-window limit on Q_r if, after passing to a subsequence,

$$V_n \rightarrow V_* \quad \text{strongly in } L^3(Q_r),$$

where $V_* \in L^\infty(Q_r)$. The limit is nonzero if $V_* \not\equiv 0$ on Q_r .

Definition 1.2 (Local Type II sequence). A represented sequence is called a local Type II sequence if it has positive local Caffarelli–Kohn–Nirenberg concentration on some fixed compact cylinder and no selected-window subsequence has a nonzero bounded selected-window limit on any compact cylinder in the sense of [Theorem 1.1](#).

Definition 1.3 (Compact single-core vanishing-dissipation branch). Fix radii $0 < r_* < R_*$, constants $M < \infty$, $\eta_0 > 0$, and a sequence of represented local rescalings (V_n, P_n, a_n, b_n) on Q_{R_*} . The sequence is a compact single-core vanishing-dissipation branch if:

- (i) each (V_n, P_n) is suitable on Q_{R_*} and solves the represented Navier–Stokes equation
- (1) $\partial_s V_n + (V_n \cdot \nabla) V_n + \nabla P_n = \nu \Delta V_n + a_n(s)(V_n + y \cdot \nabla V_n) + b_n(s) \cdot \nabla V_n, \quad \nabla \cdot V_n = 0;$

- (ii) the compact-cylinder suitable package is uniformly bounded:

$$A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \leq M;$$

- (iii) the selected core has positive CKN concentration on Q_{r_*} :

$$C_{V_n}(r_*) + D_{P_n}(r_*) \geq \eta_0 \quad \text{for all } n;$$

- (iv) the localized scale and center are nondegenerate for the selected core, so the representation theorem gives the variables in (1);

- (v) the modulation coefficients are uniformly bounded on the selected cylinder:

$$\|a_n\|_{L^\infty(-R_*^2, 0)} + \|b_n\|_{L^\infty(-R_*^2, 0)} \leq M;$$

- (vi) the branch has vanishing localized dissipation in the sense of [Theorem 1.4](#) below.

Definition 1.4 (Vanishing localized dissipation). Let $r_* < \rho < R_*$. Selected windows are first translated and rescaled to the fixed cylinders $Q_{r_*} \subseteq Q_\rho \subseteq Q_{R_*}$. The localized dissipation on Q_ρ is

$$\mathcal{D}_n(\rho) = \int_{Q_\rho} |\nabla V_n|^2 dy ds.$$

The branch has vanishing localized dissipation if, after passing to a selected-window subsequence, there is $\rho \in (r_*, R_*)$ such that

$$\mathcal{D}_n(\rho) \rightarrow 0.$$

1.3. Local analytic inputs. The proof uses the following imported local analytic statements. They are stated here in the form needed for the compact single-core argument.

Theorem 1.5 (Local concentration dichotomy). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$, and suppose $\Sigma(T) \neq \emptyset$. Then, using the pressure-normalized quantities C, D from the local concentration dichotomy, there exist $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

Moreover, at the concentration point either a local Type I scale-invariant bound holds, or the point lies in the local Type II alternative, meaning that no such local Type I bound holds.

Proof. This is the local blowup-rate dichotomy used as the entry point for the Type II analysis. $\square$

Proposition 1.6 (Bridge to local Type II sequences). *Let (x_0, T) be a concentration point in the local Type II alternative of [Theorem 1.5](#). After choosing the positive concentration scales given there and applying the local scale-translation selection used in the present section to a nondegenerate compact single core, the represented selected windows form a sequence with positive local CKN concentration on a fixed compact cylinder. In the selected-window Type II branch, no subsequence has a nonzero bounded selected-window limit on any compact cylinder. Hence the represented selected windows are a local Type II sequence in the sense of [Theorem 1.2](#).*

Proof. The local concentration dichotomy supplies the positive CKN concentration sequence and the exclusion of the local Type I alternative at the chosen concentration point. The scale-translation selection rewrites those shrinking cylinders as fixed selected windows. The last condition is precisely the selected-window formulation of the Type II branch used in this section: any branch with a nonzero bounded selected-window limit is assigned to the bounded-window alternative, not to [Theorem 1.2](#). $\square$

Proposition 1.7 (Imported local representation input). *For a nondegenerate selected single core, the local scale-translation construction gives represented variables (V_n, P_n, a_n, b_n) satisfying (1) on compact cylinders. Failure of the localized nondegeneracy condition is a multibubble, cascade, or gauge-degenerate alternative, not a compact single-core branch.*

Proposition 1.8 (Imported local pressure decomposition and stability). *For every $B_r \Subset B_R$, the represented pressure decomposes as*

$$P_n = P_{\text{loc},n} + H_n, \quad -\Delta P_{\text{loc},n} = \partial_i \partial_j (\zeta V_{n,i} V_{n,j}),$$

where $\zeta \in C_c^\infty(B_R)$ equals one on B_r , and H_n is harmonic in B_r . The local part satisfies the Calderon–Zygmund bound

$$\|P_{\text{loc},n}\|_{L^{3/2}(B_R)} \leq C \|V_n\|_{L^3(B_R)}^2$$

for a.e. time, and the harmonic part is controlled modulo spatial means by the local pressure norm.

Moreover, if $V_n \rightarrow 0$ strongly in $L^3(Q_R)$, then for every $r < R$,

$$P_n - (P_n)_{B_r} \rightarrow 0 \quad \text{strongly in } L^{3/2}(Q_r).$$

Lemma 1.9 (Localization of pressure oscillation bounds). *Let $0 < r < R$, and let $q \in L^{3/2}(Q_R)$. Then*

$$D_q(r) \leq C \left(\frac{R}{r}\right)^2 D_q(R),$$

where C is universal.

Proof. For a.e. $s \in (-r^2, 0)$, Jensen's inequality gives

$$|(q)_{B_r}(s) - (q)_{B_R}(s)|^{3/2} \leq \frac{1}{|B_r|} \int_{B_r} |q(y, s) - (q)_{B_R}(s)|^{3/2} dy.$$

Hence

$$\int_{B_r} |q - (q)_{B_r}(s)|^{3/2} dy \leq C \int_{B_r} |q - (q)_{B_R}(s)|^{3/2} dy.$$

Integrating over $(-r^2, 0)$ and using $Q_r \subset Q_R$,

$$\begin{aligned} D_q(r) &\leq Cr^{-2} \int_{-r^2}^0 \int_{B_r} |q - (q)_{B_R}(s)|^{3/2} dy ds \\ &\leq Cr^{-2} \int_{-R^2}^0 \int_{B_R} |q - (q)_{B_R}(s)|^{3/2} dy ds = C \left(\frac{R}{r}\right)^2 D_q(R). \end{aligned}$$

$\square$

1.4. Compactness and pressure stability.

Proposition 1.10 (Compactness on selected windows). *Let (V_n, P_n) be suitable on Q_{R_*} , solve (1), and satisfy*

$$A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \leq M,$$

with

$$\|a_n\|_{L^\infty(-R_*^2, 0)} + \|b_n\|_{L^\infty(-R_*^2, 0)} \leq M.$$

Then for every $r < R_*$, after passing to a subsequence there are (V, P) such that

$$V_n \xrightarrow{*} V \quad \text{in } L^\infty(-r^2, 0; L^2(B_r)),$$

$$V_n \rightharpoonup V \quad \text{in } L^2(-r^2, 0; H^1(B_r)),$$

$$V_n \rightarrow V \quad \text{strongly in } L^3(Q_r),$$

and

$$P_n - (P_n)_{B_r} \rightharpoonup P \quad \text{in } L^{3/2}(Q_r).$$

After passing to a further subsequence, a_n and b_n converge weak-* in L^∞ to bounded coefficients, and the limit satisfies the corresponding represented equation distributionally on smaller cylinders.

Proof. The bounds on A and E give weak-* compactness in $L^\infty L^2$ and weak compactness in $L^2 H^1$ on smaller cylinders. Fix $r < r_1 < R_*$, and choose m so large that

$$H_0^m(B_{r_1}) \hookrightarrow W_0^{1,\infty}(B_{r_1}).$$

We claim that $\partial_s V_n$ is uniformly bounded in

$$L^1(-r_1^2, 0; H^{-m}(B_{r_1})).$$

Let $\varphi \in H_0^m(B_{r_1})$. Pairing (1) with φ , integrating by parts in space where needed, and using the embedding above gives

$$|\langle (V_n \cdot \nabla) V_n, \varphi \rangle| = \left| \int_{B_{r_1}} V_{n,i} V_{n,j} \partial_j \varphi_i dy \right| \leq C \|V_n\|_{L^3(B_{r_1})}^2 \|\varphi\|_{H^m},$$

$$|\langle \nu \Delta V_n, \varphi \rangle| \leq C \|\nabla V_n\|_{L^2(B_{r_1})} \|\varphi\|_{H^m},$$

$$|\langle \nabla P_n, \varphi \rangle| = \left| \int_{B_{r_1}} (P_n - (P_n)_{B_{r_1}}) \nabla \cdot \varphi dy \right| \leq C \|P_n - (P_n)_{B_{r_1}}\|_{L^{3/2}(B_{r_1})} \|\varphi\|_{H^m}.$$

The modulation terms obey

$$|\langle a_n V_n, \varphi \rangle| \leq C \|V_n\|_{L^2(B_{r_1})} \|\varphi\|_{H^m},$$

$$|\langle a_n y \cdot \nabla V_n, \varphi \rangle| = \left| \int_{B_{r_1}} a_n V_{n,i} \partial_j (y_j \varphi_i) dy \right| \leq C \|V_n\|_{L^2(B_{r_1})} \|\varphi\|_{H^m},$$

and

$$|\langle b_n \cdot \nabla V_n, \varphi \rangle| = \left| \int_{B_{r_1}} V_{n,i} b_{n,j} \partial_j \varphi_i dy \right| \leq C \|V_n\|_{L^2(B_{r_1})} \|\varphi\|_{H^m},$$

where the constant uses r_1 and the uniform L^∞ bounds for a_n, b_n . Integrating these estimates in time gives a uniform bound because

$$\begin{aligned} \int_{-r_1^2}^0 \|V_n(s)\|_{L^3(B_{r_1})}^2 ds &\leq r_1^{2/3} \|V_n\|_{L^3(Q_{r_1})}^2, \\ \int_{-r_1^2}^0 \|\nabla V_n(s)\|_{L^2(B_{r_1})} ds &\leq r_1 \|\nabla V_n\|_{L^2(Q_{r_1})}, \end{aligned}$$

$$\int_{-r_1^2}^0 \|P_n(s) - (P_n)_{B_{r_1}}(s)\|_{L^{3/2}(B_{r_1})} ds \leq r_1^{2/3} \|P_n - (P_n)_{B_{r_1}}\|_{L^{3/2}(Q_{r_1})},$$

and

$$\int_{-r_1^2}^0 \|V_n(s)\|_{L^2(B_{r_1})} ds \leq r_1^2 \|V_n\|_{L^\infty(-r_1^2, 0; L^2(B_{r_1}))}.$$

The uniform C , E , D , and A bounds therefore prove the claimed time-derivative bound; the D -bound on Q_{r_1} follows from the bound on Q_{R_*} by [Theorem 1.9](#).

The compact embedding $H^1(B_{r_1}) \Subset L^2(B_r)$, together with Aubin–Lions, gives

$$V_n \rightarrow V \quad \text{strongly in } L^2(Q_r)$$

after passing to a subsequence. The same $L^\infty L^2 \cap L^2 H^1$ bounds give the parabolic interpolation estimate

$$\|V_n\|_{L^{10/3}(Q_r)} \leq C(r, M).$$

Indeed, for a.e. s , the Sobolev and Gagliardo–Nirenberg inequalities give

$$\|V_n(s)\|_{L^{10/3}(B_r)} \leq C \|V_n(s)\|_{L^2(B_r)}^{2/5} \|\nabla V_n(s)\|_{L^2(B_r)}^{3/5}.$$

Raising to the power $10/3$, integrating in time, and using the A and E bounds yields

$$\int_{-r^2}^0 \|V_n(s)\|_{L^{10/3}(B_r)}^{10/3} ds \leq C \left(\operatorname{ess\,sup}_{-r^2 < s < 0} \|V_n(s)\|_{L^2(B_r)}^2 \right)^{2/3} \int_{Q_r} |\nabla V_n|^2 dy ds \leq C(r, M).$$

The same estimate applies to V by weak lower semicontinuity, and therefore to $V_n - V$ by the triangle inequality. Interpolating between L^2 and $L^{10/3}$, one obtains

$$\|V_n - V\|_{L^3(Q_r)} \leq \|V_n - V\|_{L^2(Q_r)}^{1/6} \|V_n - V\|_{L^{10/3}(Q_r)}^{5/6}.$$

The second factor is uniformly bounded and the first tends to zero, so

$$V_n \rightarrow V \quad \text{strongly in } L^3(Q_r).$$

The pressure compactness is the weak compactness from the uniform D -bound: after subtracting spatial means,

$$P_n - (P_n)_{B_r} \rightharpoonup P \quad \text{in } L^{3/2}(Q_r).$$

We pass to the limit distributionally using the preceding convergences. The only point requiring comment is the modulation. For every compactly supported test field φ ,

$$\int a_n(s) y \cdot \nabla V_n \cdot \varphi = - \int a_n(s) V_{n,i} \partial_j (y_j \varphi_i),$$

where summation over repeated spatial indices is understood. Equivalently,

$$\partial_j (y_j \varphi_i) = 3\varphi_i + y_j \partial_j \varphi_i.$$

Hence this term passes to the limit by strong local L^2 convergence of V_n and weak-* convergence of a_n : if $F_n \rightarrow F$ in L^1 and $a_n \rightharpoonup^* a$ in L^∞ , then

$$\int a_n F_n - \int a F = \int a_n (F_n - F) + \int (a_n - a) F \rightarrow 0.$$

The term $a_n V_n$ is immediate from the same argument, while

$$\int b_n(s) \cdot \nabla V_n \cdot \varphi = - \int V_{n,i} b_{n,j}(s) \partial_j \varphi_i$$

passes to the limit by strong local L^2 convergence of V_n and weak-* convergence of b_n . These passages give the limiting represented equation in the sense of distributions. $\square$

Remark 1.11 (Use of bounded modulation). The uniform L^∞ bounds on a_n and b_n enter the compactness argument in exactly two places. First, they give the negative Sobolev time-derivative bound for the three lower-order modulation terms in (1). Second, after subsequential weak-* convergence of a_n, b_n , they allow the modulation terms to pass to the distributional limit once $V_n \rightarrow V$ strongly in L^2_{loc} .

Lemma 1.12 (Stability of local pressure decomposition). *Let $V_n \rightarrow V$ strongly in $L^3(Q_R)$, and use the pressure decomposition of Theorem 1.8. Then for every $r < R$,*

$$P_{\text{loc},n} \rightarrow P_{\text{loc}} \quad \text{strongly in } L^{3/2}(Q_r),$$

where $-\Delta P_{\text{loc}} = \partial_i \partial_j (\zeta V_i V_j)$. If $V \equiv 0$ on Q_R , then

$$P_n - (P_n)_{B_r} \rightarrow 0 \quad \text{strongly in } L^{3/2}(Q_r).$$

More locally in time, if $V_n \rightarrow 0$ strongly in

$$L^3(B_R \times (-r^2, 0)),$$

then

$$P_n - (P_n)_{B_r} \rightarrow 0 \quad \text{strongly in } L^{3/2}(Q_r).$$

Proof. Strong convergence in L^3 gives

$$V_{n,i} V_{n,j} \rightarrow V_i V_j \quad \text{strongly in } L^{3/2}(Q_R).$$

Indeed,

$$\|V_{n,i} V_{n,j} - V_i V_j\|_{L^{3/2}(Q_R)} \leq (\|V_n\|_{L^3(Q_R)} + \|V\|_{L^3(Q_R)}) \|V_n - V\|_{L^3(Q_R)}.$$

Calderon–Zygmund estimates for the compactly supported localized source yield strong convergence of the localized pressures in $L^{3/2}$ on smaller balls:

$$\|P_{\text{loc},n} - P_{\text{loc}}\|_{L^{3/2}(Q_r)} \leq C_{r,R} \|V_{n,i} V_{n,j} - V_i V_j\|_{L^{3/2}(Q_R)}.$$

Assume now that $V \equiv 0$ on Q_R . Then $P_{\text{loc}} = 0$. By the zero-limit stability part of Theorem 1.8,

$$P_n - (P_n)_{B_r} \rightarrow 0 \quad \text{strongly in } L^{3/2}(Q_r).$$

The same proof is local in time. If $V_n \rightarrow 0$ strongly in $L^3(B_R \times (-r^2, 0))$, the Calderon–Zygmund estimate and harmonic pressure control are applied only for times $s \in (-r^2, 0)$, giving the strong convergence on Q_r . $\square$

Lemma 1.13 (Zero-dissipation limit). *If*

$$V_n \rightharpoonup V \quad \text{in } L^2(I; H^1(B_\rho))$$

and

$$\int_I \int_{B_\rho} |\nabla V_n|^2 dy ds \rightarrow 0,$$

then $\nabla V = 0$ a.e. on $B_\rho \times I$. Hence for a.e. time, $V(\cdot, s)$ is spatially constant on B_ρ .

Proof. Weak lower semicontinuity gives

$$\int_I \int_{B_\rho} |\nabla V|^2 dy ds \leq \liminf_{n \rightarrow \infty} \int_I \int_{B_\rho} |\nabla V_n|^2 dy ds = 0.$$

Thus $\nabla V = 0$ a.e. on $B_\rho \times I$. By Fubini, for a.e. $s \in I$ one has $\nabla V(\cdot, s) = 0$ in $L^2(B_\rho)$. Since B_ρ is connected, every $H^1(B_\rho)$ function with zero weak gradient is a.e. equal to a constant. Hence $V(\cdot, s)$ is spatially constant for a.e. s . $\square$

Lemma 1.14 (Spatially constant limits). *Let V be a limit on Q_ρ and suppose $\nabla V = 0$ a.e. there. Then $V(y, s) = c(s)$ on Q_ρ , with $c \in L^\infty(-\rho^2, 0)$. If $c \not\equiv 0$ on Q_ρ , then the approximating selected-window subsequence has a nonzero bounded selected-window limit.*

Proof. The spatial constancy follows from [Theorem 1.13](#). Since $V(\cdot, s) = c(s)$ on B_ρ , the local $L_s^\infty L_y^2$ bound gives

$$|c(s)|^2 |B_\rho| \leq \operatorname{ess\,sup}_{-\rho^2 < s < 0} \int_{B_\rho} |V(y, s)|^2 dy,$$

so $c \in L^\infty(-\rho^2, 0)$. If $c \not\equiv 0$ on Q_ρ , then the strong L^3 convergence of [Theorem 1.10](#) gives a subsequential limit to a nonzero bounded function on Q_ρ , which is precisely a nonzero bounded selected-window limit in the sense of [Theorem 1.1](#). $\square$

1.5. Contradiction lemmas.

Lemma 1.15 (Persistence of positive core concentration). *For a compact single-core branch in the sense of [Theorem 1.3](#), there are fixed $r_* > 0$ and $\eta_0 > 0$ such that along the selected-window subsequence,*

$$C_{V_n}(r_*) + D_{P_n}(r_*) \geq \eta_0 \quad \text{for all } n.$$

Proof. This is part of the definition of the selected compact core and is preserved by passing to subsequences of the selected windows. $\square$

Lemma 1.16 (Zero constant contradicts positive CKN concentration). *Let $r_* < \rho$. Suppose the compactness limit in [Theorem 1.10](#) is spatially constant on Q_ρ and is identically zero on Q_{r_*} . Then*

$$C_{V_n}(r_*) + D_{P_n}(r_*) \rightarrow 0.$$

Consequently this case contradicts [Theorem 1.15](#).

Proof. Strong $L^3(Q_{r_*})$ convergence to zero gives

$$C_{V_n}(r_*) = r_*^{-2} \int_{Q_{r_*}} |V_n|^3 dy ds = r_*^{-2} \|V_n\|_{L^3(Q_{r_*})}^3 \rightarrow 0.$$

It remains to control the pressure term. Since V is spatially constant on Q_ρ , the identity $V = 0$ on Q_{r_*} implies

$$V = 0 \quad \text{on } B_\rho \times (-r_*^2, 0).$$

Indeed, for a.e. $s \in (-r_*^2, 0)$, the spatially constant value on B_ρ vanishes on the nonempty subball B_{r_*} , hence vanishes on all of B_ρ . The strong L_{loc}^3 convergence therefore gives

$$V_n \rightarrow 0 \quad \text{strongly in } L^3(B_\rho \times (-r_*^2, 0)).$$

Applying [Theorem 1.12](#) with $R = \rho$ and $r = r_*$, in its local-in-time form, gives

$$P_n - (P_n)_{B_{r_*}} \rightarrow 0 \quad \text{strongly in } L^{3/2}(Q_{r_*}).$$

Therefore

$$D_{P_n}(r_*) = r_*^{-2} \|P_n - (P_n)_{B_{r_*}}\|_{L^{3/2}(Q_{r_*})}^{3/2} \rightarrow 0.$$

Thus

$$C_{V_n}(r_*) + D_{P_n}(r_*) \rightarrow 0,$$

contradicting the uniform lower bound

$$C_{V_n}(r_*) + D_{P_n}(r_*) \geq \eta_0.$$

$\square$

Lemma 1.17 (Nonzero constant contradicts the Type II regime). *If the compactness limit is spatially constant and nonzero on a compact subcylinder, then the original sequence is not a local Type II sequence.*

Proof. By Theorem 1.14, the selected-window subsequence has a nonzero bounded selected-window limit. This contradicts Theorem 1.2. $\square$

1.6. Main theorem.

Theorem 1.18 (Local single-core Type II criterion). *Under the imported inputs Theorems 1.6 to 1.8, a compact single-core vanishing-dissipation branch in the sense of Theorem 1.3 cannot be a local Type II sequence.*

Proof. By the definition of vanishing localized dissipation, after passing to a selected-window subsequence there is a fixed radius $\rho \in (r_*, R_*)$ such that

$$\int_{Q_\rho} |\nabla V_n|^2 dy ds \rightarrow 0$$

and the positive core bound on Q_{r_*} remains valid along this subsequence. Apply compactness on the fixed compact cylinder Q_ρ and pass to a subsequence. The compactness proposition gives

$$V_n \rightharpoonup V \quad \text{in } L^2(-\rho^2, 0; H^1(B_\rho))$$

and strong convergence in L^3 on every smaller cylinder. Since the localized dissipation tends to zero, the zero-dissipation lemma gives

$$\nabla V = 0 \quad \text{a.e. on } Q_\rho.$$

Thus $V(y, s) = c(s)$ on Q_ρ , and in particular on the selected core cylinder Q_{r_*} .

If this spatially constant limit is zero on the core cylinder, the zero-limit contradiction lemma gives

$$C_{V_n}(r_*) + D_{P_n}(r_*) \rightarrow 0,$$

contradicting persistence of positive core concentration.

If the spatially constant limit is not identically zero on Q_{r_*} , the nonzero-limit lemma produces a nonzero bounded selected-window limit, contradicting the definition of a local Type II sequence. The two cases exhaust all spatially constant limits, so the branch cannot occur. $\square$

Remark 1.19 (Minimal local inputs). The proof uses two local analytic inputs from the Type II analysis: the localized representation for a nondegenerate selected core, the compact-ball pressure reconstruction with stability on smaller balls. The compact-cylinder A, E, C, D bounds and vanishing localized dissipation are part of the branch hypothesis, not separate analytic inputs. Multibubble, cascade, gauge-degenerate, and compactness-failure alternatives are not treated inside this single-core theorem; they are treated by the local multibubble and cascade exclusion theorem.

2. REPAIRED-GAUGE CONSTRUCTION

Abstract. This section gives the local representation theorem for the retained nondegenerate single-core Type II stratum. From explicit physical-core assumptions and the nondegenerate repaired-gauge selection defining this stratum on a compact time window we obtain:

- (1) a preliminary scale-center path and its renormalized map,
- (2) exact transformed equations with fixed sign conventions,
- (3) an explicit finite-dimensional gauge map,
- (4) a full Jacobian computation and Schur-complement invertibility criterion,
- (5) propagation of the repaired-gauge constraints along the absolutely continuous modulation path,
- (6) exact local pressure transport and a local pressure decomposition,
- (7) explicit modulation coefficients from differentiated constraints, and
- (8) preservation of local suitable structure in renormalized variables.

Failure of this local repaired-gauge selection is not treated as a defect of the single-core representation theorem. It is the gauge-degenerate/multibubble or cascade alternative in the state-space stratification. On the retained nondegenerate stratum, pressure transport, modulation formulas, suitable structure, and compact-window representation follow by the theorems below.

2.1. Assumptions and notation. Let $u : \mathbb{R}^3 \times (T - \delta, T) \rightarrow \mathbb{R}^3$, $p : \mathbb{R}^3 \times (T - \delta, T) \rightarrow \mathbb{R}$ be a suitable weak Navier–Stokes solution with viscosity $\nu > 0$. For $z = (x_0, t_0)$ and $R > 0$,

$$Q_R(z) := B_R(x_0) \times (t_0 - R^2, t_0), \quad Q_R^* := B_R \times I_*.$$

Fix a reference physical center-time $(x_\bullet, t_\star)$ around which $B_{R_\star}(x_\bullet)$ is taken. For a fixed spatial center x_0 we also use

$$Q_R(t) := Q_R(x_0, t).$$

Fix a compact window $I_* := [\tau_0 - \ell, \tau_0 + \ell]$, $\ell > 0$, and write $I_{\text{ph}} := [t_\star - \Lambda_{\max}^2 \ell, t_\star + \Lambda_{\max}^2 \ell]$.

For a cylinder $Q_R(x_0, t_0) = B_R(x_0) \times (t_0 - R^2, t_0)$, set

$$\begin{aligned} A(u; Q_R(x_0, t_0)) &:= R^{-1} \sup_{s \in (t_0 - R^2, t_0)} \int_{B_R(x_0)} |u(x, s)|^2 dx, \\ E(u; Q_R(x_0, t_0)) &:= R^{-1} \int_{Q_R(x_0, t_0)} |\nabla u|^2 dx ds, \\ C(u; Q_R(x_0, t_0)) &:= R^{-2} \int_{Q_R(x_0, t_0)} |u|^3 dx ds, \\ D(p; Q_R(x_0, t_0)) &:= R^{-2} \int_{t_0 - R^2}^{t_0} \int_{B_R(x_0)} |p(x, s) - \langle p(s) \rangle_{B_R(x_0)}|^{3/2} dx ds. \end{aligned}$$

Here $\langle p(s) \rangle_{B_R(x_0)} := |B_R|^{-1} \int_{B_R(x_0)} p(x, s) dx$; pressure is always understood modulo time-dependent constants.

Definition 2.1 (Single-core assumption package). Let $(R_\star, r_\star, M, \eta, \kappa, \Lambda_{\min}, \Lambda_{\max})$ satisfy $\Lambda_{\min} \leq \Lambda_{\max} < \infty$. We say (u, p) satisfies $\text{SC}(R_\star, r_\star, M, \eta, \kappa)$ on $Q_{R_\star}(x_\bullet, t_\star)$ if the following hold:

- (SC1) $A(u; Q_{R_\star}(x_\bullet, t)) + E(u; Q_{R_\star}(x_\bullet, t)) + C(u; Q_{R_\star}(x_\bullet, t)) + D(p; Q_{R_\star}(x_\bullet, t)) \leq M$ for every $t \in [t_\star - R_\star^2, t_\star]$.

- (SC2) There exists a physically selected core map $(x_c(t), \rho_c(t))$ on I_{ph} such that $\rho_c(t) \in (0, R_*]$, $x_c(t) \in B_{R_*/2}(x_\bullet)$, and for some $\Theta_0 > 0$,

$$\int_{B_{r_*}(0)} |\rho_c(t)^{-1} u(x_c(t) + \rho_c(t)y, t)|^3 dy \geq \eta \quad \text{for a.e. } t \in I_{\text{ph}}.$$

- (SC3) There exists $0 < r_{\text{sep}} \leq r_*/2$ such that for every $0 < r \leq r_{\text{sep}}$, every competitor pair

$$\mathcal{C}_{\text{core}}(t) := \left\{ (\bar{x}, \bar{\rho}) : |\bar{x} - x_\bullet| \leq \frac{R_*}{2}, \ 0 < \bar{\rho} \leq R_* \right\},$$

$(\bar{x}, \bar{\rho}) \in \mathcal{C}_{\text{core}}(t) \setminus \{(x_c(t), \rho_c(t))\}$ with

$$\rho_c(t)^{-1} |\bar{x} - x_c(t)| + \left| \frac{\bar{\rho}}{\rho_c(t)} - 1 \right| \geq r_{\text{sep}}$$

has separated local weighted mass in the same core-scale variables:

$$\left| \int_{B_r} |\rho_c(t)^{-1} u(x_c(t) + \rho_c(t)y, t)|^3 dy - \int_{B_r} |\bar{\rho}^{-1} u(\bar{x} + \bar{\rho}y, t)|^3 dy \right| \geq \eta.$$

This is the quantitative non-competing core condition, and is imposed on the selected physical core directly; it is not a consequence of the repaired-gauge construction.

- (SC4) For the selected core,

$$\sup_{t \in I_{\text{ph}}} \int_{B_{2R_*}} |y|^{-p} |\rho_c(t)^{-1} u(x_c(t) + \rho_c(t)y, t)|^3 dy \leq M,$$

for some $0 < p < 3$.

- (SC5) The preliminary frame associated with the selected core is the absolutely continuous representative

$$0 < \Lambda_{\min} \leq \Lambda_{\text{pre}}(\tau) \leq \Lambda_{\max}, \quad \Lambda_{\text{pre}} \in AC(I_*), \quad X_{\text{pre}} \in AC(I_*; \mathbb{R}^3).$$

It is tied to the selected physical core by the time change $dt_c/d\tau = \Lambda_{\text{pre}}(\tau)^2$, $t_c(\tau_0) = t_*$: one has

$$X_{\text{pre}}(\tau) = x_c(t_c(\tau)), \quad \Lambda_{\text{pre}}(\tau) = \rho_c(t_c(\tau))$$

for a.e. $\tau \in I_*$.

The parameter κ records the quantitative constants obtained in the local gauge estimates and is not used in items (SC1)–(SC5).

Remark 2.2. The selected core (x_c, ρ_c) and its separation/nondegeneracy conditions are specified in physical variables; no repaired-gauge constraint is used in their definition. This avoids the circularity of defining the core via the chart to be constructed.

2.2. Preliminary frame map from the selected core. This section records the transport map generated by the selected core; no new core extraction is proved here.

Define

$$\frac{dt_c}{d\tau} = \Lambda_{\text{pre}}(\tau)^2, \quad t_c(\tau_0) = t_*,$$

and use the compatibility in (SC5) to identify $\Lambda_{\text{pre}}(\tau) = \rho_c(t_c(\tau))$ and $X_{\text{pre}}(\tau) = x_c(t_c(\tau))$ a.e.

Proposition 2.3 (Preliminary frame map regularity). *Under $\text{SC}(R_*, r_*, M, \eta, \kappa)$, the map*

$$\Phi_{\text{pre}}(\tau, Y) := (x, t) = (X_{\text{pre}}(\tau) + \Lambda_{\text{pre}}(\tau)Y, t_c(\tau))$$

is well-defined and absolutely continuous in τ . Moreover, $1/\Lambda_{\text{pre}}, \Lambda_{\text{pre}} \in L^\infty(I_)$, $X_{\text{pre}} \in W_{\text{loc}}^{1,1}(I_*; \mathbb{R}^3)$, and for every $R \leq R_*$,*

$$\Phi_{\text{pre}} : Q_R^* \rightarrow \Phi_{\text{pre}}(Q_R^*) \subset Q_{2R_*}(x_\bullet, t_*)$$

is bi-Lipschitz with $\det \nabla_Y \Phi_{\text{pre}} = \Lambda_{\text{pre}}^3$, hence Jacobian and inverse-Jacobian bounds depend only on $\Lambda_{\min}, \Lambda_{\max}$.

Proof. From (SC5), $\Lambda_{\text{pre}} \in AC(I_*)$, $0 < \Lambda_{\min} \leq \Lambda_{\text{pre}} \leq \Lambda_{\max} < \infty$, and $X_{\text{pre}} \in AC(I_*; \mathbb{R}^3)$. The scalar ODE

$$\frac{dt_c}{d\tau} = \Lambda_{\text{pre}}(\tau)^2, \quad t_c(\tau_0) = t_*$$

therefore has an absolutely continuous strictly increasing solution on I_* , and

$$\frac{dt_c}{d\tau} = \Lambda_{\text{pre}}(\tau)^2 \quad \text{a.e. on } I_*.$$

The compatibility condition (SC5) identifies this absolutely continuous frame with the selected physical core a.e.; no differentiability of x_c or ρ_c beyond the frame regularity in (SC5) is used. For fixed τ , $\nabla_Y \Phi_{\text{pre}}(\tau, \cdot) = \Lambda_{\text{pre}}(\tau)I$, so

$$\det \nabla_Y \Phi_{\text{pre}}(\tau, \cdot) = \Lambda_{\text{pre}}(\tau)^3, \quad \Lambda_{\min}^3 \leq \det \nabla_Y \Phi_{\text{pre}} \leq \Lambda_{\max}^3.$$

Hence, if $Y_1, Y_2 \in B_R$,

$$\Lambda_{\min}|Y_1 - Y_2| \leq |\Phi_{\text{pre}}(\tau, Y_1) - \Phi_{\text{pre}}(\tau, Y_2)| \leq \Lambda_{\max}|Y_1 - Y_2|,$$

and therefore $\Phi_{\text{pre}}(\tau, \cdot)$ is bi-Lipschitz on B_R . Its inverse has Lipschitz constant $\Lambda_{\min}^{-1}$, so Jacobian and inverse-Jacobian bounds are controlled by $\Lambda_{\min}, \Lambda_{\max}$.

For image inclusions, if $x = X_{\text{pre}}(\tau) + \Lambda_{\text{pre}}(\tau)Y$ with $|Y| \leq R$, then

$$|x - X_{\text{pre}}(\tau)| \leq \Lambda_{\max}R, \quad X_{\text{pre}}(\tau) \in B_{R_*/2}(x_\bullet),$$

so $x \in B_{R_*/2 + \Lambda_{\max}R}(x_\bullet)$. As $\tau \in I_*$, we also have $t = t_c(\tau) \in I_{\text{ph}}$. Therefore

$$\Phi_{\text{pre}}(Q_R^*) \subset B_{R_*/2 + \Lambda_{\max}R}(x_\bullet) \times I_{\text{ph}}.$$

□

Definition 2.4 (Preliminary profile). For $(x, t) = \Phi_{\text{pre}}(\tau, Y)$, define

$$V_{\text{pre}}(Y, \tau) := \Lambda_{\text{pre}}(\tau) u(x, t), \quad P_{\text{pre}}(Y, \tau) := \Lambda_{\text{pre}}(\tau)^2 p(x, t).$$

Hence $u(x, t) = \Lambda_{\text{pre}}(\tau)^{-1} V_{\text{pre}}(Y, \tau)$ and $p(x, t) = \Lambda_{\text{pre}}(\tau)^{-2} P_{\text{pre}}(Y, \tau)$ under Φ_{pre} .

2.3. Exact renormalized change of variables. Define renormalized fields on Q_{2R}^* by

$$Y = \frac{x - X(\tau)}{\Lambda(\tau)}, \quad u(x, t) = \Lambda(\tau)^{-1} V(Y, \tau), \quad p(x, t) = \Lambda(\tau)^{-2} P(Y, \tau),$$

where $dt/d\tau = \Lambda(\tau)^2$, $X : I_* \rightarrow \mathbb{R}^3$, $\Lambda : I_* \rightarrow (0, \infty)$, and $\Lambda, X \in AC(I_*)$. Write

$$a(\tau) := -\Lambda'(\tau), \quad b(\tau) := -X'(\tau).$$

Lemma 2.5 (Exact differential identities). *For smooth data and fixed (x, t) , the chain rule gives*

$$\partial_t u = \Lambda^{-3} (\partial_\tau V - (X' + \Lambda'Y) \cdot \nabla_Y V - \Lambda'V),$$

$$\nabla_x u = \Lambda^{-2} \nabla_Y V, \quad \Delta_x u = \Lambda^{-3} \Delta_Y V, \quad \nabla_x p = \Lambda^{-3} \nabla_Y P.$$

In particular

$$\nabla_x \cdot u = \Lambda^{-2} \nabla_Y \cdot V.$$

Proof. For fixed (x, t) , $\tau = \tau(t)$ and $Y = (x - X(\tau))/\Lambda(\tau)$. Differentiate

$$u(x, t) = \Lambda(\tau)^{-1} V(Y, \tau)$$

in t :

$$\partial_t u = -\frac{\Lambda'}{\Lambda^2} V + \frac{1}{\Lambda} \partial_\tau V \frac{d\tau}{dt} + \frac{1}{\Lambda} \nabla_Y V \cdot \frac{dY}{dt}.$$

Since $d\tau/dt = \Lambda^{-2}$ and

$$Y_\tau = -\Lambda^{-1} X' - \frac{\Lambda'}{\Lambda} Y, \quad \frac{dY}{dt} = Y_\tau \frac{d\tau}{dt},$$

we have

$$\partial_t u = \Lambda^{-3} (\partial_\tau V - (X' + \Lambda' Y) \cdot \nabla_Y V - \Lambda' V).$$

For spatial derivatives, $Y = (x - X)/\Lambda$ gives $\nabla_x Y = \Lambda^{-1} I$, hence

$$\nabla_x u = \Lambda^{-2} \nabla_Y V, \quad \Delta_x u = \Lambda^{-3} \Delta_Y V.$$

Similarly $p(x, t) = \Lambda^{-2} P(Y, \tau)$ yields

$$\nabla_x p = \Lambda^{-3} \nabla_Y P.$$

Taking divergence of $u = \Lambda^{-1} V$ gives

$$\nabla_x \cdot u = \Lambda^{-2} \nabla_Y \cdot V.$$

□

Theorem 2.6 (Renormalized Navier–Stokes equation). *Assume u, p solve*

$$\partial_t u + (u \cdot \nabla_x) u + \nabla_x p - \nu \Delta_x u = 0, \quad \nabla_x \cdot u = 0$$

in distributions. Under the transformation above, V, P satisfy, on the image renormalized cylinder, the exact equation

$$\partial_\tau V + (V \cdot \nabla) V + \nabla P - \nu \Delta V + a(\tau)(V + Y \cdot \nabla V) + b(\tau) \cdot \nabla V = 0, \quad \nabla \cdot V = 0,$$

with the sign convention fixed by $a = -\Lambda'$, $b = -X'$. Conversely, any V, P satisfying this equation pull back to a distributional solution u, p of Navier–Stokes on the physical image.

Proof. To avoid hidden regularity loss, first write the physical equation tested against $\Phi \in C_c^\infty$ in the physical cylinder and then use the smooth pull-back formula (an approximation argument gives distributions). By Lemma 2.5,

$$\partial_t u + (u \cdot \nabla_x) u + \nabla_x p - \nu \Delta_x u = 0$$

becomes

$$\Lambda^{-3} (\partial_\tau V - (X' + \Lambda' Y) \cdot \nabla V - \Lambda' V) + \Lambda^{-3} (V \cdot \nabla) V + \Lambda^{-3} \nabla P - \Lambda^{-3} \nu \Delta V = 0.$$

Multiplying by Λ^3 and using $a = -\Lambda'$, $b = -X'$ gives

$$\partial_\tau V + (V \cdot \nabla) V + \nabla P - \nu \Delta V + a(V + Y \cdot \nabla V) + b \cdot \nabla V = 0.$$

The divergence condition is

$$\nabla_x \cdot u = \Lambda^{-2} \nabla \cdot V = 0 \implies \nabla \cdot V = 0.$$

Conversely, assume (V, P) solve this equation on the renormalized cylinder. Set u, p by the same change-of-variables map. Then the preceding identities run in reverse and give

$$\partial_t u + (u \cdot \nabla_x) u + \nabla_x p - \nu \Delta_x u = 0, \quad \nabla_x \cdot u = 0$$

in $\mathcal{D}'$ on the physical image.

□

2.4. Domain bookkeeping and support. Let $R > 0$ be fixed and $0 < \ell \leq \ell_0$. Set $Q_R^* = B_R \times I_*$. With Φ as above define physical support maps

$$\mathcal{Q}_R^{\text{ph}} := \Phi(Q_R^*), \quad \mathcal{Q}_{2R}^{\text{ph}} := \Phi(Q_{2R}^*).$$

Proposition 2.7 (Compact support transport). *Assume $\Lambda_{\min} \leq \Lambda \leq \Lambda_{\max}$ on I_* . Then*

$$X(\tau) \in B_{R_*/2}(x_\bullet), \quad |x - X(\tau)| \leq \Lambda_{\max} R,$$

$$\mathcal{Q}_R^{\text{ph}} \subset B_{R_*/2 + \Lambda_{\max} R}(x_\bullet) \times I_{\text{ph}},$$

$$\mathcal{Q}_{2R}^{\text{ph}} \subset B_{R_*/2 + 2\Lambda_{\max} R}(x_\bullet) \times I_{\text{ph}},$$

for every $\tau \in I_*$ and $x = X(\tau) + \Lambda(\tau)Y$ with $Y \in B_{2R}$. In particular, if $|c(\tau)| \leq \delta$ from Proposition 2.20, then $X(\tau) \in B_{R_*/2 + \Lambda_{\max} \delta}(x_\bullet)$ on I_* . Hence, $\chi_R \in C_c^\infty(B_{2R})$ pulls back to $\chi_R^\tau := \chi_R \circ \Phi^{-1} \in C_c^\infty(\mathcal{Q}_{2R}^{\text{ph}})$ with support depending only on $R, \Lambda_{\min}, \Lambda_{\max}$.

Proof. Fix $\tau \in I_*$ and $Y \in B_R$. Since $x = X(\tau) + \Lambda(\tau)Y$,

$$|x - X(\tau)| \leq |Y| \Lambda_{\max} \leq \Lambda_{\max} R, \quad \Lambda_{\max}^{-1} |x - X(\tau)| \leq |Y|.$$

Because $X(\tau) \in B_{R_*/2}(x_\bullet)$, this gives

$$x \in B_{R_*/2 + \Lambda_{\max} R}(x_\bullet).$$

Similarly, if $Y \in B_{2R}$, then $x \in B_{R_*/2 + 2\Lambda_{\max} R}(x_\bullet)$. Together with $t = t_c(\tau) \in I_{\text{ph}}$, these give the claimed inclusions for $\mathcal{Q}_R^{\text{ph}}$ and $\mathcal{Q}_{2R}^{\text{ph}}$.

For pull-backs of cutoffs, Φ is bi-Lipschitz at every τ , so composition

$$\chi_R^\tau(x, t) := \chi_R\left(\frac{x - X(\tau)}{\Lambda(\tau)}\right)$$

is smooth on $\mathcal{Q}_{2R}^{\text{ph}}$, supported where $Y \in B_{2R}$, hence supported where $(x, t) \in \mathcal{Q}_{2R}^{\text{ph}}$. The support size depends only on $R, \Lambda_{\min}, \Lambda_{\max}$ and the fixed C_c^∞ norm of χ_R , not on the particular branch. $\square$

2.5. Transport of local quantities under the chart. Let $t = t(\tau) = t_c(\tau)$ and, for $R > 0$, define

$$J_\tau^-(R) := (t(\tau) - \Lambda_{\max}^2 R^2, t(\tau)].$$

Define renormalized local quantities on $Q_R^*(\tau)$ by

$$\begin{aligned} A_V(\tau, R) &:= \sup_{s \in (\tau - R^2, \tau)} \int_{B_R} |V(\cdot, s)|^2 dY, & E_V(\tau, R) &:= \int_{\tau - R^2}^\tau \int_{B_R} |\nabla V|^2 dY ds, \\ C_V(\tau, R) &:= \int_{\tau - R^2}^\tau \int_{B_R} |V|^3 dY ds, & D_V(\tau, R) &:= \int_{\tau - R^2}^\tau \int_{B_R} |P - \langle P \rangle_{B_R}|^{3/2} dY ds. \end{aligned}$$

For interval notation and a center X , define

$$\begin{aligned} A_u(J, R; X) &:= \sup_{s \in J} \int_{B_R(X)} |u(\cdot, s)|^2 dx, & E_u(J, R; X) &:= \int_J \int_{B_R(X)} |\nabla u|^2 dx ds, \\ C_u(J, R; X) &:= \int_J \int_{B_R(X)} |u|^3 dx ds, & D_u(J, R; X) &:= \int_J \int_{B_R(X)} |p - \langle p \rangle_{B_R(X)}|^{3/2} dx ds. \end{aligned}$$

In the inequalities below we abbreviate

$$\begin{aligned} A_u(J, R) &:= A_u(J, R; X(\tau)), & E_u(J, R) &:= E_u(J, R; X(\tau)), \\ C_u(J, R) &:= C_u(J, R; X(\tau)), & D_u(J, R) &:= D_u(J, R; X(\tau)). \end{aligned}$$

Proposition 2.8 (Cylinder comparison on compact windows). *Assume $\Lambda_{\min} \leq \Lambda \leq \Lambda_{\max}$ and formulas of Lemma 2.5. Then for every $\tau \in I_*$, $R > 0$,*

$$\begin{aligned} A_V(\tau, R) &\leq \Lambda_{\min}^{-1} A_u(J_\tau^-(R), \Lambda_{\max} R), \\ E_V(\tau, R) &\leq \Lambda_{\min}^{-3} E_u(J_\tau^-(R), \Lambda_{\max} R), \\ C_V(\tau, R) &\leq \Lambda_{\min}^{-2} C_u(J_\tau^-(R), \Lambda_{\max} R), \\ D_V(\tau, R) &\leq \Lambda_{\min}^{-2} D_u(J_\tau^-(R), \Lambda_{\max} R). \end{aligned}$$

In particular, all four renormalized quantities are bounded whenever the physical quantities are bounded on the corresponding compact physical cylinders.

Proof. These are direct changes of variables under Φ_{pre} :

$$\begin{aligned} V(\cdot, \tau) &= \Lambda(\tau) u(\cdot, t(\tau)), & P(\cdot, \tau) &= \Lambda(\tau)^2 p(\cdot, t(\tau)), \\ \nabla_Y V &= \Lambda \nabla_x u, & dY d\tau &= \Lambda^{-5} dx dt, & d\tau &= \Lambda^2 dt. \end{aligned}$$

Insert these formulas into each definition and use $\Lambda_{\min} \leq \Lambda \leq \Lambda_{\max}$, plus

$$\tau - s \in (0, R^2) \implies t(\tau) - \Lambda_{\max}^2 R^2 \leq t(s) \leq t(\tau) - \Lambda_{\min}^2 R^2.$$

□

2.6. Local repaired gauge map. Fix $R > 0$, $0 < p < 3$, and $\chi_R \in C_c^\infty(B_{2R})$, $\chi_R \equiv 1$ on B_R . Let the local energy profile space be

$$\mathcal{V}_R := \left\{ V \in L^3(B_{2R}) \cap W^{1,3/2}(B_{2R}; \mathbb{R}^3) : |Y|^{-p/3} V \in L^3(B_{2R}) \right\},$$

with norm

$$\|V\|_{\mathcal{V}_R} := \|V\|_{L^3(B_{2R})} + \||Y|^{-p/3} V\|_{L^3(B_{2R})} + \|\nabla V\|_{L^{3/2}(B_{2R})}.$$

For differentiating the repaired scale and centering gauges we use the admissible gauge class

$$\mathcal{A}_{p,R} := \left\{ V : \begin{aligned} &V \in L^3(B_{4R}) \cap W_{\text{loc}}^{1,3}(B_{4R}; \mathbb{R}^3), \\ &\int_{B_{4R}} |Y|^{-p} |V|^3 dY < \infty, \quad \int_{B_{4R}} |Y|^{-p} |V| |Y \cdot \nabla V| dY < \infty, \\ &\int_{B_{4R}} |Y|^{-p-1} |V|^3 dY < \infty \end{aligned} \right\}.$$

The larger ball is only a domain buffer: all gauge integrals are supported in B_{2R} , while (λ, c) is restricted near $(1, 0)$ so that $\lambda^{-1}(B_{2R} - c) \subset B_{4R}$.

Definition 2.9 (Gauge actions). For $(\lambda, c) \in (0, \infty) \times \mathbb{R}^3$, define

$$V_{\lambda,c}(Y) := \lambda^{-1} V(\lambda^{-1}(Y - c)).$$

For pressure profiles use

$$P_{\lambda,c}(Y) := \lambda^{-2} P(\lambda^{-1}(Y - c)).$$

We also write $V^{\lambda,c} := V_{\lambda,c}$ and $P^{\lambda,c} := P_{\lambda,c}$.

Definition 2.10 (Finite-dimensional gauge map). For $V \in \mathcal{V}_R$ and $(\lambda, c) \in (0, \infty) \times \mathbb{R}^3$, set

$$\mathcal{F}(\lambda, c; V) := (\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3),$$

with

$$\mathcal{F}_0(\lambda, c; V) := \int_{\mathbb{R}^3} \chi_R(Y) |Y|^{-p} |V_{\lambda,c}(Y)|^3 dY - \Theta_0, \quad \mathcal{F}_j(\lambda, c; V) := \int_{\mathbb{R}^3} Y_j \chi_R(Y) |V_{\lambda,c}(Y)|^2 dY$$

for $j = 1, 2, 3$. The gauge conditions are $\mathcal{F}(\lambda, c; V) = 0$. Codomain is $\mathbb{R}^4$, domain $(0, \infty) \times \mathbb{R}^3 \times \mathcal{V}_R$.

Definition 2.11 (Localized modulation matrix). For $J \subset I_*$ and a profile path $W : J \rightarrow \mathcal{V}_R$, define

$$J_W(\tau) := D_{(\lambda,c)} \mathcal{F}(1, 0; W(\tau)) \in \mathbb{R}^{4 \times 4}.$$

The rows are ordered as the scale constraint followed by the three centering constraints.

Lemma 2.12 (Weighted integrability and differentiation under integrals). Assume $V, W \in \mathcal{A}_{p,R}$. Then $G_0(V) := \int \chi_R |Y|^{-p} |V|^3$ is finite and

$$\left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \int \chi_R |Y|^{-p} |V + \epsilon W|^3 dY = \int 3\chi_R |Y|^{-p} |V| V \cdot W dY.$$

Proof. For $W \in \mathcal{A}_{p,R}$, define

$$F_\epsilon(Y) := \chi_R(Y) |Y|^{-p} |V(Y) + \epsilon W(Y)|^3.$$

For each Y , $\epsilon \mapsto F_\epsilon(Y)$ is C^1 and

$$\partial_\epsilon F_\epsilon = 3\chi_R |Y|^{-p} |V + \epsilon W| (V + \epsilon W) \cdot W.$$

For $|\epsilon| \leq 1$,

$$|F_\epsilon(Y)| + |\partial_\epsilon F_\epsilon| \leq C \chi_R |Y|^{-p} (|V|^3 + |W|^3) \in L^1(B_{2R})$$

by the defining weighted L^3 condition of $\mathcal{A}_{p,R}$. Hence dominated convergence allows differentiation under the integral, and

$$\left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \int_{B_{2R}} \chi_R |Y|^{-p} |V + \epsilon W|^3 dY = \int_{B_{2R}} 3\chi_R |Y|^{-p} |V| V \cdot W dY.$$

□

Lemma 2.13 (Gauge covariance). For $V \in \mathcal{V}_R$, $(\lambda, c) \in (0, \infty) \times \mathbb{R}^3$, and $\alpha = 0, \dots, 3$,

$$\mathcal{F}_\alpha(1, 0; V_{\lambda,c}) = \mathcal{F}_\alpha(\lambda, c; V).$$

Proof. For each component this is a direct substitution:

$$\mathcal{F}_0(1, 0; V_{\lambda,c}) = \int \chi_R(Y) |Y|^{-p} |V_{\lambda,c}(Y)|^3 - \Theta_0 dY = \mathcal{F}_0(\lambda, c; V),$$

$$\mathcal{F}_j(1, 0; V_{\lambda,c}) = \int Y_j \chi_R(Y) |V_{\lambda,c}(Y)|^2 dY = \mathcal{F}_j(\lambda, c; V), \quad j = 1, 2, 3.$$

□

Proposition 2.14 (Differentiability of $\mathcal{F}$). For each fixed $V \in \mathcal{A}_{p,R}$, the map $(\lambda, c) \mapsto \mathcal{F}(\lambda, c; V)$ is C^1 in a neighborhood of $(1, 0)$. In each component $\mathcal{F}_\alpha$ all derivatives under the integral sign are legitimate.

Proof. Fix (λ, c) near $(1, 0)$, and abbreviate

$$V_{\lambda,c}(Y) := \lambda^{-1} V(\lambda^{-1}(Y - c)).$$

By direct differentiation,

$$\partial_\lambda V_{\lambda,c} = -\lambda^{-1} V_{\lambda,c} - \lambda^{-1} (Y - c) \cdot \nabla V_{\lambda,c}, \quad \partial_{c_k} V_{\lambda,c} = -\lambda^{-2} \partial_k V(\lambda^{-1}(Y - c)) = -\partial_k V_{\lambda,c}.$$

The neighborhood of $(1, 0)$ is chosen so that the arguments of V remain inside B_{4R} for $Y \in \text{supp} \chi_R$. The scale derivative is controlled by the two weighted integrability conditions

$$\int |Y|^{-p} |V|^3 < \infty, \quad \int |Y|^{-p} |V| |Y \cdot \nabla V| < \infty,$$

transported by the smooth change of variables. The translation derivative in the scale row is treated by the source identity

$$3 \int |Y|^{-p} |V| V \cdot \partial_k V = p \int Y_k |Y|^{-p-2} |V|^3,$$

obtained first for smooth compactly supported profiles and then by approximation in $\mathcal{A}_{p,R}$; the right-hand side is finite because $\int |Y|^{-p-1} |V|^3 < \infty$. The centering rows are supported in B_{2R} and are controlled by $V \in L_{\text{loc}}^2$ and $\nabla V \in L_{\text{loc}}^3$. These bounds give an integrable majorant for the difference quotients, so dominated convergence applies.

$$\mathcal{F}_0(\lambda, c; V) = \int_{\mathbb{R}^3} \chi_R(Y) |Y|^{-p} |V_{\lambda,c}|^3 dY - \Theta_0,$$

for which dominated convergence gives

$$\partial_\lambda \mathcal{F}_0 = \int_{\mathbb{R}^3} \chi_R |Y|^{-p} 3 |V_{\lambda,c}| V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c} dY,$$

$$\partial_{c_k} \mathcal{F}_0 = \int_{\mathbb{R}^3} \chi_R |Y|^{-p} 3 |V_{\lambda,c}| V_{\lambda,c} \cdot \partial_{c_k} V_{\lambda,c} dY.$$

Similarly

$$\mathcal{F}_j(\lambda, c; V) = \int_{\mathbb{R}^3} Y_j \chi_R(Y) |V_{\lambda,c}|^2 dY, \quad j = 1, 2, 3,$$

hence

$$\partial_\lambda \mathcal{F}_j = \int_{\mathbb{R}^3} 2 Y_j \chi_R V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c} dY,$$

$$\partial_{c_k} \mathcal{F}_j = \int_{\mathbb{R}^3} 2 Y_j \chi_R V_{\lambda,c} \cdot \partial_{c_k} V_{\lambda,c} dY.$$

Thus each component admits pointwise C^1 formulas and these are continuous in (λ, c) near $(1, 0)$. Therefore $(\lambda, c) \mapsto \mathcal{F}(\lambda, c; V) \in C^1$ near $(1, 0)$. $\square$

Proposition 2.15 (Full local Jacobian at $(1, 0)$). *Let $V \in \mathcal{A}_{p,R}$. Set $V_0(Y) = V(Y)$. At $(\lambda, c) = (1, 0)$, the matrix*

$$J(V) := D_{(\lambda,c)} \mathcal{F}(1, 0; V) \in \mathbb{R}^{4 \times 4}$$

has rows

$$J_\alpha = (\partial_\lambda \mathcal{F}_\alpha, \partial_{c_1} \mathcal{F}_\alpha, \partial_{c_2} \mathcal{F}_\alpha, \partial_{c_3} \mathcal{F}_\alpha), \quad \alpha = 0, \dots, 3.$$

Writing $W_{\text{scl}} = V_0 + Y \cdot \nabla V_0$, one has

$$(2) \quad \partial_\lambda \mathcal{F}_0(1, 0; V) = -3 \int_{\mathbb{R}^3} \chi_R |Y|^{-p} |V_0| V_0 \cdot W_{\text{scl}} dY,$$

$$(3) \quad \partial_{c_k} \mathcal{F}_0(1, 0; V) = -3 \int_{\mathbb{R}^3} \chi_R |Y|^{-p} |V_0| V_0 \cdot \partial_k V_0 dY,$$

$$(4) \quad \partial_\lambda \mathcal{F}_j(1, 0; V) = -2 \int_{\mathbb{R}^3} Y_j \chi_R V_0 \cdot W_{\text{scl}} dY,$$

$$(5) \quad \partial_{c_k} \mathcal{F}_j(1, 0; V) = -2 \int_{\mathbb{R}^3} \chi_R Y_j V_0 \cdot \partial_k V_0 dY,$$

for $j, k = 1, 2, 3$.

Proof. Differentiate the formulas in Definition 2.10 at $(\lambda, c) = (1, 0)$.

Since $V_{\lambda,c}(Y) = \lambda^{-1}V(\lambda^{-1}(Y - c))$,

$$\partial_\lambda V_{\lambda,c} = -\lambda^{-1}V_{\lambda,c} - \lambda^{-1}Y \cdot \nabla V_{\lambda,c}, \quad \partial_{c_k} V_{\lambda,c} = -\lambda^{-2}\partial_k V(\lambda^{-1}(Y - c)).$$

At $(1, 0)$, $\partial_\lambda V_{\lambda,c}|_{(1,0)} = -(V + Y \cdot \nabla V)$, $\partial_{c_k} V_{\lambda,c}|_{(1,0)} = -\partial_k V$.

For $\mathcal{F}_0$, write

$$\mathcal{F}_0(\lambda, c; V) = \int \chi_R(Y) |Y|^{-p} |V_{\lambda,c}(Y)|^3 dY - \Theta_0.$$

Derivative in λ :

$$\partial_\lambda \mathcal{F}_0 = \int \chi_R |Y|^{-p} 3|V_{\lambda,c}| V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c} dY$$

Evaluating at $(1, 0)$ yields (2).

For $k = 1, 2, 3$, c_k -derivative of $\mathcal{F}_0$ has no cutoff term:

$$\partial_{c_k} \mathcal{F}_0(1, 0; V) = \int \chi_R |Y|^{-p} 3|V| V \cdot (-\partial_k V) dY,$$

which is (3).

For $\mathcal{F}_j$,

$$\mathcal{F}_j(\lambda, c; V) = \int Y_j \chi_R(Y) |V_{\lambda,c}|^2 dY.$$

Hence

$$\partial_\lambda \mathcal{F}_j = \int 2Y_j \chi_R V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c} dY,$$

and at $(1, 0)$:

$$\partial_\lambda \mathcal{F}_j(1, 0; V) = -2 \int Y_j \chi_R V \cdot W_{\text{scl}} dY,$$

which is (4). Similarly,

$$\partial_{c_k} \mathcal{F}_j(1, 0; V) = \int 2Y_j \chi_R V \cdot \partial_{c_k} V_{\lambda,c}|_{(1,0)} dY = -2 \int \chi_R Y_j V \cdot \partial_k V dY,$$

giving (5). □

Proposition 2.16 (Translation block and Schur-complement invertibility). *Let $V : J \rightarrow \mathcal{A}_{p,R}$ satisfy the repaired gauge constraints*

$$\mathcal{F}(1, 0; V(\tau)) = 0 \quad \text{for a.e. } \tau \in J.$$

Write the modulation matrix in block form

$$J(V(\tau)) = \begin{pmatrix} \alpha(\tau) & u(\tau)^T \\ v(\tau) & A(\tau) \end{pmatrix},$$

where $\alpha = \partial_\lambda \mathcal{F}_0$, $u_\ell = \partial_{c_\ell} \mathcal{F}_0$, $v_j = \partial_\lambda \mathcal{F}_j$, and $A_{j\ell} = \partial_{c_\ell} \mathcal{F}_j$. Assume that for a.e. $\tau \in J$

$$m_\chi(\tau) := \int |V(Y, \tau)|^2 \chi_R(Y) dY \geq m_0 > 0,$$

$$\int_{A_R} |V(Y, \tau)|^2 dY \leq \varepsilon_0, \quad 3C_\chi \varepsilon_0 \leq \frac{m_0}{2},$$

where $A_R = \{R \leq |Y| \leq 2R\}$ and

$$C_\chi := \max_{j,\ell} \|Y_j \partial_\ell \chi_R\|_{L^\infty}.$$

Assume also

$$C_u := \text{ess sup}_{\tau \in J} \int |Y|^{-p-1} |V(Y, \tau)|^3 dY < \infty$$

and, with $C_\nabla := \| |Y| \nabla \chi_R \|_{L^\infty}$,

$$6p \left(\frac{2}{m_0} \right) C_u R C_\nabla \varepsilon_0 \leq \frac{p\Theta_0}{2}.$$

Then $J(V(\tau))$ is invertible for a.e. $\tau \in J$, and

$$\operatorname{ess\,sup}_{\tau \in J} \|J(V(\tau))^{-1}\|_{\infty \rightarrow \infty} \leq K(R, p, \Theta_0, m_0, C_u, \varepsilon_0, \chi_R).$$

Proof. The entries are finite by $V(\tau) \in \mathcal{A}_{p,R}$ and the compact support of χ_R . With the present parameter convention, $\partial_\lambda V_{\lambda,c}|_{(1,0)} = -(V + Y \cdot \nabla V)$. Hence the scale entry satisfies the signed transversality identity

$$\alpha(\tau) = -p \int |Y|^{-p} |V(Y, \tau)|^3 dY = -p\Theta_0$$

on the gauge surface. Thus $|\alpha| \geq p\Theta_0$.

For the centering block, integration by parts gives

$$A_{j\ell} = -\delta_{j\ell} \int |V|^2 \chi_R - \int |V|^2 Y_j \partial_\ell \chi_R.$$

Writing $A = -m_\chi I_3 + E$, the annular support of $\nabla \chi_R$ gives

$$|E_{j\ell}| \leq C_\chi \int_{A_R} |V|^2 \leq C_\chi \varepsilon_0, \quad \|E\|_{\infty \rightarrow \infty} \leq 3C_\chi \varepsilon_0 \leq m_0/2.$$

Thus $A = -m_\chi(I_3 - m_\chi^{-1}E)$ and $\|m_\chi^{-1}E\|_{\infty \rightarrow \infty} \leq 1/2$, so A^{-1} exists and $\|A^{-1}\|_{\infty \rightarrow \infty} \leq 2/m_0$.

For the scale-to-translation block, the repaired scale-row formula gives

$$|u_\ell| \leq p \int |Y|^{-p-1} |V|^3 dY \leq p \int |Y|^{-p-1} |V|^3 dY,$$

so $\|u\|_\infty \leq pC_u$. For the translation-to-scale block,

$$v_j = - \int |V|^2 Y_j (Y \cdot \nabla \chi_R) dY$$

because the centering constraint cancels the interior term. Therefore

$$\|v\|_\infty \leq 2R C_\nabla \varepsilon_0.$$

The Schur complement $S = \alpha - u^T A^{-1} v$ satisfies

$$|u^T A^{-1} v| \leq 3\|u\|_\infty \|A^{-1}\|_{\infty \rightarrow \infty} \|v\|_\infty \leq 6p \left(\frac{2}{m_0} \right) C_u R C_\nabla \varepsilon_0 \leq \frac{p\Theta_0}{2}.$$

Since $|\alpha| = p\Theta_0$, this gives $|S| \geq p\Theta_0/2$. The block inverse formula

$$J^{-1} = \begin{pmatrix} S^{-1} & -S^{-1} u^T A^{-1} \\ -A^{-1} v S^{-1} & A^{-1} + A^{-1} v S^{-1} u^T A^{-1} \end{pmatrix}$$

then gives the displayed bound. $\square$

2.7. Repaired gauge by implicit inversion in time.

Lemma 2.17 (Time chain rule for preliminary profile). *For each $\alpha = 0, \dots, 3$, set $G_\alpha(\tau) := \mathcal{F}_\alpha(1, 0; V_{\text{pre}}(\tau))$. Then $G_\alpha \in W_{\text{loc}}^{1,1}(I_*)$ and for a.e. τ ,*

$$G'_\alpha(\tau) = D_V \mathcal{F}_\alpha(1, 0; V_{\text{pre}}(\tau)) [\partial_\tau V_{\text{pre}}(\tau)].$$

The right-hand side is obtained from the profile formulas of Proposition 2.14 and is in $L_{\text{loc}}^1(I_)$.*

Proof. Let $V_{\text{pre}}^\varepsilon$ be a spatial and temporal Steklov mollification inside a slightly larger compact cylinder. For the smoothed profiles the ordinary chain rule gives

$$\frac{d}{d\tau} \mathcal{F}_\alpha(1, 0; V_{\text{pre}}^\varepsilon(\tau)) = D_V \mathcal{F}_\alpha(1, 0; V_{\text{pre}}^\varepsilon(\tau)) [\partial_\tau V_{\text{pre}}^\varepsilon(\tau)].$$

The transformed equation expresses $\partial_\tau V_{\text{pre}}^\varepsilon$ as the mollification of $\nu \Delta V_{\text{pre}} - (V_{\text{pre}} \cdot \nabla) V_{\text{pre}} - \nabla P_{\text{pre}}$ plus the chart transport terms. Pairing this distribution with the compactly supported Gateaux gradients below is legitimate by the local suitable bounds, the pressure bound, and the weighted L^3 control in (SC4). Passing $\varepsilon \downarrow 0$ gives the asserted $W_{\text{loc}}^{1,1}$ identity. For $W \in \mathcal{V}_R$, these derivative formulas hold:

$$\begin{aligned} D_V \mathcal{F}_0(1, 0; V)[W] &= \int_{\mathbb{R}^3} 3\chi_R |Y|^{-p} |V| V \cdot W \, dY, \\ D_V \mathcal{F}_j(1, 0; V)[W] &= \int_{\mathbb{R}^3} 2Y_j \chi_R V \cdot W \, dY, \quad j = 1, 2, 3, \end{aligned}$$

where both right-hand sides are finite on $\mathcal{V}_R$ by compact support, $p < 3$, and $V, W \in L^3 \cap W_{\text{loc}}^{1,3/2}$. The right-hand side is finite by the explicit formulas above and by the distributional equation paired with compactly supported test gradients; the local energy, pressure, and weighted L^3 bounds give an $L_{\text{loc}}^1(I_*)$ majorant. $\square$

Lemma 2.18 (AC implicit inversion with parameters). *Let $U \subset \mathbb{R}^4$ be open and $G : I_* \times U \rightarrow \mathbb{R}^4$. Assume:*

- (i) *for each $s \in U$, $\tau \mapsto G(\tau, s) \in W_{\text{loc}}^{1,1}(I_*; \mathbb{R}^4)$, and $D_\tau G \in L_{\text{loc}}^1(I_*; L^\infty(U; \mathbb{R}^4))$;*
- (ii) *for a.e. $\tau \in I_*$, $s \mapsto G(\tau, s)$ is C^1 on U , with $D_s G$ continuous in (τ, s) ;*
- (iii) *for some $(\tau_0, s_0) \in I_* \times U$, $G(\tau_0, s_0) = 0$, and there is $\rho, C > 0$ such that*

$$\sup_{(\tau, s) \in I_* \times B_\rho(s_0)} \|D_s G(\tau, s)^{-1}\|_{\mathcal{L}(\mathbb{R}^4, \mathbb{R}^4)} \leq C.$$

Then after possibly shrinking I_ to a subinterval $J \ni \tau_0$, there is a unique map $s \in AC(J; U)$, $s(\tau_0) = s_0$, with*

$$G(\tau, s(\tau)) = 0 \quad \text{a.e. in } J.$$

Moreover

$$s'(\tau) = -D_s G(\tau, s(\tau))^{-1} D_\tau G(\tau, s(\tau)) \quad \text{a.e. on } J.$$

Proof. By (ii), the matrix field $D_s G$ is continuous in s , so there is $r_0 \in (0, \rho)$ such that $\inf_{(\tau, s) \in I_* \times B_{r_0}(s_0)} |\det D_s G(\tau, s)| > 0$ after possibly shrinking I_* ; this keeps $D_s G^{-1}$ bounded and continuous on $B_{r_0}(s_0)$.

Set

$$F(\tau, s) := -D_s G(\tau, s)^{-1} D_\tau G(\tau, s).$$

Then $F(\tau, \cdot)$ is continuous on $B_{r_0}(s_0)$ for a.e. τ and $F(\cdot, s) \in L_{\text{loc}}^1(I_*; \mathbb{R}^4)$ by (i), hence F is a Carathéodory right-hand side. Since the gauge map is C^1 in the finite variables, after shrinking r_0 there is $L \in L^1(J)$ such that

$$|F(\tau, s_1) - F(\tau, s_2)| \leq L(\tau) |s_1 - s_2| \quad \text{for } s_1, s_2 \in B_{r_0}(s_0).$$

Choose $\delta > 0$ so small that

$$\int_{\tau_0 - \delta}^{\tau_0 + \delta} \sup_{s \in B_{r_0}(s_0)} |F(\sigma, s)| \, d\sigma < r_0.$$

On $J = [\tau_0 - \delta, \tau_0 + \delta] \cap I_*$ define the Picard operator

$$(\mathcal{T}s)(\tau) := s_0 + \int_{\tau_0}^{\tau} F(\sigma, s(\sigma)) \, d\sigma$$

on $C(J; B_{r_0}(s_0))$. The preceding bound keeps $\mathcal{T}$ in the same closed ball, and the Lipschitz estimate gives

$$\|\mathcal{T}s_1 - \mathcal{T}s_2\|_{C(J)} \leq \left(\int_J L(\sigma) d\sigma \right) \|s_1 - s_2\|_{C(J)}.$$

After decreasing δ once more, the coefficient is < 1 , so Banach's fixed-point theorem gives a unique solution

$$s(\tau) = s_0 + \int_{\tau_0}^{\tau} F(\sigma, s(\sigma)) d\sigma, \quad \tau \in J,$$

which stays in $B_{r_0}(s_0) \subset U$. This solution is absolutely continuous.

Define $H(\tau) := G(\tau, s(\tau))$. Since $G(\cdot, s(\cdot)) \in W_{\text{loc}}^{1,1}$ from (i), the chain rule gives for a.e. $\tau \in J$

$$H'(\tau) = D_{\tau}G(\tau, s(\tau)) + D_sG(\tau, s(\tau))s'(\tau) = D_{\tau}G(\tau, s(\tau)) + D_sG(\tau, s(\tau))F(\tau, s(\tau)) = 0.$$

Hence $H \equiv G(\tau_0, s_0) = 0$ on J , which is the claimed constraint. Uniqueness follows from Grönwall on the difference of two AC solutions on $B_{r_0}(s_0)$, using Lipschitz continuity of $F(\tau, \cdot)$ on that ball. $\square$

Definition 2.19 (Retained nondegenerate repaired-gauge stratum). Let $J \subset I_*$ be compact. The retained nondegenerate repaired-gauge stratum over the preliminary profile V_{pre} consists of branches for which there are

$$\lambda \in AC(J; (0, \infty)), \quad c \in AC(J; \mathbb{R}^3),$$

such that, for a.e. $\tau \in J$,

$$V(\tau) := V_{\text{pre}}^{\lambda(\tau), c(\tau)} \in \mathcal{A}_{p,R}, \quad \mathcal{F}(1, 0; V(\tau)) = 0.$$

The path is quantitatively nondegenerate if the translation-block, annular, weighted first-moment, and Schur-complement bounds in Proposition 2.16 hold on J . If no such local AC selection exists, or if the quantitative nondegeneracy package fails persistently, the branch is not in this stratum; it is routed to the gauge-degenerate, multibubble, or cascade alternatives.

Proposition 2.20 (Gauge regularity on the retained stratum). *Let (λ, c) be the repaired-gauge selection of the retained nondegenerate stratum on J . Then*

$$\mathcal{F}(\lambda(\tau), c(\tau); V_{\text{pre}}(\tau)) = 0 \quad \text{for a.e. } \tau \in J,$$

and $(\lambda, c) \in AC(J; \mathbb{R}^4)$ by definition. If the path is quantitatively nondegenerate, then

$$\text{ess sup}_{\tau \in J} \|J(V(\tau))^{-1}\| < \infty.$$

Proof. The constraint follows from Lemma 2.13 applied to $V(\tau) = V_{\text{pre}}^{\lambda(\tau), c(\tau)}$. The absolute continuity is part of Definition 2.19. The final assertion is exactly Proposition 2.16. $\square$

Definition 2.21 (Repaired chart). Define

$$\begin{aligned} \Lambda(\tau) &:= \Lambda_{\text{pre}}(\tau)\lambda(\tau), & X(\tau) &:= X_{\text{pre}}(\tau) + \Lambda_{\text{pre}}(\tau)c(\tau), \\ V(\tau) &:= V_{\text{pre}}^{\lambda(\tau), c(\tau)}, & P(\tau) &:= P_{\text{pre}}^{\lambda(\tau), c(\tau)} \text{ (same physical pressure pull-back under } \Phi). \end{aligned}$$

Theorem 2.22 (AC repaired parameters). *Assume $\text{SC}(R_*, r_*, M, \eta, \kappa)$, and suppose the branch lies in the retained nondegenerate repaired-gauge stratum on J , with selection (λ, c) . Then the corrected chart (Λ, X) is absolutely continuous on J , satisfies*

$$a(\tau) = -\Lambda'(\tau) \in L_{\text{loc}}^1(J), \quad b(\tau) = -X'(\tau) \in L_{\text{loc}}^1(J),$$

and the gauge constraints $\mathcal{F}(1, 0; V(\tau)) = 0$ hold a.e. Equivalently,

$$\mathcal{F}(\lambda(\tau), c(\tau); V_{\text{pre}}(\tau)) = 0$$

for the repaired-gauge selection (λ, c) .

Proof. By Definition 2.19, $(\lambda, c) \in AC(J)$ and the repaired gauge constraints hold for $V(\tau)$ a.e. By definition $V(\tau) = V_{\text{pre}}^{\lambda(\tau), c(\tau)}$, so Lemma 2.13 applied at $(\lambda, c) = (\lambda(\tau), c(\tau))$ gives

$$\mathcal{F}(1, 0; V_{\text{pre}}^{\lambda, c}) = \mathcal{F}(\lambda, c; V_{\text{pre}})$$

implies $\mathcal{F}(1, 0; V(\tau)) = 0$ a.e.

Since $\Lambda = \Lambda_{\text{pre}}\lambda$ and $X = X_{\text{pre}} + \Lambda_{\text{pre}}c$, and $\Lambda_{\text{pre}}, X_{\text{pre}}$ are AC by (SC5), the products and sums preserve AC, hence $(\Lambda, X) \in AC(I_*)$. Differentiating,

$$a = -\Lambda' = -(\Lambda'_{\text{pre}}\lambda + \Lambda_{\text{pre}}\lambda') \in L_{\text{loc}}^1, \quad b = -X' = -(X'_{\text{pre}} + \Lambda'_{\text{pre}}c + \Lambda_{\text{pre}}c') \in L_{\text{loc}}^1.$$

This proves the claimed regularity. $\square$

Remark 2.23 (Gauge degeneracy). Failure of invertibility of J is the gauge-degenerate exit from the retained single-core stratum for this fixed renormalization mechanism. In the Type II state-space stratification this exit is handled with the multibubble/cascade and scale-rigidity alternatives, not inside the present representation theorem.

2.8. Pressure transport and decomposition.

Lemma 2.24 (Renormalized pressure equation). *If V, P satisfy the renormalized Navier–Stokes equation in Theorem 2.6, then*

$$-\Delta P = \partial_i \partial_j (V_i V_j) \quad \text{in } \mathcal{D}'(Q_{2R}^*).$$

Proof. Take test $\varphi \in C_c^\infty(Q_{2R}^*)$. Since $V, P \in L_{\text{loc}}^2 \times L_{\text{loc}}^{3/2}$ on Q_{2R}^* , all pairings below are legitimate. Expand

$$\nabla \cdot ((V \cdot \nabla)V) = \sum_{i,j} \partial_i \partial_j (V_i V_j) - \sum_{i,j} \partial_i V_i \partial_j V_j.$$

Because $\nabla \cdot V = 0$, this reduces to $\partial_i \partial_j (V_i V_j)$. Taking divergence of

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu \Delta V + a(V + Y \cdot \nabla V) + b \cdot \nabla V = 0$$

and using $\nabla \cdot V = 0$, $\nabla \cdot (V + Y \cdot \nabla V) = 0$, $\nabla \cdot (b \cdot \nabla V) = b \cdot \nabla (\nabla \cdot V) = 0$, and $\nabla \cdot \Delta V = \Delta (\nabla \cdot V) = 0$, we get in $\mathcal{D}'$:

$$\Delta P + \partial_i \partial_j (V_i V_j) = 0.$$

Hence $-\Delta P = \partial_i \partial_j (V_i V_j)$ on Q_{2R}^* . $\square$

Theorem 2.25 (Localized pressure decomposition). *Let $\zeta \in C_c^\infty(B_{2R})$, $\zeta \equiv 1$ on B_R , and assume $V \in L_{\text{loc}}^1(I_*; L^3(B_{2R}))$, $P \in L_{\text{loc}}^1(I_*; L^{3/2}(B_{2R}))$. For a.e. $\tau \in I_*$ there exists*

$$P(\cdot, \tau) = P_{\text{loc}}(\cdot, \tau) + H(\cdot, \tau)$$

on B_R such that

$$\begin{cases} -\Delta P_{\text{loc}}(\cdot, \tau) = \partial_i \partial_j (\zeta V_i V_j), \\ -\Delta H(\cdot, \tau) = 0 \text{ on } B_R, \quad \int_{B_R} H(\cdot, \tau) dY = 0. \end{cases}$$

Moreover

$$\|P_{\text{loc}}(\tau)\|_{L^{3/2}(B_R)} \lesssim \|V(\tau)\|_{L^3(B_{2R})}^2, \quad \|H(\tau)\|_{L^q(B_r)} \lesssim_{q,R,r} \|P(\tau)\|_{L^{3/2}(B_{2R})} + \|V(\tau)\|_{L^3(B_{2R})}^2$$

for each $1 \leq q < \infty$ and each $0 < r < R$, with constants depending only on R, r , and q .

Proof. For fixed τ , set

$$P_{\text{loc}} = \mathcal{R}_i \mathcal{R}_j (\zeta V_i V_j), \quad H := P - P_{\text{loc}}.$$

$\zeta V_i V_j \in L^1_{\text{loc}}(I_*; L^{3/2}(B_{2R}))$, so the Calderón–Zygmund operator $f \mapsto \mathcal{R}_i \mathcal{R}_j f$ gives

$$P_{\text{loc}}(\cdot, \tau) = \mathcal{R}_i \mathcal{R}_j (\zeta V_i V_j)(\cdot, \tau), \quad \|P_{\text{loc}}(\tau)\|_{L^{3/2}(B_R)} \lesssim \|V(\tau)\|_{L^3(B_{2R})}^2.$$

Because $\mathcal{R}_i \mathcal{R}_j$ is linear bounded on $L^{3/2}$,

$$P_{\text{loc}} \in L^1_{\text{loc}}(I_*; L^{3/2}(B_R)).$$

Since $-\Delta P_{\text{loc}} = \partial_i \partial_j (\zeta V_i V_j)$ and $\zeta \equiv 1$ on B_R ,

$$-\Delta H(\cdot, \tau) = \partial_i \partial_j ((1 - \zeta) V_i V_j) = 0 \quad \text{in } B_R,$$

so $H(\cdot, \tau)$ is harmonic on B_R . Imposing $\oint_{B_R} H = 0$ fixes the time-dependent additive constant uniquely in each time slice. For $q \in [1, \infty)$ and $0 < r < R$, harmonicity and local L^p -to- L^q interior estimates imply

$$\|H(\tau)\|_{L^q(B_r)} \lesssim_{q,R,r} \|H(\tau)\|_{L^{3/2}(B_R)}.$$

Finally, $\|H(\tau)\|_{L^{3/2}(B_R)} \leq \|P(\tau)\|_{L^{3/2}(B_R)} + \|P_{\text{loc}}(\tau)\|_{L^{3/2}(B_R)}$, giving the announced bound in terms of P and V . $\square$

Remark 2.26 (Measurability and normalization). If V, P are measurable in τ , then $\tau \mapsto P_{\text{loc}}(\cdot, \tau)$ is measurable in $L^{3/2}(B_R)$ by boundedness of $\mathcal{R}_i \mathcal{R}_j$, hence $\tau \mapsto H(\cdot, \tau)$ is measurable in each $L^q(B_r)$, $0 < r < R$. The normalization $\oint_{B_R} H(\cdot, \tau) dY = 0$ fixes the time-dependent additive constant in H uniquely for each τ , and therefore the decomposition is a legitimate renormalized pressure splitting.

2.9. Modulation from differentiated constraints.

Lemma 2.27 (Differentiation of constraints). *For the repaired profile V constructed above, with $|Y|^{-p}|V|^3 \in L^1_{\text{loc}}(I_*; L^1)$, Then $G_\alpha(\tau) := \mathcal{F}_\alpha(1, 0; V(\tau))$ are absolutely continuous and*

$$\frac{d}{d\tau} G_\alpha(\tau) = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \partial_\tau V dY$$

for a.e. τ . Here $\mathbf{D}_\alpha$ is the L^1 -integrable Gateaux gradient associated to $\mathcal{F}_\alpha$.

Proof.

$$G_\alpha(\tau) = \mathcal{F}_\alpha(1, 0; V(\tau))$$

is first evaluated on spatial-temporal mollifications V^ε . For those smooth profiles the ordinary chain rule gives

$$\frac{d}{d\tau} G_\alpha^\varepsilon(\tau) = D_V \mathcal{F}_\alpha(1, 0; V^\varepsilon(\tau)) [\partial_\tau V^\varepsilon(\tau)].$$

The local suitable bounds, pressure estimates, and compact support of the Gateaux gradients below allow passage to the limit $\varepsilon \downarrow 0$, which gives the displayed derivative formula for V . For $\mathcal{F}_0$,

$$\mathbf{D}_0(V, \nabla V, \nabla \chi_R) = 3\chi_R |Y|^{-p} |V| V;$$

for $\alpha = 1, 2, 3$,

$$\mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) = 2\chi_R Y_\alpha V.$$

Each coefficient is $L^1(B_{2R})$: $3\chi_R |Y|^{-p} |V| V \in L^{3/2}(B_{2R}) \subset L^1(B_{2R})$ from $p < 3$ and $|Y|^{-p} |V|^3 \in L^1$, and $2\chi_R Y_\alpha V \in L^2(B_{2R}) \subset L^1(B_{2R})$ by compact support.

$$\frac{d}{d\tau} G_\alpha(\tau) = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \cdot \partial_\tau V dY,$$

Since $\mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \in L^1(B_{2R})$ and $\partial_\tau V \in L^1_{\text{loc}}(I_*; L^2(B_{2R}))$, the right-hand side is in $L^1_{\text{loc}}(I_*)$, so $G_\alpha \in W^{1,1}_{\text{loc}}(I_*) \subset AC_{\text{loc}}(I_*)$. $\square$

Lemma 2.28 (Directional derivatives in the profile variable). *For $\mathcal{A}_\tau = V + Y \cdot \nabla V$ and $\alpha = 0, 1, 2, 3$,*

$$D_V \mathcal{F}_\alpha(1, 0; V)[\mathcal{A}_\tau] = -\partial_\lambda \mathcal{F}_\alpha(1, 0; V), \quad D_V \mathcal{F}_\alpha(1, 0; V)[\partial_k V] = -\partial_{c_k} \mathcal{F}_\alpha(1, 0; V).$$

Indeed, from $V_{\lambda,c}(Y) = \lambda^{-1}V(\lambda^{-1}(Y - c))$,

$$\partial_\lambda V_{\lambda,c}|_{(1,0)} = -(V + Y \cdot \nabla V) = -\mathcal{A}_\tau, \quad \partial_{c_k} V_{\lambda,c}|_{(1,0)} = -\partial_k V.$$

Substituting these into the Gateaux formulas from Proposition 2.14 gives the two identities.

Theorem 2.29 (Modulation system). *Let (V, P) be the repaired pair on the compact window under consideration. Writing $\theta := (a, b)$, define*

$$\mathcal{A}_\tau(Y) := V(\tau, Y) + Y \cdot \nabla V(\tau, Y).$$

For $\alpha = 0, 1, 2, 3$, define

$$\mathcal{R}_\alpha(\tau) := - \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \cdot ((V \cdot \nabla)V + \nabla P - \nu \Delta V) \, dY.$$

Then for a.e. $\tau \in I_*$,

$$\partial_\lambda \mathcal{F}_\alpha(1, 0; V) a(\tau) + \sum_{k=1}^3 \partial_{c_k} \mathcal{F}_\alpha(1, 0; V) b_k(\tau) = \mathcal{R}_\alpha(\tau),$$

Equivalently,

$$J(\tau)\theta(\tau) = \mathcal{R}(\tau).$$

Hence

$$\theta(\tau) = J(\tau)^{-1} \mathcal{R}(\tau) \in L^1_{\text{loc}}(I_*)^4, \quad |\theta(\tau)| \leq \kappa |\mathcal{R}(\tau)|.$$

Proof. Set $G_\alpha(\tau) := \mathcal{F}_\alpha(1, 0; V(\tau))$. By Theorem 2.22, $G_\alpha(\tau) = 0$ a.e.; by Lemma 2.27, $G_\alpha \in W^{1,1}_{\text{loc}}$. Hence for a.e. τ ,

$$0 = \frac{d}{d\tau} G_\alpha(\tau) = D_V \mathcal{F}_\alpha(1, 0; V(\tau)) [\partial_\tau V(\tau)].$$

Insert the equation from Theorem 2.6,

$$\partial_\tau V = -(V \cdot \nabla)V - \nabla P + \nu \Delta V - a \mathcal{A}_\tau - b \cdot \nabla V.$$

This gives

$$0 = -D_V \mathcal{F}_\alpha[(V \cdot \nabla)V + \nabla P - \nu \Delta V] - a D_V \mathcal{F}_\alpha[\mathcal{A}_\tau] - \sum_{k=1}^3 b_k D_V \mathcal{F}_\alpha[\partial_k V].$$

By Lemma 2.27, for any profile direction W ,

$$D_V \mathcal{F}_\alpha(1, 0; V)[W] = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \cdot W \, dY.$$

By Lemma 2.28,

$$D_V \mathcal{F}_\alpha[\mathcal{A}_\tau] = -\partial_\lambda \mathcal{F}_\alpha(1, 0; V), \quad D_V \mathcal{F}_\alpha[\partial_k V] = -\partial_{c_k} \mathcal{F}_\alpha(1, 0; V).$$

Substituting into the previous line yields the linear system

$$\partial_\lambda \mathcal{F}_\alpha(1, 0; V) a(\tau) + \sum_{k=1}^3 \partial_{c_k} \mathcal{F}_\alpha(1, 0; V) b_k(\tau) = \mathcal{R}_\alpha(\tau),$$

which is equivalent to $J(\tau)\theta(\tau) = \mathcal{R}(\tau)$. By definition of $\mathbf{D}_\alpha$, each $\mathcal{R}_\alpha \in L^1_{\text{loc}}(I_*)$, and Proposition 2.16 gives $|J^{-1}(\tau)| \leq K$ a.e., hence

$$\theta(\tau) = J^{-1}(\tau)\mathcal{R}(\tau), \quad \theta \in L^1_{\text{loc}}(I_*), \quad |\theta| \leq K|\mathcal{R}| \text{ a.e.}$$

□

Corollary 2.30 (Regularity of a, b). *For the repaired pair of Theorem 2.29, $a, b \in L^1_{\text{loc}}(I_*)$. If in addition $\mathcal{R} \in L^\infty_{\text{loc}}(I_*)$, then $a, b \in L^\infty_{\text{loc}}(I_*)$.*

Proof. By Theorem 2.29, $\theta = (a, b) = J^{-1}\mathcal{R}$ a.e., and $J^{-1} \in L^\infty_{\text{loc}}(I_*)$ by Theorem 2.16. The first claim is therefore immediate from $\mathcal{R} \in L^\infty_{\text{loc}}$, while the second follows if $\mathcal{R} \in L^\infty_{\text{loc}}$. □

2.10. Repaired suitable structure in renormalized variables.

Proposition 2.31 (Transport of suitable local energy inequality). *Suppose (u, p) is suitable on $\mathcal{Q}_{2R}^{\text{ph}}$, and (V, P) is obtained by the repaired map Φ . Set*

$$\tilde{a}(\tau) := -\Lambda(\tau)^{-1}\Lambda'(\tau), \quad \tilde{b}(\tau) := -\Lambda(\tau)^{-1}X'(\tau).$$

Then for every nonnegative $\phi \in C_c^\infty(Q_R^)$ and $\tau_1 < \tau_2$,*

$$\begin{aligned} \mathcal{E}_\phi(V, P) &:= \frac{1}{2}|V|^2\partial_\tau(\phi^2) + \frac{\nu}{2}|V|^2\Delta_Y\phi^2 + \frac{1}{2}|V|^2V \cdot \nabla(\phi^2) \\ &\quad + (P_{\text{loc}} + H)V \cdot \nabla(\phi^2) + \frac{\tilde{a}}{2}|V|^2\phi^2 + \frac{\tilde{a}}{2}|V|^2Y \cdot \nabla(\phi^2) + \frac{1}{2}|V|^2\tilde{b} \cdot \nabla(\phi^2). \end{aligned}$$

$$\begin{aligned} \frac{1}{2} \int_{B_R} |V(\tau_2)|^2 \phi(\cdot, \tau_2)^2 + \nu \int_{\tau_1}^{\tau_2} \int_{B_R} |\nabla V|^2 \phi^2 dY d\tau &\leq \frac{1}{2} \int_{B_R} |V(\tau_1)|^2 \phi(\cdot, \tau_1)^2 \\ &\quad + \int_{\tau_1}^{\tau_2} \int_{B_R} \mathcal{E}_\phi(V, P) dY d\tau. \end{aligned}$$

The right-hand side is controlled on compact windows by $\|\tilde{a}\|_{L^1}, \|\tilde{b}\|_{L^1}, M$, and the local bounds from Theorem 2.25, and support bounds from Section 2.4. Hence (V, P) is locally suitable on Q_R^ .*

Proof. Apply the physical local energy inequality on $\mathcal{Q}_{2R}^{\text{ph}}$: for nonnegative $\psi \in C_c^\infty(\mathcal{Q}_{2R}^{\text{ph}})$,

$$\frac{1}{2} \int |u|^2 \psi|_{t_1}^{t_2} + \nu \int_{t_1}^{t_2} \int |\nabla_x u|^2 \psi dx dt \leq \int_{t_1}^{t_2} \int \left(\frac{1}{2}|u|^2(\partial_t \psi + \nu \Delta_x \psi) + \left(\frac{|u|^2}{2}u + pu \right) \cdot \nabla_x \psi \right) dx dt.$$

Choose

$$\psi(x, t) := \Lambda(t)^{-1} \phi\left(\frac{x - X(t)}{\Lambda(t)}, \tau(t)\right)^2, \quad \tau = \tau(t), \quad d\tau/dt = \Lambda^{-2},$$

let $t_i = t(\tau_i)$, $i = 1, 2$. By Proposition 2.3 and Section 2.4, $\psi \in C_c^\infty(\mathcal{Q}_{2R}^{\text{ph}})$.

$$\nabla_x \psi = \Lambda^{-2} \nabla_Y \phi^2, \quad \Delta_x \psi = \Lambda^{-3} \Delta_Y \phi^2, \quad \partial_t \psi = -\Lambda^{-4} \Lambda' \phi^2 + \Lambda^{-3} \partial_\tau(\phi^2) - \Lambda^{-4} (X' + \Lambda' Y) \cdot \nabla_Y \phi^2.$$

Using $u = \Lambda^{-1}V$, $p = \Lambda^{-2}P$, $\nabla_x u = \Lambda^{-2} \nabla_Y V$, $dx dt = \Lambda^5 dY d\tau$, each term in the physical inequality becomes

$$\begin{aligned} \frac{1}{2} \int |u|^2 \psi|_{t_1}^{t_2} &= \frac{1}{2} \int |V|^2 \phi^2|_{\tau_1}^{\tau_2}, \\ \nu \int |\nabla_x u|^2 \psi dx dt &= \nu \int |\nabla V|^2 \phi^2 dY d\tau, \\ \frac{1}{2} \int |u|^2 \partial_t \psi dx dt &= \frac{1}{2} \int |V|^2 \left(\partial_\tau(\phi^2) + \tilde{a}\phi^2 + \tilde{a}Y \cdot \nabla_Y \phi^2 + \tilde{b} \cdot \nabla_Y \phi^2 \right) dY d\tau, \end{aligned}$$

$$\begin{aligned} \frac{\nu}{2} \int |u|^2 \Delta_x \psi \, dx dt &= \frac{\nu}{2} \int |V|^2 \Delta_Y \phi^2 \, dY \, d\tau, \\ \int \left(\frac{|u|^2}{2} u + p u \right) \cdot \nabla_x \psi \, dx dt &= \int \left(\frac{1}{2} |V|^2 V + (P_{\text{loc}} + H) V \right) \cdot \nabla_Y \phi^2 \, dY \, d\tau. \end{aligned}$$

Substituting these identities and grouping terms gives the proposition's right-hand side and coefficients $\tilde{a}, \tilde{b}$. $\square$

2.11. Stability under subsequences.

Proposition 2.32. *Let $(u^{(n)}, p^{(n)})$ satisfy $\text{SC}(R_*, r_*, M, \eta, \kappa)$ with the same data parameters and let $(\Lambda^{(n)}, X^{(n)}, V^{(n)}, P^{(n)}, a^{(n)}, b^{(n)})$ be obtained from retained quantitatively nondegenerate repaired-gauge selections on the same I_* , with the constants in Proposition 2.16 uniform in n . After extraction $n_j \rightarrow \infty$, there exist subsequences and limits (Λ, X, V, P, a, b) such that*

$$\begin{aligned} \Lambda^{(n_j)} &\rightarrow \Lambda \text{ in } L_{\text{loc}}^1, & X^{(n_j)} &\rightarrow X \text{ in } L_{\text{loc}}^1, \\ \Lambda^{(n_j)'} &\rightharpoonup \Lambda' \text{ in } L_{\text{loc}}^1, & X^{(n_j)'} &\rightharpoonup X' \text{ in } L_{\text{loc}}^1, \\ V^{(n_j)} &\rightarrow V \text{ in } L_{\text{loc}}^2(I_*; L_{\text{loc}}^2), & V^{(n_j)} &\rightharpoonup V \text{ in } L_{\text{loc}}^2(I_*; H_{\text{loc}}^1), \\ P^{(n_j)} &\rightharpoonup P \text{ in } L_{\text{loc}}^{3/2}(Q_R^*), & (a^{(n_j)}, b^{(n_j)}) &\rightharpoonup (a, b) \text{ in } L_{\text{loc}}^1(I_*)^4. \end{aligned}$$

Moreover $a = -\Lambda'$, $b = -X'$ a.e. on I_* , and the limit obeys all conclusions below.

Proof. From SC and the retained repaired-gauge selections, $(V^{(n)}, P^{(n)})$ are uniformly bounded in $L_\tau^\infty L^2(B_{2R}) \cap L_\tau^2 H^1(B_{2R})$ and $L_{\text{loc}}^{3/2}(Q_R^*)$. Theorem 2.16 gives uniform invertibility bounds for $J^{(n)}$, and Theorem 2.29 gives $(a^{(n)}, b^{(n)})$ uniformly in L_{loc}^1 (since $\mathcal{R}^{(n)} \in L_{\text{loc}}^1$). Hence $\Lambda^{(n)}$ and $X^{(n)}$ are uniformly bounded in $W_{\text{loc}}^{1,1}$, so by BV compactness and 1D subsequences,

$$\Lambda^{(n_j)} \rightarrow \Lambda \text{ in } L_{\text{loc}}^1, \quad X^{(n_j)} \rightarrow X \text{ in } L_{\text{loc}}^1,$$

and $a^{(n_j)} \rightharpoonup -\Lambda'$, $b^{(n_j)} \rightharpoonup -X'$ in L_{loc}^1 . From the renormalized equation,

$$\partial_\tau V^{(n)} \in L_\tau^1(H_{\text{loc}}^{-1}) + L_\tau^{4/3}(L_{\text{loc}}^{12/11}),$$

uniformly on compact windows. Aubin–Lions compactness on bounded balls gives

$$V^{(n_j)} \rightarrow V \text{ in } L_{\text{loc}}^2(I_*; L_{\text{loc}}^2), \quad V^{(n_j)} \rightharpoonup V \text{ in } L_{\text{loc}}^2(I_*; H_{\text{loc}}^1).$$

By Calderón–Zygmund and Theorem 2.25, $P^{(n_j)} \rightharpoonup P$ in $L_{\text{loc}}^{3/2}(Q_R^*)$, and with normalized pressure split $P = P_{\text{loc}} + H$, the same for P_{loc} and H . Passing to limits in $\mathcal{F}(1, 0; V^{(n)}(\tau)) = 0$, Theorem 2.6, and the linear modulation system yields the corresponding limit constraints and equations. $\square$

2.12. Final representation package.

Theorem 2.33 (Chart and repaired-gauge representation). *Assume $\text{SC}(R_*, r_*, M, \eta, \kappa)$, and suppose the branch lies in the retained repaired-gauge stratum with selection (λ, c) on $J = [\tau_0 - \ell, \tau_0 + \ell] \subset I_*$. Then there are fields (V, P) on Q_{2R}^J , with $(\Lambda, X) \in AC(J)$, $a = -\Lambda'$, $b = -X' \in L_{\text{loc}}^1(J)$, such that*

$$u(x, t) = \Lambda(\tau)^{-1} V\left(\frac{x - X(\tau)}{\Lambda(\tau)}, \tau\right), \quad \frac{dt}{d\tau} = \Lambda(\tau)^2,$$

and $\mathcal{F}(1, 0; V(\tau)) = 0$ for $\tau \in J$.

Proof. From (SC5) and Proposition 2.3, $(\Lambda_{\text{pre}}, X_{\text{pre}})$ are AC on I_* , so V_{pre} is defined on Q_{2R}^* . The retained repaired-gauge stratum supplies $(\lambda, c) \in AC(J; (0, \infty) \times \mathbb{R}^3)$ and $\mathcal{F}(1, 0; V_{\text{pre}}^{\lambda, c}) = 0$ a.e. By Definition 2.21 and Theorem 2.22, setting

$$\Lambda(\tau) = \Lambda_{\text{pre}}(\tau)\lambda(\tau), \quad X(\tau) = X_{\text{pre}}(\tau) + \Lambda_{\text{pre}}(\tau)c(\tau), \quad V(\tau) = V_{\text{pre}}^{\lambda(\tau), c(\tau)},$$

and $P(\tau) = P_{\text{pre}}^{\lambda(\tau), c(\tau)}$, gives $\Lambda, X \in AC(J)$ and $a = -\Lambda'$, $b = -X' \in L_{\text{loc}}^1(J)$, and

$$\mathcal{F}(1, 0; V(\tau)) = 0 \text{ for a.e. } \tau \in J.$$

Finally $u = \Lambda^{-1}V(\cdot, \tau) \circ \Phi^{-1}$ by definition of the repaired map, hence the physical identity with $\frac{dt}{d\tau} = \Lambda^2$ follows. $\square$

Theorem 2.34 (Pressure and modulation). *Assume, in addition to Theorem 2.33, that the retained repaired-gauge selection is quantitatively nondegenerate in the sense of Proposition 2.16. Then (V, P) satisfies the renormalized Navier–Stokes equation with the sign convention of Theorem 2.6; there is a decomposition $P = P_{\text{loc}} + H$ on Q_R^* as in Theorem 2.25; and (a, b) is the unique solution of*

$$J(\tau) \begin{pmatrix} a(\tau) \\ b(\tau) \end{pmatrix} = \mathcal{R}(\tau), \quad \begin{pmatrix} a \\ b \end{pmatrix} \in L_{\text{loc}}^1(J)^4,$$

with bounds $|(a, b)| \leq K|\mathcal{R}|$, where K depends on $R, p, \Theta_0, m_0, C_u, \varepsilon_0, \chi_R$.

Proof. Equation and sign convention are inherited directly from Theorem 2.6 applied to (Λ, X) from Theorem 2.33 and the definition of V . By Theorem 2.25 applied at each time slice, the decomposition on Q_R^J is available and unique once $\oint_{B_R} H = 0$ is imposed. The modulation identity is exactly Theorem 2.29; uniqueness follows from invertibility of J (Theorem 2.16) and the pointwise identity $(a, b) = J^{-1}\mathcal{R}$. Uniform bounds on (a, b) are obtained by the bound on J^{-1} there and the stated bound on $\mathcal{R}$. $\square$

Theorem 2.35 (Theorem C: compact-window representation output). *Under the assumptions of Theorems 2.33 and 2.34, for every compact window $J \subset I_*$ and each fixed $R > 0$, setting $Q_R^J := B_R \times J$, the following are available for later sections:*

- (1) the exact renormalized equation on Q_{2R}^J ,
- (2) the four repaired-gauge constraints $\mathcal{F}(1, 0; V(\tau)) = 0$,
- (3) bounds on $a, b \in L_{\text{loc}}^1(J)$ (and L_{loc}^∞ under $\mathcal{R} \in L_{\text{loc}}^\infty$),
- (4) local pressure decomposition $P = P_{\text{loc}} + H$ with $L^{3/2}$ local bounds and harmonic remainder controls,
- (5) renormalized local suitable energy inequality on Q_R^J ,
- (6) stability under subsequences in the topologies above.

No step beyond these items is used in later sections.

Proof. From Theorem 2.33 and the choice of $J \subset I_*$, all quantities are defined on Q_{2R}^J . Items (1) and (2) are Theorems 2.6 and 2.22; item (3) is Theorem 2.29 together with the inverse bound from Proposition 2.16; item (4) is Theorem 2.25; item (5) is Proposition 2.31; item (6) is Proposition 2.32. This is the package exported by the previous theorems. $\square$

Remark 2.36. The results above separate two alternatives cleanly:

- (1) nondegenerate single-core repaired-gauge regime, where Theorems A–C apply;
- (2) gauge degeneracy ($\det J = 0$) or compact-structure failure, which exits the retained stratum and is routed to the multibubble, cascade, or scale-rigidity alternatives.

3. MULTIBUBBLE AND CASCADE EXCLUSION

Abstract. We prove a local multibubble and cascade reduction for the conditional Type II Navier–Stokes exclusion problem. The setting is a positive local Caffarelli–Kohn–Nirenberg concentration sequence at a possible singular point. All arguments are local in parabolic cylinders. Compact-cylinder failure, splitting of active CKN mass, strict same-point scale cascades, separated active physical points, and localized gauge degeneracy are reduced by explicit active-frame decoupling. Under the single-core, scale-rigidity, and terminal active-frame admissibility inputs stated below, exterior profiles and remainders are perturbative in every selected active frame, and the remaining branch is single-core, scale-rigid, or no longer an independent multibubble mechanism. Thus no local multibubble, cascade, or gauge-degenerate Type II configuration remains under these inputs. This section uses the classical theory of suitable weak solutions of the Navier–Stokes equations, including Leray [Ler34], Caffarelli–Kohn–Nirenberg [CKN82], Scheffer [Sch76], and Lin [Lin98].

Imported inputs. This section is a local reduction theorem. It uses the analytic inputs already established in Sections 1, 2, 4 and 5 and proves only the multibubble, cascade, and terminal-frame decoupling steps.

- (i) From Section 1, the compact single-core criterion Theorem 1.18: a compact single-core finite-cost Type II branch with positive local CKN concentration, nondegenerate scale-center selection, pressure reconstruction, bounded compact-cylinder suitable package, bounded modulation, and vanishing localized dissipation is excluded.
- (ii) From Section 2, the renormalized equation Theorem 2.6, pressure decomposition Theorem 2.25, modulation system Theorem 2.29, and repaired-gauge representation Theorems 2.33 to 2.35: the renormalized equation, pressure decomposition modulo functions of time, bounded modulation coefficients, renormalized suitable structure, and stability of these structures under subsequences.
- (iii) From Section 4, the compact-cylinder Caccioppoli estimate Theorem 4.4 and rough-core exclusion Theorem 4.7.
- (iv) From Section 5, the collapse cost exclusion Theorems 5.6 and 5.7 and autonomous reduced-limit theorem Theorem 5.19 wherever a scale-rigid or scale-collapsing branch is produced.

3.1. Local setting and active frames.

Normalised local Navier–Stokes setting. All quantities are written in rescaled local variables $(x, t) \in Q_R = B_R \times (-R^2, 0)$ with viscosity $\nu = 1$ (critical scaling has absorbed ν). A local pair (u_n, p_n) on such a cylinder satisfies

$$\partial_t u_n - \Delta u_n + (u_n \cdot \nabla) u_n + \nabla p_n = 0, \quad \nabla \cdot u_n = 0,$$

and pressure is normalised through spatial averaging

$$(p_n)_{B_\rho}(t) := |B_\rho|^{-1} \int_{B_\rho} p_n(x, t) dx$$

to factor out time-dependent constants.

For $r > 0$, $x_0 \in \mathbb{R}^3$, and $t_0 \in \mathbb{R}$, let

$$Q_r(x_0, t_0) := B_r(x_0) \times (t_0 - r^2, t_0), \quad Q_r := Q_r(0, 0) = B_r \times (-r^2, 0).$$

Let (u_n, p_n) be parabolic rescalings around a potential singular point with positive local Caffarelli–Kohn–Nirenberg concentration on compact cylinders. Concretely, for a base triple

(x_n, t_n, λ_n) ,

$$u_n(y, s) = \lambda_n u(x_n + \lambda_n y, t_n + \lambda_n^2 s), \quad p_n(y, s) = \lambda_n^2 p(x_n + \lambda_n y, t_n + \lambda_n^2 s).$$

For each n and $R > 0$, define the scaling-invariant local quantities

$$A_n(R) := R^{-1} \operatorname{ess\,sup}_{-R^2 < t < 0} \int_{B_R} |u_n(x, t)|^2 dx,$$

$$E_n(R) := R^{-1} \int_{Q_R} |\nabla u_n(x, t)|^2 dx dt,$$

$$C_n(R) := R^{-2} \int_{Q_R} |u_n(x, t)|^3 dx dt, \quad D_n(R) := R^{-2} \int_{-R^2}^0 \int_{B_R} |p_n - (p_n)_{B_R}(t)|^{3/2} dx dt,$$

where

$$(p_n)_{B_R}(t) := |B_R|^{-1} \int_{B_R} p_n(x, t) dx.$$

For an active frame (x_n, λ_n) , $\lambda_n > 0$, define the critical rescaling

$$\mathcal{T}_{x_n, \lambda_n} f(y) := \lambda_n f(x_n + \lambda_n y).$$

If $\phi \in L_\sigma^3(\mathbb{R}^3)$ is observed in this frame, its contribution is

$$\phi_n(y) = \mathcal{T}_{x_n, \lambda_n} \phi(y) = \lambda_n \phi(x_n + \lambda_n y).$$

The critical Jacobian identity is exact:

$$\|\mathcal{T}_{x_n, \lambda_n} f\|_{L^3(E)}^3 = \int_{x_n + \lambda_n E} |f(x)|^3 dx$$

for every measurable $E \subset \mathbb{R}^3$. In particular

$$\|\mathcal{T}_{x_n, \lambda_n} f\|_{L^3(\mathbb{R}^3)}^3 = \|f\|_{L^3(\mathbb{R}^3)}^3,$$

so $\mathcal{T}_{x_n, \lambda_n}$ is an isometry on $L^3(\mathbb{R}^3)$.

Definition 3.1 (Bounded selected-window limit). A selected-window subsequence has a bounded selected-window limit on a compact cylinder Q_R if there exists $u_\infty \in L^\infty(Q_R)$ such that, after extracting a subsequence,

$$\forall \varepsilon > 0 \exists N : \forall n \geq N, \|u_n - u_\infty\|_{L^3(Q_R)} < \varepsilon.$$

The limit is nonzero if $u_\infty \not\equiv 0$ on Q_R .

Definition 3.2 (Local Type II sequence). A positive local concentration sequence is called a local Type II sequence if every selected-window rescaling has zero bounded selected-window limit on every compact cylinder Q_R :

$$\forall R > 0, \quad \lim_{n \rightarrow \infty} \|u_n\|_{L^3(Q_R)} = 0.$$

Definition 3.3 (Active frame). Fix a concentrating time sequence $t_n \uparrow T$ and a compact time interval $I \Subset (-\infty, 0]$. An active frame is a tuple

$$\mathfrak{F}^j = (x_n^j, \lambda_n^j, I, U_n^j, P_n^j), \quad \lambda_n^j \downarrow 0,$$

where $x_n^j \in \mathbb{R}^3$ and

$$U_n^j(y, s) := \lambda_n^j u(x_n^j + \lambda_n^j y, t_n + (\lambda_n^j)^2 s), \quad P_n^j(y, s) := (\lambda_n^j)^2 p(x_n^j + \lambda_n^j y, t_n + (\lambda_n^j)^2 s)$$

on $B_{2R} \times I$ for every fixed compact cylinder under consideration. The frame is active at threshold $\eta_* > 0$ if for some fixed $R_0 < \infty$

$$\liminf_{n \rightarrow \infty} \int_{Q_{R_0}} |U_n^j|^3 dy ds \geq \eta_*^3.$$

Definition 3.4 (Relations between active frames). Let $\mathfrak{F}^i$ and $\mathfrak{F}^j$ be two active frames. After passing to a subsequence exactly one of the following parameter regimes holds.

(i) They are *same-point comparable-scale* if

$$0 < c \leq \frac{\lambda_n^i}{\lambda_n^j} \leq C < \infty, \quad \frac{|x_n^i - x_n^j|}{\max(\lambda_n^i, \lambda_n^j)} \leq C$$

for all large n .

(ii) They form a *same-point cascade* if, after relabeling,

$$\frac{\lambda_n^i}{\lambda_n^j} \rightarrow 0, \quad \frac{|x_n^i - x_n^j|}{\lambda_n^j} = O(1).$$

The frame i is then the inner frame and j is an outer frame.

(iii) They are *separated-point* if

$$\frac{|x_n^i - x_n^j|}{\lambda_n^i} \rightarrow \infty \quad \text{and} \quad \frac{|x_n^i - x_n^j|}{\lambda_n^j} \rightarrow \infty.$$

Definition 3.5 (Terminal frame and perturbative remainder). Given a finite active family, a terminal frame is chosen by selecting an active scale with minimal order:

$$\lambda_n^{j*} \leq C_j \lambda_n^j \quad \text{for every active } j$$

after passing to a subsequence. Comparable frames at the same physical point are grouped before this choice; if several grouped frames remain tied, one is chosen by a fixed lexicographic ordering of their labels. A remainder r_n in a terminal frame is perturbative on $B_{2R} \times I$ if

$$\|r_n\|_{L_I^\infty L^3(B_{2R})} \rightarrow 0$$

and its equation residual and pressure satisfy

$$\|f_n\|_{L_I^1 H^{-1}(B_R)} + \|\nabla \pi_n^r\|_{L_I^1 H^{-1}(B_R)} \rightarrow 0$$

for every compact $B_R \times I$.

Definition 3.6 (Local multibubble/cascade branch). A local multibubble/cascade branch is a positive-concentration local Type II branch for which at least one of the following alternatives occurs:

- (i) the compact-cylinder suitable package $A_n + E_n + C_n + D_n$ is unbounded on some fixed rescaled cylinder;
- (ii) positive CKN mass splits into two or more comparable active cores;
- (iii) a strict same-point scale cascade appears inside bounded rescaled cylinders;
- (iv) separated physical points remain active in different local rescaled frames;
- (v) the localized scale or centering selection degenerates (no unique core selected).

Definition 3.7 (Active profile package). An active profile package is a family

$$\{(\phi^j, x_n^j, \lambda_n^j)\}_{j \in J}, \quad \phi^j \in L_\sigma^3(\mathbb{R}^3),$$

with finite or countable index set J , extracted along a positive local concentration sequence. In physical variables write

$$\Lambda_{\lambda, x_0} \phi(x) := \lambda^{-1} \phi\left(\frac{x - x_0}{\lambda}\right).$$

For every finite $J_0 \subset J$ the package gives an expansion

$$u(\cdot, t_n) = \sum_{j \in J_0} \Lambda_{\lambda_n^j, x_n^j} \phi^j + r_n^{J_0}$$

on the compact terminal windows under consideration. After the finite active subfamily has been selected, the corresponding remainder satisfies the perturbative bounds of [Theorem 3.5](#). The parameters are considered only after grouping same-point comparable-scale profiles into compound profiles as in [Theorem 3.8](#). A profile is active if

$$\|\phi^j\|_{L^3(\mathbb{R}^3)} \geq \eta_* > 0.$$

The package has finite critical mass when

$$\sum_{j \in J} \|\phi^j\|_{L^3(\mathbb{R}^3)}^3 \leq M_* < \infty.$$

It captures positive local concentration if every compact terminal-window sequence with

$$\liminf_{n \rightarrow \infty} \int_{Q_R} |u_n|^3 > 0$$

either contains an active profile observed in that window or is absorbed into the selected terminal frame; equivalently, no positive terminal-window concentration remains in the perturbative remainder.

Definition 3.8 (Compound profile). Let $J_k \subset J$ be a same-point comparable-scale class. Choose one representative $(x_n^{j_k}, \lambda_n^{j_k})$. After passing to a subsequence,

$$\rho_j := \lim_{n \rightarrow \infty} \frac{\lambda_n^j}{\lambda_n^{j_k}} \in (0, \infty), \quad z_j := \lim_{n \rightarrow \infty} \frac{x_n^j - x_n^{j_k}}{\lambda_n^{j_k}} \in \mathbb{R}^3.$$

The compound profile in the representative frame is the L_σ^3 -sum

$$\Phi^k(y) := \sum_{j \in J_k} \rho_j^{-1} \phi^j\left(\frac{y - z_j}{\rho_j}\right),$$

whenever J_k is finite; finiteness follows for active classes from [Theorem 3.9](#). If $\|\Phi^k\|_{L^3} < \varepsilon_*$, where ε_* is the critical small-data threshold, the class is perturbative. Otherwise Φ^k is treated as one active object in its representative frame.

Lemma 3.9 (Finite active profile count). *Every active profile package with finite critical mass contains only finitely many active profiles. More precisely,*

$$\#J \leq M_* \eta_*^{-3}.$$

Proof. For each active profile $j \in J$,

$$\|\phi^j\|_{L^3(\mathbb{R}^3)}^3 \geq \eta_*^3.$$

Let $J_0 \subset J$ be finite. Summing over J_0 gives

$$\#J_0 \eta_*^3 \leq \sum_{j \in J_0} \|\phi^j\|_{L^3(\mathbb{R}^3)}^3 \leq \sum_{j \in J} \|\phi^j\|_{L^3(\mathbb{R}^3)}^3 \leq M_*.$$

Hence $\#J_0 \leq M_* \eta_*^{-3}$. Taking the supremum over all finite subsets J_0 yields the same bound for the cardinality of J , so J is finite and

$$\#J \leq M_* \eta_*^{-3}.$$

□

Lemma 3.10 (Exhaustive parameter partition). *Let $\mathfrak{F}^i, \mathfrak{F}^j$ be two active frames. After passing to a subsequence, either the pair is same-point comparable-scale, same-point cascade, or separated-point in the sense of [Theorem 3.4](#); if none of these relations holds in a selected compact frame, then one of the two contributions is locally invisible there in L^3 .*

Proof. Pass to a subsequence so that

$$\frac{\lambda_n^i}{\lambda_n^j} \rightarrow \ell \in [0, \infty]$$

in the extended half-line. If $0 < \ell < \infty$, then either

$$\frac{|x_n^i - x_n^j|}{\max(\lambda_n^i, \lambda_n^j)}$$

is bounded along a further subsequence, giving the same-point comparable-scale case, or it tends to infinity, giving separated points because the two scales are comparable.

If $\ell = 0$, then either

$$\frac{|x_n^i - x_n^j|}{\lambda_n^j}$$

is bounded, giving a same-point cascade with i inner and j outer, or it tends to infinity. In the latter case the j -profile is locally invisible in the i -frame by [Theorem 3.13](#). The case $\ell = \infty$ is the same after interchanging i and j . These alternatives exhaust all subsequential limits of the scale ratio and normalized center distance. □

Theorem 3.11 (Minimal bad-configuration extraction). *Let a positive local Type II branch admit an active profile package with finite critical mass, perturbative remainder in the sense of [Theorem 3.5](#), and concentration capture in the sense of [Theorem 3.7](#). If the branch is not already a single active terminal frame satisfying the single-core hypotheses, then after passing to a subsequence one of the following occurs:*

- (i) compact-cylinder suitable control fails, and [Theorem 3.15](#) produces a bounded-cylinder concentration core;
- (ii) there are at least two active frames in a same-point comparable-scale class, hence a compound profile in the sense of [Theorem 3.8](#);
- (iii) there is a same-point cascade;
- (iv) there are separated active physical points;
- (v) the scale-center selection of the selected compound or terminal frame is degenerate.

In every case the positive local concentration is carried by an active frame or compound profile, not by the perturbative remainder.

Proof. If compact-cylinder suitable control fails, item (i) follows from [Theorem 3.15](#). Assume therefore that the compact-cylinder package is bounded on the windows under consideration.

By concentration capture, positive local concentration cannot remain solely in the perturbative remainder. Hence either one selected active frame carries the terminal-window concentration, or another active frame remains non-negligible. If there is only one nondegenerate selected active frame and no competing active object, the branch is precisely the single-core regime. Since this

case is excluded by hypothesis, either the selected scale-center map is degenerate, giving item (v), or at least two active frames remain.

For two active frames, apply [Theorem 3.10](#). Same-point comparable-scale frames give item (ii), same-point scale separation gives item (iii), and separated points give item (iv). The only remaining alternative in [Theorem 3.10](#) is local invisibility of one contribution in the selected compact frame; such a contribution is absorbed into the perturbative remainder and cannot carry the retained positive local concentration by the capture property. This proves the extraction statement. $\square$

3.2. Elementary local decoupling estimates.

Lemma 3.12 (Critical rescalings that escape are locally invisible). *Let $\phi \in L^3(\mathbb{R}^3)$, let $R < \infty$, and define*

$$W_n(y) = \rho_n \phi(z_n + \rho_n y),$$

where $\rho_n > 0$. Assume that the sets

$$z_n + \rho_n B_R$$

either have measure tending to zero, or escape to spatial infinity in the sense that for every compact set $K \subset \mathbb{R}^3$,

$$(z_n + \rho_n B_R) \cap K = \emptyset$$

for all sufficiently large n . Then

$$\|W_n\|_{L^3(B_R)} \rightarrow 0.$$

Proof. By the critical change of variables,

$$\|W_n\|_{L^3(B_R)}^3 = \int_{B_R} \rho_n^3 |\phi(z_n + \rho_n y)|^3 dy = \int_{z_n + \rho_n B_R} |\phi(x)|^3 dx.$$

Set

$$m_n := \int_{z_n + \rho_n B_R} |\phi(x)|^3 dx.$$

Since $\|W_n\|_{L^3(B_R)}^3 = m_n$, it is enough to show $m_n \rightarrow 0$.

Fix $\varepsilon > 0$. We split into the two structural alternatives.

Case 1: shrinking measure. If $|z_n + \rho_n B_R| \rightarrow 0$, then by absolute continuity of the integral $|\phi|^3 \in L^1(\mathbb{R}^3)$ there exists $\delta > 0$ such that

$$|E| < \delta \implies \int_E |\phi|^3 dx < \varepsilon.$$

Choose N with $|z_n + \rho_n B_R| < \delta$ for all $n \geq N$. Then

$$m_n = \int_{z_n + \rho_n B_R} |\phi|^3 dx < \varepsilon, \quad n \geq N.$$

Case 2: escape to infinity. Assume now that for every compact set $K \subset \mathbb{R}^3$, $(z_n + \rho_n B_R) \cap K = \emptyset$ for all sufficiently large n . By absolute continuity at infinity, choose compact $K \subset \mathbb{R}^3$ such that

$$\int_{\mathbb{R}^3 \setminus K} |\phi|^3 < \varepsilon.$$

For every such compact K , there is $N(K)$ with $(z_n + \rho_n B_R) \cap K = \emptyset$ for all $n \geq N(K)$. For these indices,

$$z_n + \rho_n B_R \subset \mathbb{R}^3 \setminus K,$$

and therefore, for $n \geq N(K)$,

$$m_n = \int_{z_n + \rho_n B_R} |\phi|^3 dx \leq \int_{\mathbb{R}^3 \setminus K} |\phi|^3 < \varepsilon.$$

In either case, for every $\varepsilon > 0$, $m_n < \varepsilon$ for n sufficiently large. Hence $m_n \rightarrow 0$, and therefore

$$\|W_n\|_{L^3(B_R)} \rightarrow 0.$$

□

Lemma 3.13 (Separated active profiles vanish in the selected frame). *Let (x_n^p, λ_n^p) and (x_n^q, λ_n^q) be two active frames with separated physical centers in the sense that*

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} \rightarrow \infty.$$

If $\phi^q \in L^3(\mathbb{R}^3)$, then the q -profile observed in the p -frame converges to zero locally in L^3 : for every $R < \infty$,

$$\left\| \frac{\lambda_n^p}{\lambda_n^q} \phi^q \left(\frac{x_n^p - x_n^q}{\lambda_n^q} + \frac{\lambda_n^p}{\lambda_n^q} y \right) \right\|_{L^3(B_R)} \rightarrow 0.$$

Proof. The displayed function has the form in [Theorem 3.12](#) with

$$z_n = \frac{x_n^p - x_n^q}{\lambda_n^q}, \quad \rho_n = \frac{\lambda_n^p}{\lambda_n^q}.$$

The corresponding physical set in the q -profile variables is

$$z_n + \rho_n B_R = \frac{x_n^p - x_n^q}{\lambda_n^q} + \frac{\lambda_n^p}{\lambda_n^q} B_R.$$

For every $y \in B_R$,

$$|z_n + \rho_n y| \geq |z_n| - \rho_n R.$$

Set $E_n := z_n + \rho_n B_R$. Then

$$\text{dist}(E_n, 0) = |z_n| - \rho_n R = \frac{\lambda_n^p}{\lambda_n^q} \left(\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R \right).$$

Since $\frac{|x_n^q - x_n^p|}{\lambda_n^p} \rightarrow \infty$, there exists $N_0(R)$ such that

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R \geq 1, \quad n \geq N_0(R),$$

and hence for such n , $\text{dist}(E_n, 0) \geq \rho_n$.

There are two cases.

(a) If $\rho_n \rightarrow 0$, then

$$|z_n + \rho_n B_R| = \rho_n^3 |B_R| \rightarrow 0.$$

Then the first alternative in [Theorem 3.12](#) applies.

(b) Otherwise, after passing to a subsequence, $\rho_n \geq \rho_* > 0$. Then

$$\text{dist}(z_n + \rho_n B_R, 0) \geq \rho_* \left(\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R \right) \rightarrow \infty,$$

so the sets escape every fixed compact subset of $\mathbb{R}^3$. The second alternative in [Theorem 3.12](#) applies.

In both subcases the $L^3(B_R)$ -norm of the displayed expression converges to zero. □

Lemma 3.14 (Strict outer scales vanish in the innermost frame). *Let several active profiles have the same physical center and scales $\lambda_n^1, \dots, \lambda_n^J$. Suppose λ_n^1 is an innermost scale and*

$$\frac{\lambda_n^1}{\lambda_n^j} \rightarrow 0 \quad (j \neq 1).$$

Then every outer profile $\phi^j \in L^3(\mathbb{R}^3)$, viewed in the λ_n^1 -frame, converges to zero in $L^3(B_R)$ for each fixed $R < \infty$. Any constant ambient velocity arising from the regular part of the outer flow is absorbed into the translation modulation.

Proof. In the innermost variables, the j -th outer contribution has the form

$$W_n^j(y) = \frac{\lambda_n^1}{\lambda_n^j} \phi^j \left(\frac{x_n^1 - x_n^j}{\lambda_n^j} + \frac{\lambda_n^1}{\lambda_n^j} y \right).$$

Set

$$\rho_n := \frac{\lambda_n^1}{\lambda_n^j} \rightarrow 0, \quad z_n^j := \frac{x_n^1 - x_n^j}{\lambda_n^j}.$$

Then

$$W_n^j(y) = \rho_n \phi^j(z_n^j + \rho_n y),$$

and by the critical change of variables,

$$\|W_n^j\|_{L^3(B_R)}^3 = \int_{z_n^j + \rho_n B_R} |\phi^j(x)|^3 dx.$$

Since $\rho_n \rightarrow 0$, the translated domains have

$$|z_n^j + \rho_n B_R| = \rho_n^3 |B_R| \rightarrow 0.$$

By absolute continuity of the integral for $\phi^j \in L^3(\mathbb{R}^3)$,

$$\|W_n^j\|_{L^3(B_R)} \rightarrow 0.$$

To treat a general ϕ^j , let $\phi_\varepsilon^j \in C_c^\infty(\mathbb{R}^3)$ with $\|\phi^j - \phi_\varepsilon^j\|_{L^3} < \varepsilon$. By critical isometry,

$$\|W_n^j - (W_n^j)_\varepsilon\|_{L^3(B_R)} \leq \|\phi^j - \phi_\varepsilon^j\|_{L^3} < \varepsilon,$$

where $(W_n^j)_\varepsilon$ is the same scaling of ϕ_ε^j . The compactly supported term has vanishing local norm by the shrinking-domain argument above, and then letting first $n \rightarrow \infty$, then $\varepsilon \downarrow 0$ implies the claimed convergence. A constant vector field from the regular part of an outer flow contributes only a Galilean drift in the selected frame; it is therefore incorporated into the translation modulation and does not remain as an exterior perturbation. $\square$

Lemma 3.15 (Compactness failure produces another core). *If the compact-cylinder suitable package is unbounded on a fixed rescaled cylinder, then after passing to a subsequence the local $C_n + D_n$ quantity is unbounded on a comparable cylinder. Consequently the branch contains an additional local CKN concentration core.*

Proof. Fix $r > 0$. Apply the local Caccioppoli inequality with outer radius $2r$. For each index n , suitable weak solutions satisfy, with C_r independent of n ,

$$A_n(r) + E_n(r) \leq C_r(1 + C_n(2r) + C_n(2r)^{2/3} + D_n(2r)),$$

where the pressure is normalized by subtracting spatial means $(p_n)_{B_{2r}}(t)$.

Suppose that $A_n(r) + E_n(r) + C_n(r) + D_n(r)$ is unbounded along a subsequence. If $C_n(2r) + D_n(2r)$ were bounded on this subsequence, the preceding estimate would give uniform boundedness

of $A_n(r) + E_n(r)$. Monotonicity of the local integrals also gives $C_n(r) + D_n(r) \leq C_n(2r) + D_n(2r)$, contradicting unboundedness of the full package on Q_r . Therefore, after extracting a subsequence,

$$C_n(2r) + D_n(2r) \rightarrow \infty.$$

Define the localized CKN concentration at variable center and radius

$$G_n(x, t, \rho) := \rho^{-2} \int_{Q_\rho(x, t)} |u_n|^3 dx ds + \rho^{-2} \int_{Q_\rho(x, t)} |p_n - (p_n)_{B_\rho}(s)|^{3/2} dx ds.$$

Set

$$\mathbf{M}_n := \sup\{G_n(x, t, \rho) : 0 < \rho \leq 2r, Q_\rho(x, t) \subset Q_{4r}\}.$$

The choice $(x, t, \rho) = (0, 0, 2r)$ is admissible because $Q_{2r}(0, 0) \subset Q_{4r}$, so

$$\mathbf{M}_n \geq C_n(2r) + D_n(2r) \rightarrow \infty.$$

After extracting a further subsequence, we may assume $\mathbf{M}_n \geq 1$ for all n . Choose (x_n, t_n, ρ_n) with $0 < \rho_n \leq 2r$, $Q_{\rho_n}(x_n, t_n) \subset Q_{4r}$, such that

$$G_n(x_n, t_n, \rho_n) \geq \frac{1}{2} \mathbf{M}_n \geq \frac{1}{2}.$$

Rescale around (x_n, t_n, ρ_n) :

$$v_n(y, s) := \rho_n u_n(x_n + \rho_n y, t_n + \rho_n^2 s), \quad q_n(y, s) := \rho_n^2 p_n(x_n + \rho_n y, t_n + \rho_n^2 s).$$

Using the critical Jacobian identity,

$$\int_{Q_1} |v_n|^3 dy ds + \int_{Q_1} |q_n - (q_n)_{B_1}(s)|^{3/2} dy ds = G_n(x_n, t_n, \rho_n) \geq \frac{1}{2}.$$

Hence this new frame carries positive local critical mass, i.e. a new local CKN concentration core. This proves the asserted additional core extraction. $\square$

3.3. Terminal active-frame decoupling.

Definition 3.16 (Admissible terminal active package). Fix a compact cylinder $B_{2R} \times I$ and a selected active frame. A terminal active package is admissible on this cylinder if the following hold.

- (i) The active profiles satisfy the finite-mass and active-mass conditions of [Theorem 3.7](#).
- (ii) Every nonselected active profile is either separated from the selected physical point in the sense of [Theorem 3.13](#), or has the same physical point and a strictly outer scale in the sense of [Theorem 3.14](#), after comparable same-point scales have been grouped.
- (iii) The rescaled remainder r_n satisfies

$$\|r_n\|_{L_T^\infty L^3(B_{2R})} \rightarrow 0,$$

and there exist fields (π_n^r, f_n) such that on $B_{2R} \times I$,

$$\partial_t r_n - \Delta r_n + a_n(r_n + y \cdot \nabla r_n) + b_n \cdot \nabla r_n + \nabla \pi_n^r = f_n, \quad \nabla \cdot r_n = 0,$$

with

$$\|f_n\|_{L_T^1 H^{-1}(B_R)} \rightarrow 0,$$

$$\|\nabla \pi_n^r\|_{L_T^1 H^{-1}(B_R)} \rightarrow 0.$$

- (iv) The selected velocity satisfies

$$\sup_n \|U_n\|_{L_T^\infty L^3(B_{2R})} < \infty.$$

- (v) The selected modulation coefficients satisfy

$$\sup_n (\|a_n\|_{L^\infty(I)} + \|b_n\|_{L^\infty(I)}) < \infty.$$

- (vi) Local pressure decomposition is stable under the splitting: if tensor coefficients tend to zero in $L_I^1 L^{3/2}(B_R)$, then the corresponding pressure gradients tend to zero in $L_I^1 H^{-1}(B_R)$, after subtracting spatial means.
- (vii) Cutoff commutators generated by localizing to B_R are controlled by the $L_I^\infty L^3$, $L_I^1 L^{3/2}$, and pressure-gradient norms appearing above. In particular, for any compact $\chi_R \in C_c^\infty(B_{2R})$ with $\chi_R \equiv 1$ on B_R , the commutators satisfy

$$\|[\Delta, \chi_R]u\|_{H^{-1}(B_R)} + \|[\nabla \chi_R, \nabla \cdot](u \otimes u)\|_{H^{-1}(B_R)} \leq C_R \|u\|_{L^3(B_{2R})}^2.$$

uniformly for $u \in L^3(B_{2R})$ and the same type of estimate for S_n -quadratic commutator terms.

Theorem 3.17 (Terminal profile-evolution decoupling). *Consider an admissible terminal active package on $B_{2R} \times I$. Let U_n be the selected-frame velocity and let S_n be the sum, in that frame, of all nonselected active profiles together with the rescaled remainder. Then on $B_R \times I$:*

- (i) $\nabla \cdot S_n = 0$ distributionally;
- (ii) $S_n \rightarrow 0$ in $L_I^\infty L^3(B_{2R})$;
- (iii) $U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \rightarrow 0$ in $L_I^1 L^{3/2}(B_R)$;
- (iv)

$$\exists \pi_n, F_n : \quad \partial_t S_n - \Delta S_n + a_n(y \cdot \nabla S_n + S_n) + b_n \cdot \nabla S_n + \nabla \pi_n = F_n,$$

$$\|F_n\|_{L_I^1 H^{-1}(B_R)} \rightarrow 0, \quad \nabla \cdot S_n = 0 \quad \text{in } \mathcal{D}'(B_R \times I).$$

$$\|\nabla \pi_n\|_{L_I^1 H^{-1}(B_R)} \rightarrow 0.$$

- (v) every pressure source containing at least one factor of S_n has vanishing pressure gradient in $L_I^1 H^{-1}(B_R)$;
- (vi) localization errors introduced by compact cutoffs vanish in $L_I^1 H^{-1}(B_R)$;
- (vii) the effective modulation parameters of the selected frame remain bounded;
- (viii) after discarding S_n , the selected frame remains a positive finite-mass local Type II branch with pressure reconstruction and the compact-cylinder bounds required by the single-core analysis.

Proof. Each profile in the package is divergence-free, and the rescaled remainder is divergence-free. Since divergence commutes with the Navier–Stokes critical rescalings, finite sums remain divergence-free in distributions. This proves (i).

We prove (ii). By admissibility, every nonselected active profile is either separated from the selected physical point or is a same-point strictly outer scale. The separated profiles vanish locally by [Theorem 3.13](#), and the same-point outer profiles vanish locally by [Theorem 3.14](#). The remainder tends to zero in $L_I^\infty L^3(B_{2R})$ by [Theorem 3.16](#). The number of active profiles is finite by [Theorem 3.9](#). Since every summand in S_n tends to zero in $L_I^\infty L^3(B_{2R})$, the triangle inequality gives

$$\|S_n\|_{L_I^\infty L^3(B_{2R})} \leq \sum_{j \neq j_*} \|S_n^j\|_{L_I^\infty L^3(B_{2R})} + \|r_n\|_{L_I^\infty L^3(B_{2R})} \rightarrow 0.$$

This proves (ii).

For (iii), admissibility gives

$$\sup_n \|U_n\|_{L_I^\infty L^3(B_{2R})} < \infty.$$

Hence, by Hölder's inequality,

$$\|U_n \otimes S_n\|_{L_I^1 L^{3/2}(B_R)} \leq |I| \|U_n\|_{L_I^\infty L^3(B_R)} \|S_n\|_{L_I^\infty L^3(B_R)} \rightarrow 0,$$

and the same estimate applies to $S_n \otimes U_n$. Finally,

$$\|S_n \otimes S_n\|_{L_I^1 L^{3/2}(B_R)} \leq |I| \|S_n\|_{L_I^\infty L^3(B_R)}^2 \rightarrow 0.$$

For (iv), write the represented Navier–Stokes equation for the full profile sum $U_n + S_n$ and subtract the represented equation for the selected component U_n . The remaining equation for S_n has the displayed linear left-hand side

$$\partial_t S_n - \Delta S_n + a_n(y \cdot \nabla S_n + S_n) + b_n \cdot \nabla S_n.$$

Collect all right-hand terms into

$$F_n := -\nabla \cdot (U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n) + f_n + E_n^{\text{cut}} + E_n^{\text{mod}},$$

where f_n is the remainder forcing from item (iii) of admissibility, and $E_n^{\text{cut}}, E_n^{\text{mod}}$ are localization and modulation errors. The nonlinear stresses tend to zero in $L_I^1 L^{3/2}(B_R)$, hence in $L_I^1 H^{-1}(B_R)$, because $L^{3/2}(B_R) \hookrightarrow H^{-1}(B_R)$ on a bounded three-dimensional ball. The residual term vanishes by [Theorem 3.16](#). The cutoff terms vanish by the localization part of admissibility, recorded explicitly in (vi). This proves the displayed convergence in $L_I^1 H^{-1}(B_R)$.

For (v), decompose the pressure source of $U_n + S_n$ into the pure selected source, the pure exterior source, and the two mixed sources. The mixed sources are

$$\partial_i \partial_j (U_{n,i} S_{n,j} + S_{n,i} U_{n,j}) \quad \text{and} \quad \partial_i \partial_j (S_{n,i} S_{n,j}).$$

By (iii), their tensor coefficients tend to zero in $L_I^1 L^{3/2}(B_R)$. The pressure-stability part of admissibility, equivalently local Calderon–Zygmund estimates plus harmonic pressure control modulo spatial means, gives convergence of the corresponding pressure gradients to zero in $L_I^1 H^{-1}(B_R)$. Pure exterior sources are generated by profiles that are locally invisible in the selected frame; applying the same pressure stability after [Theorems 3.13](#) and [3.14](#) gives their vanishing pressure gradients.

For (vi), let χ_R be a cutoff equal to one on B_R and supported in B_{2R} . The commutators are finite sums of terms of the following types:

$$(\nabla \chi_R) \cdot \nabla S_n, \quad (\Delta \chi_R) S_n, \quad (\nabla \chi_R) \cdot (U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n),$$

plus analogous pressure cutoff terms. Their supports lie in the fixed annulus $B_{2R} \setminus B_R$. By the localization part of admissibility, these commutators are controlled by the norms already shown to vanish in (ii), (iii), and (v). Hence all localization errors vanish in $L_I^1 H^{-1}(B_R)$.

For (vii), the only exterior contribution that can remain nonzero at first order in a compact selected frame is a constant ambient velocity. This has already been absorbed into the translation modulation. The remaining exterior profiles vanish locally, so their contribution to the modulation equations is small in the same local L^3 topology used above. The selected modulation coefficients are bounded by admissibility; hence the effective parameters remain bounded.

For (viii), the positive CKN mass of the selected profile is unchanged by subtracting a term that tends to zero in local L^3 and whose mixed pressure sources vanish by (v). The pressure reconstruction is stable under the same decomposition by the pressure-stability condition in [Theorem 3.16](#). The compact-cylinder bounds pass to the selected component because the full package is bounded and the discarded terms vanish on compact cylinders. Thus the selected frame remains a positive finite-mass local Type II branch with the analytic inputs needed for the single-core analysis. $\square$

3.4. Same-point and separated-point reductions.

Theorem 3.18 (Same-point cascade reduction). *Every same-point multibubble or strict same-point cascade reduces, in the selected innermost or compound frame, to one of the following: the single-core criterion, scale-rigidity, or a perturbative terminal branch covered by [Theorem 3.17](#),*

provided the terminal package generated in the selected innermost frame is admissible in the sense of [Theorem 3.16](#).

Proof. Step 1: scale-comparability classes. Split active same-point profiles into equivalence classes by

$$i \sim j \iff \exists c, C \in (0, \infty) : c \leq \frac{\lambda_n^i}{\lambda_n^j} \leq C \quad \text{for all } n$$

after passing to a subsequence. Thus within each class there are constants c_{ij}, C_{ij} with $c_{ij} \leq \lambda_n^i / \lambda_n^j \leq C_{ij}$ for all n , so one may define a single common rescaling sequence (choose $\lambda_n^{(k)} = \lambda_n^{j_k}$) and write all class- k profiles in the same coordinates. Within each class, finite sums of divergence-free profiles remain divergence-free.

Step 2: compound profile reduction inside each class. For each class set

$$\Phi_n^k := \sum_{j \in J_k} \lambda_n^j \phi^j(x_n^j + \lambda_n^j y).$$

If the L^3 -size of this rescaled sum satisfies

$$\left\| \sum_{j \in J_k} \phi^j \right\|_{L^3(\mathbb{R}^3)} < \varepsilon_\star,$$

with $\varepsilon_\star$ the perturbative small-data threshold, then standard local small-data theory for Navier–Stokes in the critical space removes this whole class from the active list.

If the class is above threshold, it is retained as a single compound active core; no longer than one active core is needed from that class for local selection.

Step 3: local alternatives. If the retained compound core is unique and its scale-centering Jacobian is nondegenerate, we obtain the single selected core alternative. If this Jacobian is degenerate, the localized choice of active center or scale is not unique, which is the gauge-degenerate alternative in [Theorem 3.6](#).

Step 4: strict same-point scale separation. Assume now the branch is not reduced to comparable scales and choose the innermost active scale λ_n^1 in that physical point. By [Theorem 3.14](#), every profile with larger scale vanishes in the λ_n^1 -frame in local L^3 , up to constant drift terms. These constant drifts are absorbed into translation modulation by admissibility. Let S_n be the sum of all such outer profiles plus the rescaled remainder r_n . Then, by Step 2 and [Theorem 3.14](#),

$$S_n \rightarrow 0 \quad \text{in } L_I^\infty L^3(B_{2R}).$$

Subtracting the equation for the innermost selected profile from the full-frame equation gives a perturbation equation for the selected profile whose forcing belongs to $L_I^1 H^{-1}(B_R)$ and vanishes. This is the setting of [Theorem 3.17](#). Therefore one of three outcomes holds:

- (a) **Single-core branch:** after discarding S_n , the branch is governed by one active component, so it lands in the compact single-core branch of [Theorem 3.20](#);
- (b) **Scale-rigid branch:** the relative Jacobian between comparable scales becomes rigid in the sense of [Theorem 3.21](#), so [Theorem 3.21](#) excludes it;
- (c) **Purely perturbative branch:** $S_n \rightarrow 0$ and the selected profile satisfies a vanishing forcing perturbation in $L_I^1 H^{-1}(B_R)$. This branch carries no independent same-point cascade mechanism; after discarding S_n , the selected component is passed to the single-core or scale-rigidity analysis.

These are the strict same-point scale-separation alternatives, so the same-point reduction is exhausted. $\square$

Theorem 3.19 (Separated-point decoupling). *Separated active physical points are locally invisible in each other's active rescaled frames. Consequently every separated-point multibubble reduces in each selected active frame to a single-core, scale-rigid, or perturbative terminal branch, provided the terminal package in that selected active frame is admissible in the sense of Theorem 3.16.*

Proof. Fix an active point p with frame (x_n^p, λ_n^p) . Let $q \neq p$ be another active point. By separation,

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} \rightarrow \infty.$$

Thus Theorem 3.13 gives local L^3 convergence to zero of the q -profile in the p -frame on every compact ball. The same conclusion holds for each profile centered at any physical point different from p . Since only finitely many active profiles remain after the critical mass threshold is imposed, the sum of all profiles outside the p -frame tends to zero locally in L^3 .

The pure pressure generated by a separated profile is locally invisible by the same change of variables whenever its profile pressure belongs to $L_{\text{loc}}^{3/2}$ with the standard Calderon–Zygmund control. Mixed pressure terms contain at least one exterior velocity factor, hence vanish in $L^1 H_{\text{loc}}^{-1}$ by the pressure part of Theorem 3.17. Localization and modulation commutators are localized to $B_{2R} \setminus B_R$, so their H^{-1} -norm is controlled by the L^3 -smallness already proved in Theorems 3.13 and 3.14 and by admissibility item (vii).

Therefore, in the selected p -frame, all other physical points contribute only a perturbative forcing tending to zero in $L^1 H^{-1}$ on compact cylinders. The selected frame remains a positive finite-mass branch with the same compact-cylinder and pressure reconstruction properties. Hence the branch reduces to the single-core case, the scale-rigid case, or the perturbative terminal branch. Since p was arbitrary, the same reduction holds in every active frame. $\square$

3.5. Multibubble and cascade exclusion. The assumptions in this section isolate the analytic inputs needed for the multibubble reduction: the single-core exclusion, the scale-rigidity reduction, and terminal active-frame admissibility. They are stated explicitly so that the theorem below uses no unstated compactness, pressure, or profile-completeness conclusion.

Assumption 3.20 (Local single-core exclusion). Every compact single-core branch with positive local CKN concentration, nondegenerate scale-center selection, pressure reconstruction, bounded compact-cylinder suitable package, bounded modulation, and vanishing localized dissipation is excluded from the local Type II regime.

Assumption 3.21 (Local scale-rigidity exclusion). Every scale-rigid local branch produced by the same-point or separated-point reductions, including a branch with degenerate relative scale-center Jacobian, either has a nonzero bounded selected-window limit or reduces to the single-core branch in Theorem 3.20. Consequently no such branch is compatible with the local Type II condition.

Assumption 3.22 (Terminal admissibility in active frames). Every terminal package produced by the same-point and separated-point reductions is admissible on each compact selected cylinder in the sense of Theorem 3.16.

Theorem 3.23 (Local multibubble and cascade exclusion). *Assume the local single-core and scale-rigidity exclusions Theorems 3.20 and 3.21, and assume terminal admissibility in active frames, Theorem 3.22. Let a positive local Type II branch admit an active package satisfying the hypotheses of Theorem 3.11. Then no local multibubble, cascade, or gauge-degenerate Type II case can occur in this class.*

Proof. Let a local Type II branch be in the multibubble/cascade class of [Theorem 3.6](#). Apply [Theorem 3.11](#). The positive local concentration is carried by an active frame or compound profile, and one of the alternatives listed there occurs.

Case 1: compactness-failure reduction. Compactness failure yields, after changing radius by a fixed factor and rescaling as in [Theorem 3.15](#), a bounded-cylinder concentration branch. Relabel this branch as an active core. After passing to a subsequence, the new core and every previously retained active core have one of three relations: comparable scale at the same physical point, strictly separated scale at the same physical point, or separated physical center. Comparable same-point cores are grouped into compound profiles. A perturbative compound profile is removed by small-data stability; an active compound profile is treated as one selected core in its active frame. If the scale-center selection inside such a compound core is degenerate, the branch is gauge-degenerate, giving the non-uniqueness alternative in [Theorem 3.6](#) rather than a new independent core dynamics.

Case 1a: gauge degeneracy. If the scale-center selection of the selected compound or terminal frame is degenerate, then the branch is in the gauge-degenerate alternative of [Theorem 3.6](#). By the repaired-gauge representation in [Section 2](#), nondegeneracy is exactly the hypothesis needed to enter the single-core branch. Persistent degeneracy is therefore routed to the scale-rigidity alternative in [Theorem 3.21](#); if the degeneracy disappears after passing to a subsequence, the branch enters the single-core or terminal-frame alternatives below.

Case 1b: same-point comparable scales. Same-point comparable-scale active frames are grouped into the compound profile of [Theorem 3.8](#). If the compound profile is below the critical small-data threshold, it is part of the perturbative remainder. If it is above threshold and its scale-center selection is nondegenerate, it is one selected active core and enters the single-core branch. If the corresponding scale-center selection is degenerate, it is covered by Case 1a. Thus comparable same-point profiles do not produce an additional multibubble mechanism.

Case 2: same-point cascading. Strict same-point scale cascades are handled by [Theorem 3.18](#). In the innermost frame, all outer scales are locally invisible after constant drifts are absorbed. The branch therefore reduces to one of the three terminal alternatives

(A) single-core, (B) scale-rigid, (C) terminal perturbative.

Case 3: separated-point interactions. For separated physical points, [Theorem 3.19](#) gives the same three terminal alternatives as in Case 2:

(A) single-core, (B) scale-rigid, (C) terminal perturbative.

The single-core alternative is excluded by [Theorem 3.20](#). The scale-rigid alternative is excluded by [Theorem 3.21](#), which packages the scale-collapse and autonomous-limit exclusions in [Section 5](#) in this setting. The terminal perturbative alternative cannot sustain an independent multibubble obstruction because [Theorem 3.17](#) removes all exterior profiles and remainders from the selected active equation in $L^1 H_{\text{loc}}^{-1}$, preserves the positive finite-mass selected branch, and leaves only the single-core or scale-rigid possibilities already excluded. Consequently every alternative in [Theorem 3.6](#) is impossible in the local Type II regime. $\square$

Corollary 3.24 (Removal of the multibubble alternative). *In the local Type II analysis, every failure of the single nondegenerate compact core alternative is excluded by [Theorem 3.23](#) under the hypotheses stated there.*

Proof. By [Theorem 3.6](#), failure of the single nondegenerate compact core alternative means that at least one of the following occurs: the compact-cylinder suitable package is unbounded; positive local CKN mass splits between active cores; a strict same-point scale cascade occurs; separated physical points remain active in distinct local frames; or the localized scale or center selection

is degenerate. The compactness-failure case produces an additional core by [Theorem 3.15](#). Same-point and separated cases are reduced by [Theorems 3.18](#) and [3.19](#). The final exclusion is [Theorem 3.23](#). Hence no failure of the single nondegenerate compact core alternative remains. $\square$

4. LOCAL CACCIOPPOLI AND ROUGH-CORE EXCLUSION

Abstract. We present a fully explicit local Caccioppoli estimate for renormalized Navier–Stokes fields. All terms produced by the localized energy inequality are estimated separately, and all absorptions are written explicitly. The argument uses standard PDE notation and the usual integrability classes of suitable weak solutions.

In particular, we derive local windowed H^1 control on compact space–time cylinders from local compact Caffarelli–Kohn–Nirenberg quantities. The result gives the rough-core alternative as the negation of local compact-CKN control.

This section is written in the classical setting of suitable weak solutions (Leray [Ler34], Caffarelli–Kohn–Nirenberg [CKN82], Scheffer [Sch76], Lin [Lin98]).

4.1. Imported inputs and notation. This section uses the renormalized local energy inequality obtained in Section 2. It does not rederive the renormalized suitable structure or the pressure reconstruction. All estimates below are conditional on the following local hypotheses on the compact renormalized cylinders under consideration.

Assumption 4.1 (Renormalized local energy inequality). Let (V, P, a, b) be defined on a compact renormalized cylinder $Q_{2R,J} := B_{2R} \times J$, where $J \subset \mathbb{R}$ is a bounded interval. Assume

$$(6) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu \Delta V = a(\tau)V + a(\tau)(y \cdot \nabla V) + b(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0,$$

with $\nu > 0$, $\nabla \cdot V = 0$, and $a, b \in L^\infty(J)$. For every nonnegative $\phi \in C_c^\infty(Q_{2R,J})$ and every $\tau_1 < \tau_2$ in J ,

$$(7) \quad \begin{aligned} & \int_{B_{2R}} \frac{|V(\cdot, \tau_2)|^2}{2} \phi(\cdot, \tau_2) - \int_{B_{2R}} \frac{|V(\cdot, \tau_1)|^2}{2} \phi(\cdot, \tau_1) \\ & + \nu \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \phi |\nabla V|^2 \leq \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \frac{|V|^2}{2} (\partial_\tau \phi + \nu \Delta \phi) \\ & + \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \frac{|V|^2}{2} V \cdot \nabla \phi + \int_{\tau_1}^{\tau_2} \int_{B_{2R}} P V \cdot \nabla \phi \\ & + \int_{\tau_1}^{\tau_2} \int_{B_{2R}} a \frac{|V|^2}{2} (-\phi - y \cdot \nabla \phi) \\ & - \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \frac{|V|^2}{2} b \cdot \nabla \phi. \end{aligned}$$

The pressure is understood modulo functions of time, and the mean-normalized pressure on compact balls belongs to the local class used below.

Throughout, for a space–time set $E \subset \mathbb{R}^3 \times \mathbb{R}$, write

$$\int_E f := \int_E f(y, \tau) dy d\tau.$$

For $r > 0$ and a bounded interval $J \subset \mathbb{R}$, define

$$Q_{r,J} := B_r \times J, \quad |Q_{r,J}| := |B_r| |J|.$$

For a ball $B \subset \mathbb{R}^3$, denote the spatial mean of pressure by

$$(P)_B(\tau) := \frac{1}{|B|} \int_B P(y, \tau) dy.$$

The local quantities are

$$\begin{aligned}\mathcal{A}_J(r) &:= \operatorname{ess\,sup}_{\tau \in J} \int_{B_r} |V(y, \tau)|^2 dy, \\ \mathcal{H}_J(r) &:= \int_J \int_{B_r} |\nabla V|^2 dy d\tau, \\ \mathcal{C}_J(r) &:= \int_J \int_{B_r} |V|^3 dy d\tau, \quad \mathcal{D}_J(r) := \int_J \int_{B_r} |P - (P)_{B_r}(\tau)|^{3/2} dy d\tau.\end{aligned}$$

Lemma 4.2 (Finite-measure conversion). *For every bounded interval J and every $r > 0$,*

$$\int_J \int_{B_r} |V|^2 \leq |Q_{r,J}|^{1/3} \mathcal{C}_J(r)^{2/3} \leq |Q_{r,J}|^{1/3} (1 + \mathcal{C}_J(r)).$$

Proof. Hölder's inequality on $Q_{r,J}$ gives

$$\int_J \int_{B_r} |V|^2 \leq |Q_{r,J}|^{1/3} \left(\int_J \int_{B_r} |V|^3 \right)^{2/3}.$$

The second inequality follows from $s^{2/3} \leq 1 + s$ for $s \geq 0$. $\square$

Lemma 4.3 (Comparison of pressure normalizations). *Let $1 \leq p < \infty$ and $0 < r \leq R$. For almost every τ such that $P(\tau) \in L^p(B_R)$,*

$$\|P(\tau) - (P)_{B_r}(\tau)\|_{L^p(B_r)} \leq 2\|P(\tau) - (P)_{B_R}(\tau)\|_{L^p(B_R)}.$$

Consequently,

$$\mathcal{D}_J(r) \leq 2^{3/2} \mathcal{D}_J(R)$$

whenever the right-hand side is finite.

Proof. Set $c = (P)_{B_R}(\tau)$. Then

$$\|P - (P)_{B_r}\|_{L^p(B_r)} \leq \|P - c\|_{L^p(B_r)} + |B_r|^{1/p} |(P)_{B_r} - c|.$$

Since

$$|(P)_{B_r} - c| \leq |B_r|^{-1} \int_{B_r} |P - c| \leq |B_r|^{-1/p} \|P - c\|_{L^p(B_r)},$$

the displayed pointwise estimate follows. Raising to the power $3/2$ and integrating in time gives the last assertion. $\square$

4.2. Compact-cylinder Caccioppoli estimate.

Theorem 4.4 (Compact-cylinder Caccioppoli estimate). *Let $R > 0$, let $J \subset \mathbb{R}$ be a bounded open interval, and let $J' \Subset J$ be a compact subinterval. Assume [Theorem 4.1](#) on $Q_{2R,J}$ and assume*

$$\mathcal{C}_J(2R) + \mathcal{D}_J(2R) < \infty.$$

Then there is a constant

$$C = C(R, |J|, \nu, \|a\|_{L^\infty(J)}, \|b\|_{L^\infty(J)}, \operatorname{dist}(J', \partial J))$$

such that

$$\mathcal{A}_{J'}(R) + \mathcal{H}_{J'}(R) \leq C \left(1 + \mathcal{C}_J(2R) + \mathcal{D}_J(2R) \right).$$

The estimate is outer-to-inner: all right-hand quantities are measured on $B_{2R} \times J$, while the conclusion is on $B_R \times J'$.

Proof. Step 1: localized cut-off and the two energy inequalities. Set

$$\delta := \text{dist}(J', \partial J) > 0.$$

Choose $\eta \in C_c^\infty(J)$ and $\zeta \in C_c^\infty(B_{2R})$ with

$$0 \leq \eta \leq 1, \quad 0 \leq \zeta \leq 1, \quad \eta \equiv 1 \text{ on } J', \quad \zeta \equiv 1 \text{ on } B_R,$$

and

$$\|\eta'\|_{L^\infty(J)} \leq C_\eta \delta^{-1}, \quad \|\nabla \zeta\|_{L^\infty(B_{2R})} \leq C_\zeta R^{-1}, \quad \|\Delta \zeta\|_{L^\infty(B_{2R})} \leq C_\zeta R^{-2}.$$

Set

$$\phi(y, \tau) := \eta(\tau) \zeta(y)^2.$$

Then $\phi \in C_c^\infty(Q_{2R,J})$, $0 \leq \phi \leq 1$, and

$$\phi \equiv 1 \text{ on } B_R \times J'.$$

Choose $\tau_- < \inf J'$ and $\tau_+ > \sup J'$ in J with $\eta(\tau_-) = \eta(\tau_+) = 0$. Applying (7) on (τ_-, τ_+) gives

$$\nu \int_{\tau_-}^{\tau_+} \int_{B_{2R}} \phi |\nabla V|^2 \leq |R_\tau| + |R_\Delta| + |R_{\text{conv}}| + |R_{\text{pres}}| + |R_{\text{mod}}|,$$

where

$$\begin{aligned} R_\tau &:= \int_J \int_{B_{2R}} \frac{|V|^2}{2} \partial_\tau \phi, & R_\Delta &:= \nu \int_J \int_{B_{2R}} \frac{|V|^2}{2} \Delta \phi, \\ R_{\text{conv}} &:= \int_J \int_{B_{2R}} \left(\frac{|V|^2}{2} V \right) \cdot \nabla \phi, & R_{\text{pres}} &:= \int_J \int_{B_{2R}} (P - (P)_{B_{2R}}(\tau)) V \cdot \nabla \phi, \\ R_{\text{mod}} &:= \int_J \int_{B_{2R}} a \frac{|V|^2}{2} (-\phi - y \cdot \nabla \phi) - \int_J \int_{B_{2R}} \frac{|V|^2}{2} b \cdot \nabla \phi. \end{aligned}$$

For almost every $\sigma \in J'$, applying (7) on (τ_-, σ) gives the same right-hand estimates and leaves the terminal term $\frac{1}{2} \int_{B_R} |V(\sigma)|^2$ on the left. Therefore the bounds below control both $\mathcal{H}_{J'}(R)$ and $\mathcal{A}_{J'}(R)$. Define

$$L := \int_J \int_{B_{2R}} |V|^2, \quad C_3 := C_J(2R), \quad D_{3/2} := \mathcal{D}_J(2R).$$

By Theorem 4.2,

$$L \leq |Q_{2R,J}|^{1/3} C_3^{2/3} \leq C(R, |J|)(1 + C_3).$$

Since $\phi(y, \tau) = \eta(\tau) \zeta(y)^2$,

$$\partial_\tau \phi = \eta'(\tau) \zeta(y)^2, \quad \nabla \phi = 2\eta(\tau) \zeta(y) \nabla \zeta(y), \quad \Delta \phi = \eta(\tau) \Delta(\zeta(y)^2),$$

hence

$$\|\partial_\tau \phi\|_{L^\infty(Q_{2R,J})} \leq C_\eta \delta^{-1}, \quad \|\nabla \phi\|_{L^\infty(Q_{2R,J})} \leq 2C_\zeta R^{-1}, \quad \|\Delta \phi\|_{L^\infty(Q_{2R,J})} \leq C_\zeta R^{-2}.$$

Step 2: derivative terms from the cut-off.

$$|R_\tau| \leq \frac{1}{2} \|\partial_\tau \phi\|_{L^\infty(Q_{2R,J})} L \leq C \delta^{-1} (1 + C_3),$$

$$|R_\Delta| \leq \frac{\nu}{2} \|\Delta \phi\|_{L^\infty(Q_{2R,J})} L \leq C \nu R^{-2} (1 + C_3).$$

Step 3: convection term. Using $|\nabla \phi| \leq C R^{-1}$,

$$|R_{\text{conv}}| \leq \frac{1}{2} \|\nabla \phi\|_{L^\infty(Q_{2R,J})} \int_J \int_{B_{2R}} |V|^3 \leq C R^{-1} C_3.$$

Step 4: pressure term. For each time τ ,

$$\int_{B_{2R}} (P - (P)_{B_{2R}}(\tau)) V \cdot \nabla \phi = \int_{B_{2R}} P V \cdot \nabla \phi,$$

with

$$\int_{B_{2R}} (P)_{B_{2R}}(\tau) V \cdot \nabla \phi = (P)_{B_{2R}}(\tau) \int_{B_{2R}} V \cdot \nabla \phi = -(P)_{B_{2R}}(\tau) \int_{B_{2R}} \phi \nabla \cdot V = 0$$

(using $\phi \in C_c^\infty(B_{2R})$ and $\nabla \cdot V = 0$). Thus subtracting any time-dependent pressure constant is legitimate in the pressure flux term. Hence

$$|R_{\text{pres}}| \leq \|\nabla \phi\|_{L^\infty(Q_{2R,J})} \int_J \|P - (P)_{B_{2R}}(\tau)\|_{L_x^{3/2}(B_{2R})} \|V(\tau, \cdot)\|_{L_x^3(B_{2R})} d\tau \leq CR^{-1} D_{3/2}^{2/3} C_3^{1/3}.$$

Young's inequality gives

$$|R_{\text{pres}}| \leq C(R)(C_3 + D_{3/2}).$$

Step 5: modulation terms. The imported local energy inequality already contains the modulation terms after the spatial integrations by parts from the repaired-gauge equation:

$$\int_{B_{2R}} (y \cdot \nabla V) \cdot V \phi = \frac{1}{2} \int_{B_{2R}} y \cdot \nabla (|V|^2) \phi = -\frac{1}{2} \int_{B_{2R}} |V|^2 \nabla \cdot (y \phi),$$

and

$$\int_{B_{2R}} (b(\tau) \cdot \nabla V) \cdot V \phi = \frac{1}{2} \int_{B_{2R}} b(\tau) \cdot \nabla (|V|^2) \phi = -\frac{1}{2} \int_{B_{2R}} |V|^2 b(\tau) \cdot \nabla \phi.$$

Thus we estimate the imported terms directly. Split

$$R_{\text{mod}} = R_a + R_{ay} + R_b,$$

where

$$R_a := - \int_J \int_{B_{2R}} a \frac{|V|^2}{2} \phi, \quad R_{ay} := - \int_J \int_{B_{2R}} a \frac{|V|^2}{2} y \cdot \nabla \phi, \quad R_b := - \int_J \int_{B_{2R}} \frac{|V|^2}{2} b \cdot \nabla \phi.$$

$$M_a := \|a\|_{L^\infty(J)}, \quad M_b := \|b\|_{L^\infty(J)}.$$

For R_a ,

$$|R_a| \leq \frac{1}{2} M_a L \leq C M_a (1 + C_3).$$

For the scale-gradient part, $|y| \leq 2R$ on B_{2R} and $\|\nabla \phi\|_{L^\infty(Q_{2R,J})} \leq CR^{-1}$, hence

$$|y \cdot \nabla \phi| \leq C.$$

Therefore

$$|R_{ay}| \leq C M_a L \leq C M_a (1 + C_3).$$

For R_b ,

$$|R_b| \leq C M_b R^{-1} L \leq C M_b R^{-1} (1 + C_3).$$

Step 6: absorption and conclusion. Collecting estimates from Steps 2–5 yields

$$\nu \int_{\tau_-}^{\tau_+} \int_{B_{2R}} \phi |\nabla V|^2 \leq C(1 + C_3 + D_{3/2}),$$

where C depends only on

$$R, |J|, \nu, \|a\|_{L^\infty(J)}, \|b\|_{L^\infty(J)}, \delta.$$

Since $\phi \equiv 1$ on $B_R \times J'$,

$$\mathcal{H}_{J'}(R) \leq C(1 + \mathcal{C}_J(2R) + \mathcal{D}_J(2R)).$$

Repeating the same estimates with terminal time $\sigma \in J'$ gives

$$\frac{1}{2} \int_{B_R} |V(y, \sigma)|^2 dy \leq C(1 + \mathcal{C}_J(2R) + \mathcal{D}_J(2R))$$

for almost every $\sigma \in J'$. Taking the essential supremum over $\sigma \in J'$ and renaming C proves the theorem. $\square$

Definition 4.5 (Local compact-cylinder control). Local Caffarelli–Kohn–Nirenberg control on $Q_{r,J}$ means

$$\mathcal{C}_J(r) + \mathcal{D}_J(r) < \infty.$$

Local windowed H^1 control on $Q_{r,J}$ means

$$\mathcal{H}_J(r) < \infty.$$

Corollary 4.6 (Outer control implies inner suitable control). *Let $R > 0$, let $J \subset \mathbb{R}$ be a bounded open interval, and let $J' \Subset J$ be a compact subinterval. Assume [Theorem 4.1](#) on $Q_{2R,J}$ and $\mathcal{C}_J(2R) + \mathcal{D}_J(2R) < \infty$. Then*

$$\mathcal{A}_{J'}(R) + \mathcal{H}_{J'}(R) < \infty.$$

More quantitatively, the bound is the one in [Theorem 4.4](#). If a later argument uses pressure means on a smaller ball, [Theorem 4.3](#) converts those normalizations to the outer-ball normalization used here.

Proof. This is [Theorem 4.4](#). The final sentence follows from [Theorem 4.3](#). $\square$

Corollary 4.7 (Rough-core exclusion on compact windows). *Let $R > 0$, let $J \subset \mathbb{R}$ be a bounded open interval, and let $J' \Subset J$ be a compact subinterval. Assume [Theorem 4.1](#) on $Q_{2R,J}$. If*

$$\mathcal{C}_J(2R) + \mathcal{D}_J(2R) < \infty,$$

then

$$\mathcal{H}_{J'}(R) < \infty.$$

Equivalently, if $\mathcal{H}_{J'}(R) = +\infty$ for an inner cylinder $Q_{R,J'}$, then for every enclosing cylinder $Q_{2R,J}$ with $J' \Subset J$ on which the imported local energy inequality and bounded modulation hypotheses hold,

$$\mathcal{C}_J(2R) = +\infty \quad \text{or} \quad \mathcal{D}_J(2R) = +\infty.$$

Proof. The direct implication is the gradient part of [Theorem 4.4](#). The last assertion is its contrapositive with the same outer-to-inner geometry: finite outer $\mathcal{C} + \mathcal{D}$ on $Q_{2R,J}$ forces finite inner $\mathcal{H}$ on $Q_{R,J'}$. $\square$

Remark 4.8 (Meaning of rough core). In later applications, a rough core means failure of local windowed H^1 control on compact inner renormalized cylinders while the modulation coefficients are bounded. [Theorem 4.7](#) shows that such a failure is not independent of compact-cylinder Caffarelli–Kohn–Nirenberg control: it can occur only if the velocity-pressure package $\mathcal{C} + \mathcal{D}$ fails on every relevant larger enclosing compact cylinder.

4.3. Inputs exported to later sections. The following consequences are used later in the document.

- (1) On every compact renormalized window satisfying the imported local energy inequality and bounded modulation, finite $\mathcal{C} + \mathcal{D}$ on $Q_{2R,J}$ implies finite $\mathcal{A} + \mathcal{H}$ on $Q_{R,J'}$ whenever $J' \Subset J$.
- (2) The pressure normalization may be taken on any comparable ball, after using [Theorem 4.3](#).
- (3) Failure of local windowed H^1 control on an inner compact cylinder forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ package on each larger enclosing cylinder to which [Theorem 4.4](#) applies.

5. SCALE-COLLAPSE COST AND AUTONOMOUS LIMITS

Abstract. We work with suitable weak solutions of the three-dimensional Navier–Stokes equations and give an explicit step-by-step version of the scale-collapse exclusions. Under a renormalized gauge, the logarithmic scale satisfies $\partial_\tau \log \lambda = a$. If a renormalized branch has a nontrivial localized core and the scale collapses, then $\int a_-(\tau) M_R(\tau) d\tau = \infty$; thus finite scale-collapsing cost or finite absolute cost is incompatible with collapse. In addition, we record the autonomous-window limit passage under an explicitly stated compactness input; that input is not derived from the cost hypotheses in this section.

Notation and functional conventions. For a bounded spatial region $K \subset \mathbb{R}^3$, we use

$$(f, g)_{L^2(K)} := \int_K f \cdot g \, dx, \quad \langle F, \phi \rangle_{H^{-1}(K), H_0^1(K)} := \int_0^T \langle F(\cdot, s), \phi(\cdot, s) \rangle_{H^{-1}(K), H_0^1(K)} \, ds$$

whenever $F \in L^1(0, T; H^{-1}(K))$ and $\phi \in L^\infty(0, T; H_0^1(K))$. Weak limits and weak convergences are taken in the distributional sense on C_c^∞ -test functions.

$$\|f\|_{L_t^p L_x^q} := \left(\int_0^T \|f(\cdot, s)\|_{L^q(\mathbb{R}^3)}^p \, ds \right)^{1/p},$$

for any fixed $T > 0$, with the usual modification $p = \infty$.

5.1. Renormalized scale-collapse setting.

Position of the result. We split the scale-collapse analysis into two separate levels:

- (1) Part A (Sections 5.2–5.4): unconditional cost exclusion arguments that do not use compactness.
- (2) Part B (Section 5.5): a conditional compactness statement under explicitly stated compactness hypotheses.

Throughout this section, we assume the renormalized representation established earlier: on the time interval under consideration, the fields (V, P) satisfy (8) and its distributional weak formulation. In particular, the solenoidal formulation is obtained by restricting the test fields below to divergence-free fields.

Let (u, p) be a divergence-free suitable weak solution of

$$\partial_t u + (u \cdot \nabla) u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0,$$

in $\mathbb{R}^3$, with viscosity $\nu > 0$. Let $\lambda(\tau) > 0$, $X(\tau) \in \mathbb{R}^3$, and a measurable gauge function $c(t)$ be given, and define the renormalized fields by

$$u(x, t) = \lambda(\tau)^{-1} V \left(\frac{x - X(\tau)}{\lambda(\tau)}, \tau \right), \quad p(x, t) = \lambda(\tau)^{-2} P \left(\frac{x - X(\tau)}{\lambda(\tau)}, \tau \right) + c(t).$$

Pressure is defined only up to an arbitrary function of time; $c(t)$ has no effect on ∇p , so it is irrelevant for all subsequent estimates. Assume that (V, P) satisfies on $(0, \infty) \times \mathbb{R}^3$

$$(8) \quad \partial_\tau V + (V \cdot \nabla) V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0.$$

For every $\psi \in C_c^\infty((0, \infty) \times \mathbb{R}^3; \mathbb{R}^3)$ with $\nabla \cdot \psi = 0$, the pair (V, P) satisfies the weak formulation

$$\begin{aligned} 0 &= \int_0^\infty \int_{\mathbb{R}^3} (-V \cdot \partial_\tau \psi - (V \otimes V) : \nabla \psi + \nu \nabla V : \nabla \psi) \, dy \, d\tau \\ &\quad + \int_0^\infty \int_{\mathbb{R}^3} (-a(\tau)(V + y \cdot \nabla V) \cdot \psi - (b(\tau) \cdot \nabla V) \cdot \psi + P \nabla \cdot \psi) \, dy \, d\tau. \end{aligned}$$

For every $0 < T < \infty$, restricting the τ -integration to $(0, T)$ gives the same identity on $(0, T) \times \mathbb{R}^3$. Equivalently, for every $\psi \in C_c^\infty((0, \infty) \times \mathbb{R}^3; \mathbb{R}^3)$, not necessarily divergence free,

$$\begin{aligned} 0 &= \int_0^\infty \int_{\mathbb{R}^3} (-V \cdot \partial_\tau \psi - (V \otimes V) : \nabla \psi \nabla V : \nabla \psi - P \nabla \cdot \psi) dy d\tau \\ &\quad + \int_0^\infty \int_{\mathbb{R}^3} (-a(\tau)(V + y \cdot \nabla V) \cdot \psi - (b(\tau) \cdot \nabla V) \cdot \psi) dy d\tau. \end{aligned}$$

Define positive and negative parts of the scalar field a by

$$a_+(\tau) := \max\{a(\tau), 0\}, \quad a_-(\tau) := \max\{-a(\tau), 0\}.$$

Then for a.e. τ , $a = a_+ - a_-$, $a_+, a_- \geq 0$, and $a_+ a_- = 0$. Assume

$$a \in L_{\text{loc}}^1([\tau_0, \infty)), \quad b \in L_{\text{loc}}^1([\tau_0, \infty); \mathbb{R}^3),$$

and the scale convention

$$(9) \quad \frac{d}{d\tau} \log \lambda(\tau) = a(\tau)$$

for a.e. $\tau \geq \tau_0$, with $\log \lambda$ absolutely continuous on compact intervals.

All integrals in this section are interpreted in $[0, \infty]$. When we write $\int_{\tau_0}^\infty$, partial integrals are taken over $[\tau_0, T]$ and monotone convergence is used as $T \rightarrow \infty$.

Fix $R > 0$. Choose $\Phi \in C_c^\infty(\mathbb{R}^3; [0, 1])$ satisfying

$$(10) \quad \Phi(y) = 1 \text{ if } |y| \leq R, \quad \text{supp}_y \Phi \subset B_{2R}.$$

Define the localized kinetic mass

$$M_R(\tau) := \int_{\mathbb{R}^3} |V(y, \tau)|^2 \Phi(y) dy.$$

Since Φ is fixed in τ , M_R is measurable in τ , and only this measurability is used below.

Definition 5.1 (Nonvanishing localized core). A renormalized branch has nonvanishing localized core if there exist constants $m_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$M_R(\tau) \geq m_0 \quad \text{for a.e. } \tau \in [\tau_1, \infty).$$

In applications, this lower bound is supplied by the selected-core construction; here it is stated as an explicit hypothesis.

Definition 5.2 (Scale-collapse and absolute scale-drift densities). Define

$$\mathcal{C}_{\text{sc}}(\tau) := a_-(\tau) M_R(\tau), \quad \mathcal{C}_{\text{abs}}(\tau) := \nu \int_{\mathbb{R}^3} |\nabla V(y, \tau)|^2 \Phi(y) dy + |a(\tau)| M_R(\tau).$$

$\mathcal{C}_{\text{sc}}$ is the negative drift weighted by localized core mass, while $\mathcal{C}_{\text{abs}}$ controls both collapse drift and local dissipation. Their total costs are extended real-valued by

$$\int_{\tau_0}^\infty \mathcal{C}_{\text{sc}}(\tau) d\tau, \quad \int_{\tau_0}^\infty \mathcal{C}_{\text{abs}}(\tau) d\tau.$$

Definition 5.3 (Finite-cost branch). A branch has finite scale-collapsing cost if

$$\int_{\tau_0}^\infty \mathcal{C}_{\text{sc}}(\tau) d\tau < \infty.$$

It has finite absolute cost if

$$\int_{\tau_0}^\infty \mathcal{C}_{\text{abs}}(\tau) d\tau < \infty.$$

5.2. Elementary alternative.

Lemma 5.4 (Finite-or-infinite alternative). *Let $f \geq 0$ be measurable on $[\tau_0, \infty)$. Then exactly one of the following holds:*

$$\int_{\tau_0}^{\infty} f(\tau) d\tau < \infty \quad \text{or} \quad \int_{\tau_0}^{\infty} f(\tau) d\tau = \infty.$$

Proof. (1) For $T > \tau_0$, define partial costs

$$F(T) := \int_{\tau_0}^T f(\tau) d\tau \in [0, \infty].$$

(2) If $T_2 > T_1$, then by nonnegativity of f ,

$$F(T_2) - F(T_1) = \int_{T_1}^{T_2} f(\tau) d\tau \geq 0,$$

so F is monotone nondecreasing.

(3) A monotone function on $[\tau_0, \infty)$ has an extended limit $L \in [0, \infty]$,

$$L := \lim_{T \rightarrow \infty} F(T).$$

By definition of improper integral, this limit equals $\int_{\tau_0}^{\infty} f(\tau) d\tau$.

(4) Therefore L is either finite or equal to $+\infty$, and these cases are mutually exclusive. Hence exactly one item in the lemma holds. $\square$

Lemma 5.5 (Logarithmic scale identity). *Under (9), for all τ_1, τ_2 with $\tau_0 \leq \tau_1 < \tau_2 < \infty$,*

$$\int_{\tau_1}^{\tau_2} a(\tau) d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

Proof. (1) Since $a \in L^1_{\text{loc}}$, the map $\log \lambda$ is absolutely continuous on $[\tau_1, \tau_2]$, hence has a representative with a.e. derivative equal to a .

(2) By the fundamental theorem of calculus for absolutely continuous functions, for every $\tau \in [\tau_1, \tau_2]$,

$$\log \lambda(\tau) = \log \lambda(\tau_1) + \int_{\tau_1}^{\tau} a(s) ds.$$

(3) Setting $\tau = \tau_2$ yields the identity. $\square$

5.3. Scale collapse versus localized core lower bounds.

Theorem 5.6 (Collapse with nonvanishing localized core forces infinite scale-collapsing cost). *Assume:*

- (1) $\lambda(\tau) \rightarrow 0$ as $\tau \rightarrow \infty$;
- (2) the branch has nonvanishing localized core.

Then

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\text{sc}}(\tau) d\tau = \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau = \infty.$$

Proof. (1) Choose $m_0 > 0$ and $\tau_1 \geq \tau_0$ from the nonvanishing core assumption so that

$$M_R(\tau) \geq m_0 \quad \text{for a.e. } \tau \in [\tau_1, \infty).$$

(2) By Lemma 5.5, for any $\tau_2 > \tau_1$,

$$\int_{\tau_1}^{\tau_2} a(\tau) d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

Taking $\tau_2 \rightarrow \infty$ and using $\lambda(\tau_2) \rightarrow 0$,

$$\lim_{\tau_2 \rightarrow \infty} \int_{\tau_1}^{\tau_2} a(\tau) d\tau = -\infty.$$

(3) Decompose $a = a_+ - a_-$. Then for each $\tau_2 > \tau_1$,

$$\int_{\tau_1}^{\tau_2} a(\tau) d\tau = \int_{\tau_1}^{\tau_2} a_+(\tau) d\tau - \int_{\tau_1}^{\tau_2} a_-(\tau) d\tau \geq - \int_{\tau_1}^{\tau_2} a_-(\tau) d\tau.$$

Hence

$$\int_{\tau_1}^{\tau_2} a_-(\tau) d\tau \geq - \int_{\tau_1}^{\tau_2} a(\tau) d\tau.$$

(4) Taking limits and using Step 2 yields

$$\int_{\tau_1}^{\infty} a_-(\tau) d\tau = \infty.$$

Indeed, by Step 2,

$$- \int_{\tau_1}^{\tau_2} a(\tau) d\tau \rightarrow +\infty.$$

Since the partial integrals of a_- dominate this quantity, they are unbounded. Because $a_- \geq 0$, monotone convergence gives $\int_{\tau_1}^{\infty} a_-(\tau) d\tau = \infty$.

(5) Split the cost into early and late times:

$$\int_{\tau_0}^{\infty} a_- M_R d\tau = \int_{\tau_0}^{\tau_1} a_- M_R d\tau + \int_{\tau_1}^{\infty} a_- M_R d\tau.$$

The first term is finite or infinite independent from the argument; the second term satisfies

$$\int_{\tau_1}^{\infty} a_-(\tau) M_R(\tau) d\tau \geq m_0 \int_{\tau_1}^{\infty} a_-(\tau) d\tau = \infty$$

by the bound in (1) and Step 4. Hence the total integral is infinite. $\square$

5.4. Finite-cost exclusions.

Theorem 5.7 (Scale-collapse cost exclusion). *No branch with finite scale-collapsing cost (Definition 5.3) can satisfy simultaneously $\lambda(\tau) \rightarrow 0$ and nonvanishing localized core.*

Proof. Assume, for contradiction, that such a branch exists. The two hypotheses are exactly those of Theorem 5.6; therefore

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\text{sc}}(\tau) d\tau = \infty.$$

This contradicts finite scale-collapsing cost. Therefore no such branch exists. $\square$

Lemma 5.8 (Absolute-cost domination). *For a.e. $\tau \in [\tau_0, \infty)$,*

$$\mathcal{C}_{\text{abs}}(\tau) \geq \mathcal{C}_{\text{sc}}(\tau).$$

Hence, pointwise and after integrating in τ ,

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\text{sc}}(\tau) d\tau \leq \int_{\tau_0}^{\infty} \mathcal{C}_{\text{abs}}(\tau) d\tau,$$

as extended real numbers. In particular, finite absolute cost implies finite scale-collapsing cost, and equivalently

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\text{sc}}(\tau) d\tau = \infty \implies \int_{\tau_0}^{\infty} \mathcal{C}_{\text{abs}}(\tau) d\tau = \infty.$$

Proof. At each time τ ,

$$\mathcal{C}_{\text{abs}}(\tau) = \nu \int |\nabla V|^2 \Phi + (a_+ + a_-)M_R \geq (a_+ + a_-)M_R \geq a_-M_R = \mathcal{C}_{\text{sc}}(\tau),$$

since $a_+, a_- \geq 0$ and $\nu \int |\nabla V|^2 \Phi \geq 0$. Integrating over $[\tau_0, \infty)$ yields the claimed implication. $\square$

Corollary 5.9 (Absolute scale-cost exclusion). *No branch with finite absolute cost in the sense of Definition 5.3 can satisfy simultaneously $\lambda(\tau) \rightarrow 0$ and nonvanishing localized core.*

Proof. Assume instead such a branch exists. By Theorem 5.6, $\int \mathcal{C}_{\text{sc}} = \infty$. By Lemma 5.8, the absolute cost is pointwise above $\mathcal{C}_{\text{sc}}$, so its integral must also be infinite, contradicting finite absolute cost. $\square$

Remark 5.10. The exclusions in this section are purely integral and use neither compactness nor any autonomous-limit argument.

5.5. Autonomous reduced-limit theorem from the selected-window estimates. The exclusions above are integral statements and do not rely on compactness. We now give the standard local compactness argument on a fixed finite autonomous window. The input used here is not the cost bound alone; it is the selected-window suitable package supplied by the renormalized representation, pressure reconstruction, and local Caccioppoli estimates in Sections 2 and 4.

Dependencies used in this section. In standard PDE terms, Sections 2 and 4 supply the following ingredients on selected compact renormalized windows:

- (1) the repaired-gauge renormalized equation and weak formulation for (V, P) ;
- (2) pressure reconstruction modulo spatial means, with local $L^{3/2}$ bounds on compact cylinders;
- (3) local windowed bounds $V \in L_t^\infty L_x^2 \cap L_t^2 H_x^1$ on compact cylinders;
- (4) a retained compact core lower bound preventing the selected profile from vanishing.

Together with the thick-window convergence of the modulation coefficients, these estimates imply the compactness and limit passage below.

Definition 5.11 (Thick autonomous windows). A branch has *thick autonomous windows* if there exist constants $T > 0$, $c > 0$, $M_1 \in (0, \infty)$, a sequence (τ_n) with $\tau_n \rightarrow \infty$, and limits $a_\infty \in \mathbb{R}$, $b_\infty \in \mathbb{R}^3$, such that for every n :

(i)

$$\frac{1}{T} \int_{\tau_n}^{\tau_n+T} a_-(\tau) M_R(\tau) d\tau \geq c;$$

(ii)

$$0 \leq M_R(\tau) \leq M_1 \quad \text{for a.e. } \tau \in [\tau_n, \tau_n + T];$$

(iii)

$$a(\tau_n + \cdot) \rightarrow a_\infty \text{ in } L^1(0, T), \quad b(\tau_n + \cdot) \rightarrow b_\infty \text{ in } L^1(0, T; \mathbb{R}^3).$$

Remark 5.12. This notion is used only to isolate a limiting drift coefficient $a_\infty < 0$. The compactness to extract a profile limit is obtained from the selected-window suitable estimates, not from the scalar cost inequalities of Sections 5.2–5.4.

Lemma 5.13 (Thick drift implies negative autonomous limit parameter). *On thick autonomous windows, the limiting autonomous coefficient is negative:*

$$a_\infty \leq -\frac{c}{M_1} < 0.$$

Proof. (1) From (ii), for a.e. $\tau \in [\tau_n, \tau_n + T]$, $0 \leq M_R(\tau) \leq M_1$. Therefore

$$\frac{a_-(\tau)M_R(\tau)}{M_1} \leq a_-(\tau).$$

Therefore

$$\frac{1}{T} \int_{\tau_n}^{\tau_n+T} a_-(\tau) d\tau \geq \frac{1}{M_1 T} \int_{\tau_n}^{\tau_n+T} a_-(\tau) M_R(\tau) d\tau \geq \frac{c}{M_1}.$$

(2) Set $a_n(s) := a(\tau_n + s)$. By (iii), $a_n \rightarrow a_\infty$ in $L^1(0, T)$. The map $r \mapsto r^-$ is 1-Lipschitz, so $a_n^- \rightarrow a_\infty^-$ in $L^1(0, T)$:

$$\|a_n^- - a_\infty^-\|_{L^1(0, T)} \leq \|a_n - a_\infty\|_{L^1(0, T)} \rightarrow 0.$$

Thus

$$\lim_{n \rightarrow \infty} \frac{1}{T} \int_{\tau_n}^{\tau_n+T} a_-(\tau) d\tau = \frac{1}{T} \int_0^T (a_\infty)^- ds = (a_\infty)^-.$$

(3) Combine Step 1 and Step 2: $(a_\infty)^- \geq c/M_1$, i.e. $a_\infty \leq -c/M_1 < 0$. □

Assumption 5.14 (Selected-window suitable estimates). Let (τ_n) be a thick autonomous-window sequence and set

$$\begin{aligned} V_n(y, s) &:= V(y, \tau_n + s), & P_n(y, s) &:= P(y, \tau_n + s), \\ \alpha_n(s) &:= a(\tau_n + s), & \beta_n(s) &:= b(\tau_n + s), \end{aligned} \quad s \in (0, T).$$

For every bounded Lipschitz domain $K \Subset \mathbb{R}^3$, assume there is a constant $C_K < \infty$, independent of n , such that

$$\|V_n\|_{L^\infty(0, T; L^2(K))} + \|V_n\|_{L^2(0, T; H^1(K))} + \|P_n - (P_n)_K\|_{L^{3/2}((0, T) \times K)} \leq C_K,$$

where

$$(P_n)_K(s) := \frac{1}{|K|} \int_K P_n(y, s) dy.$$

Assume also that each V_n is divergence free and that there exist $\chi \in C_c^\infty(B_R)$, $\chi \geq 0$, $\eta > 0$, and $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\int_{\mathbb{R}^3} |V_n(y, s)|^2 \chi(y) dy \geq \eta \quad \text{for a.e. } s \in (0, T).$$

Proposition 5.15 (Derivation of the selected-window estimates). *Let a repaired-gauge branch satisfy the renormalized representation and pressure reconstruction of the repaired-gauge representation theorem [Theorem 2.35](#) on the windows $[\tau_n, \tau_n + T]$. Assume that on each compact cylinder $(0, T) \times K$ the local compact-cylinder Caffarelli–Kohn–Nirenberg quantities are finite uniformly in n , and that the selected core has the localized lower bound stated in [Assumption 5.14](#). Then [Assumption 5.14](#) holds.*

Proof. The renormalized representation theorem supplies the shifted equation, divergence-free condition, and pressure reconstruction modulo functions of time. The pressure reconstruction gives, after subtracting spatial means,

$$\sup_n \|P_n - (P_n)_K\|_{L^{3/2}((0, T) \times K)} < \infty$$

on every compact K . The local Caccioppoli estimate applied on a slightly larger compact cylinder gives the inner bounds

$$\sup_n \|V_n\|_{L^\infty(0,T;L^2(K))} + \sup_n \|V_n\|_{L^2(0,T;H^1(K))} < \infty.$$

These are precisely the local velocity and pressure estimates in Assumption 5.14. The retained selected-core lower bound is the last condition in that assumption. Hence all items in Assumption 5.14 follow from the repaired-gauge representation, pressure, and local Caccioppoli results. $\square$

Lemma 5.16 (Local time regularity on autonomous windows). *Assume thick autonomous windows and Assumption 5.14. Let $K_0 \Subset K_1 \Subset \mathbb{R}^3$ be bounded Lipschitz domains, and choose an integer m so large that $H_0^m(K_1) \hookrightarrow W_0^{1,\infty}(K_1)$. Then*

$$\{\partial_s V_n\}_{n \geq 1} \text{ is bounded in } L^1(0,T;H^{-m}(K_1)).$$

Proof. Fix $\zeta \in C_c^\infty(K_1)$ with $\zeta \equiv 1$ on a neighborhood of K_0 . Let $\phi \in H_0^m(K_1; \mathbb{R}^3)$. Since $H_0^m(K_1) \hookrightarrow W_0^{1,\infty}(K_1)$, there is C_m such that

$$\|\phi\|_{L^\infty(K_1)} + \|\nabla \phi\|_{L^\infty(K_1)} \leq C_m \|\phi\|_{H^m(K_1)}.$$

The shifted equation is

$$\partial_s V_n = -(V_n \cdot \nabla) V_n - \nabla P_n + \nu \Delta V_n + \alpha_n (V_n + y \cdot \nabla V_n) + \beta_n \cdot \nabla V_n$$

in distributions. We estimate each term against ϕ .

For the convection term, integration by parts gives

$$|\langle (V_n \cdot \nabla) V_n, \phi \rangle| = \left| \int_{K_1} V_{n,i} V_{n,j} \partial_j \phi_i dy \right| \leq C_m \|V_n(s)\|_{L^2(K_1)}^2 \|\phi\|_{H^m(K_1)}.$$

For the viscous term,

$$|\langle \nu \Delta V_n, \phi \rangle| = \nu \left| \int_{K_1} \nabla V_n : \nabla \phi dy \right| \leq \nu C_m \|\nabla V_n(s)\|_{L^2(K_1)} |K_1|^{1/2} \|\phi\|_{H^m(K_1)}.$$

For the pressure term, subtracting the spatial mean does not change the gradient:

$$|\langle \nabla P_n, \phi \rangle| = \left| \int_{K_1} (P_n - (P_n)_{K_1}) \nabla \cdot \phi dy \right| \leq C_m |K_1|^{1/3} \|P_n - (P_n)_{K_1}\|_{L^{3/2}(K_1)} \|\phi\|_{H^m(K_1)}.$$

For the V_n -part of the scaling drift,

$$|\alpha_n(s)| \left| \int_{K_1} V_n \cdot \phi dy \right| \leq C_m |\alpha_n(s)| |K_1|^{1/2} \|V_n(s)\|_{L^2(K_1)} \|\phi\|_{H^m(K_1)}.$$

For the $y \cdot \nabla V_n$ -part, integrate by parts:

$$\int_{K_1} (y \cdot \nabla V_n) \cdot \phi dy = - \int_{K_1} V_n \cdot (y \cdot \nabla \phi + 3\phi) dy,$$

because ϕ has zero trace. Hence

$$|\alpha_n(s)| \left| \int_{K_1} (y \cdot \nabla V_n) \cdot \phi dy \right| \leq C_{K_1,m} |\alpha_n(s)| \|V_n(s)\|_{L^2(K_1)} \|\phi\|_{H^m(K_1)}.$$

Similarly,

$$\left| \int_{K_1} (\beta_n \cdot \nabla V_n) \cdot \phi dy \right| = \left| \int_{K_1} V_n \cdot (\beta_n \cdot \nabla \phi) dy \right| \leq C_m |\beta_n(s)| |K_1|^{1/2} \|V_n(s)\|_{L^2(K_1)} \|\phi\|_{H^m(K_1)}.$$

Combining the estimates and taking the supremum over $\|\phi\|_{H^m(K_1)} \leq 1$, we obtain

$$\begin{aligned} \|\partial_s V_n(s)\|_{H^{-m}(K_1)} &\leq C_{K_1, m} (\|V_n(s)\|_{L^2(K_1)}^2 + \|\nabla V_n(s)\|_{L^2(K_1)} \\ &\quad + \|P_n(s) - (P_n)_{K_1}(s)\|_{L^{3/2}(K_1)} + (|\alpha_n(s)| + |\beta_n(s)|) \|V_n(s)\|_{L^2(K_1)}). \end{aligned}$$

Integrating in $s \in (0, T)$, using Assumption 5.14, and using $\alpha_n \rightarrow a_\infty$, $\beta_n \rightarrow b_\infty$ in L^1 , hence uniform L^1 -bounds for α_n, β_n , proves the claimed $L^1(0, T; H^{-m})$ bound. $\square$

Lemma 5.17 (Local compactness on autonomous windows). *Assume thick autonomous windows and Assumption 5.14. Then after passing to a subsequence there exist*

$$U \in L^\infty(0, T; L_{\text{loc}}^2(\mathbb{R}^3)) \cap L^2(0, T; H_{\text{loc}}^1(\mathbb{R}^3)), \quad \Pi \in L_{\text{loc}}^{3/2}((0, T) \times \mathbb{R}^3),$$

such that, for every compact $K \Subset \mathbb{R}^3$,

(a)

$$V_n \rightarrow U \quad \text{strongly in } L^2(0, T; L^2(K));$$

(b)

$$V_n \rightharpoonup U \quad \text{weakly in } L^2(0, T; H^1(K));$$

(c)

$$V_n \rightharpoonup^* U \quad \text{weakly-}^* \text{ in } L^\infty(0, T; L^2(K));$$

(d)

$$P_n - (P_n)_K \rightharpoonup \Pi_K \quad \text{weakly in } L^{3/2}((0, T) \times K),$$

where the local distributions $\nabla \Pi_K$ agree on overlaps and define a pressure gradient $\nabla \Pi$ modulo functions of time.

Proof. Fix $K_0 \Subset K_1 \Subset \mathbb{R}^3$. Assumption 5.14 gives boundedness of V_n in $L^2(0, T; H^1(K_1))$, and Lemma 5.16 gives boundedness of $\partial_s V_n$ in $L^1(0, T; H^{-m}(K_1))$ for m large. Since

$$H^1(K_1) \Subset L^2(K_0) \hookrightarrow H^{-m}(K_1),$$

the Aubin–Lions–Simon compactness theorem implies, after extracting a subsequence,

$$V_n \rightarrow U \quad \text{strongly in } L^2(0, T; L^2(K_0)).$$

The local $L^\infty L^2$ and $L^2 H^1$ bounds also give, after extraction,

$$V_n \rightharpoonup^* U \quad \text{in } L^\infty(0, T; L^2(K_0)), \quad V_n \rightharpoonup U \quad \text{in } L^2(0, T; H^1(K_0)).$$

The pressure bound in Assumption 5.14 gives weak compactness of $P_n - (P_n)_{K_0}$ in $L^{3/2}((0, T) \times K_0)$. A diagonal extraction over balls B_j gives a single subsequence for every compact K . On overlaps, two pressure limits can differ only by a function of s , because both represent the same distributional pressure gradient in the limit equation. Thus the gradients assemble into $\nabla \Pi$, with Π defined modulo functions of time. $\square$

Lemma 5.18 (Compatibility of local pressure limits). *Let $K_1, K_2 \Subset \mathbb{R}^3$ be bounded Lipschitz domains with nonempty connected overlap $K_{12} := K_1 \cap K_2$. Suppose that, after passing to a common subsequence,*

$$P_n - (P_n)_{K_i} \rightharpoonup \Pi_i \quad \text{in } L^{3/2}((0, T) \times K_i), \quad i = 1, 2.$$

Then there exists $c_{12} \in L^{3/2}(0, T)$ such that

$$\Pi_1(y, s) - \Pi_2(y, s) = c_{12}(s) \quad \text{for a.e. } (y, s) \in K_{12} \times (0, T).$$

Consequently the pressure gradient $\nabla \Pi$ obtained from the local limits is globally well-defined on $(0, T) \times \mathbb{R}^3$, with the pressure itself defined modulo functions of time.

Proof. For each n ,

$$(P_n - (P_n)_{K_1}) - (P_n - (P_n)_{K_2}) = (P_n)_{K_2}(s) - (P_n)_{K_1}(s)$$

on K_{12} . Hence its spatial gradient is zero in $\mathcal{D}'((0, T) \times K_{12})$. Passing to the weak limit gives

$$\nabla_y(\Pi_1 - \Pi_2) = 0 \quad \text{in } \mathcal{D}'((0, T) \times K_{12}).$$

Since K_{12} is connected, the standard distributional characterization of functions with zero spatial gradient implies that $\Pi_1 - \Pi_2 = c_{12}(s)$ for some $c_{12} \in L^{3/2}(0, T)$. The last assertion follows because adding a time-dependent scalar to pressure does not change $\nabla \Pi$. $\square$

Theorem 5.19 (Autonomous reduced-limit theorem). *Assume that a branch has thick autonomous windows and satisfies Assumption 5.14 on the same window sequence. Let (U, Π) be the limit extracted in Lemma 5.17. Then*

$$\nabla \cdot U = 0,$$

$$U \not\equiv 0$$

and (U, Π) satisfies

$$(11) \quad \partial_s U + (U \cdot \nabla)U + \nabla \Pi = \nu \Delta U + a_\infty(U + y \cdot \nabla U) + b_\infty \cdot \nabla U, \quad \nabla \cdot U = 0,$$

in $\mathcal{D}'((0, T) \times \mathbb{R}^3)$. Equivalently, for every $\varphi \in C_c^\infty((0, T) \times \mathbb{R}^3; \mathbb{R}^3)$,

$$\int_0^T \int (-U \cdot \partial_s \varphi - (U \otimes U) : \nabla \varphi + \nu \nabla U : \nabla \varphi - \Pi \nabla \cdot \varphi - a_\infty(U + y \cdot \nabla U) \cdot \varphi - (b_\infty \cdot \nabla U) \cdot \varphi) dy ds = 0.$$

Remark 5.20. The theorem does not assert an autonomous Liouville or rigidity contradiction. It proves the compact finite-window autonomous limit from the selected-window estimates. Any later rigidity step must apply a separate Liouville theorem to the class containing (U, Π) .

Proof. Step 1: divergence-free property of the limit.

Each V_n is divergence free by Assumption 5.14, so for any compactly supported scalar $\psi \in C_c^\infty((0, T) \times \mathbb{R}^3)$,

$$\int_{(0, T) \times \mathbb{R}^3} V_n \cdot \nabla \psi dy ds = 0.$$

Hence, for every such ψ , since $\nabla \psi \in C_c^\infty((0, T) \times \mathbb{R}^3)$ and $V_n \rightarrow U$ in $L^2(0, T; L_{\text{loc}}^2)$,

$$\int_{(0, T) \times \mathbb{R}^3} (V_n - U) \cdot \nabla \psi dy ds \rightarrow 0,$$

we deduce

$$\int_{(0, T) \times \mathbb{R}^3} U \cdot \nabla \psi dy ds = 0.$$

Hence $\nabla \cdot U = 0$ in $\mathcal{D}'((0, T) \times \mathbb{R}^3)$.

Step 2: nontriviality of the limit profile. Define for $s \in (0, T)$

$$f_n(s) := \int_{\mathbb{R}^3} |V_n(y, s)|^2 \chi(y) dy, \quad f(s) := \int_{\mathbb{R}^3} |U(y, s)|^2 \chi(y) dy.$$

Here $K_\chi := \text{supp } \chi$, so $\chi^{1/2} V_n, \chi^{1/2} U \in L^2(0, T; L^2(K_\chi))$. Then

$$f_n - f = \int_{\mathbb{R}^3} (V_n - U) \cdot (V_n + U) \chi dy.$$

Applying Cauchy–Schwarz in space,

$$|f_n(s) - f(s)| \leq \|\chi^{1/2}(V_n(s) - U(s))\|_{L^2(\mathbb{R}^3)} \|\chi^{1/2}(V_n(s) + U(s))\|_{L^2(\mathbb{R}^3)}.$$

Integrating in s and applying Cauchy–Schwarz in time gives

$$\|f_n - f\|_{L^1(0,T)} \leq \|\chi^{1/2}(V_n - U)\|_{L^2(0,T;L^2)} \|\chi^{1/2}(V_n + U)\|_{L^2(0,T;L^2)}.$$

Since K_χ is compact, Assumption 5.14 and the membership $U \in L^2(0,T;L^2(K_\chi))$ give

$$\sup_n \|\chi^{1/2}(V_n + U)\|_{L^2(0,T;L^2)} \leq C_\chi \left(\sup_n \|V_n\|_{L^2(0,T;L^2(K_\chi))} + \|U\|_{L^2(0,T;L^2(K_\chi))} \right) < \infty.$$

Moreover

$$\|\chi^{1/2}(V_n - U)\|_{L^2(0,T;L^2)} \leq \|\chi\|_{L^\infty}^{1/2} \|V_n - U\|_{L^2(0,T;L^2(K_\chi))} \rightarrow 0.$$

Therefore

$$\|f_n - f\|_{L^1(0,T)} \rightarrow 0.$$

Hence $f_n \rightarrow f$ in $L^1(0,T)$.

Assumption 5.14 gives for large n , $f_n(s) \geq \eta$ for a.e.s. Therefore

$$\int_0^T f(s) ds = \lim_{n \rightarrow \infty} \int_0^T f_n(s) ds \geq \eta T > 0.$$

Hence $f \neq 0$, so $U \neq 0$.

Step 3: shifted weak formulation for V_n . Take any test field $\varphi \in C_c^\infty((0,T) \times \mathbb{R}^3; \mathbb{R}^3)$, and set

$$K_\varphi := \overline{\{x \in \mathbb{R}^3 : \exists s \in (0,T), \varphi(x,s) \neq 0\}} \subset \mathbb{R}^3, \quad K_\varphi \text{ bounded.}$$

Since (V,P) satisfies (8) in distributions, the change of variables $\tau = \tau_n + s$ gives the shifted weak formulation

$$(12) \quad \begin{aligned} 0 &= \int_0^T \int_{\mathbb{R}^3} \left(-V_n \cdot \partial_s \varphi - (V_n \otimes V_n) : \nabla \varphi + \nu \nabla V_n : \nabla \varphi - P_n \nabla \cdot \varphi \right) dy ds \\ &\quad + \int_0^T \int_{\mathbb{R}^3} \left(-a(\tau_n + s)(V_n + y \cdot \nabla V_n) \cdot \varphi - (b(\tau_n + s) \cdot \nabla V_n) \cdot \varphi \right) dy ds. \end{aligned}$$

Step 4: term-by-term passage to the limit. Write $\alpha_n(s) := a(\tau_n + s)$, $\beta_n(s) := b(\tau_n + s)$.

(i) Time derivative. By strong convergence in $L^2(0,T;L^2(K_\varphi))$,

$$\left| \int_0^T \int_{\mathbb{R}^3} (V_n - U) \cdot \partial_s \varphi dy ds \right| \leq \|V_n - U\|_{L^2(0,T;L^2(K_\varphi))} \|\partial_s \varphi\|_{L^2(0,T;L^2(K_\varphi))} \rightarrow 0.$$

Hence

$$\int V_n \cdot \partial_s \varphi \rightarrow \int U \cdot \partial_s \varphi.$$

(ii) Convection term. Decompose

$$V_n \otimes V_n - U \otimes U = (V_n - U) \otimes V_n + U \otimes (V_n - U).$$

Then

$$\begin{aligned} &\left| \int_0^T \int_{\mathbb{R}^3} ((V_n \otimes V_n - U \otimes U) : \nabla \varphi) dy ds \right| \\ &\leq \|\nabla \varphi\|_{L_{x,t}^\infty} \|V_n - U\|_{L^2(0,T;L^2(K_\varphi))} (\|V_n\|_{L^2(0,T;L^2(K_\varphi))} + \|U\|_{L^2(0,T;L^2(K_\varphi))}) \rightarrow 0, \end{aligned}$$

using boundedness of V_n in $L^2(0,T;L^2(K_\varphi))$. Therefore $\int (V_n \otimes V_n) : \nabla \varphi \rightarrow \int (U \otimes U) : \nabla \varphi$.

(iii) Viscous term. From weak convergence of V_n in $L^2(0,T;H_{\text{loc}}^1)$,

$$\int_0^T \int_{\mathbb{R}^3} \nabla V_n : \nabla \varphi dy ds \rightarrow \int_0^T \int_{\mathbb{R}^3} \nabla U : \nabla \varphi dy ds.$$

(iv) Pressure term. Let K be a bounded Lipschitz domain with $K_\varphi \Subset K$. Since φ is compactly supported in K ,

$$\int_K (P_n)_K(s) \nabla \cdot \varphi(y, s) dy = (P_n)_K(s) \int_K \nabla \cdot \varphi(y, s) dy = 0.$$

Therefore

$$\int_0^T \int_{\mathbb{R}^3} P_n \nabla \cdot \varphi dy ds = \int_0^T \int_K (P_n - (P_n)_K) \nabla \cdot \varphi dy ds.$$

By Lemma 5.17,

$$P_n - (P_n)_K \rightharpoonup \Pi_K \quad \text{in } L^{3/2}((0, T) \times K),$$

and $\nabla \cdot \varphi \in L^3((0, T) \times K)$. The compatibility Lemma 5.18 permits us to write the resulting pressure distribution as Π , modulo functions of time. Hence

$$\int_0^T \int_{\mathbb{R}^3} P_n \nabla \cdot \varphi dy ds \rightarrow \int_0^T \int_K \Pi_K \nabla \cdot \varphi dy ds.$$

The value of the last integral is independent of the chosen K , because replacing Π_K by $\Pi_K + c(s)$ changes the integral by $\int_0^T c(s) \int_K \nabla \cdot \varphi dy ds = 0$. This is the pressure term $\int \Pi \nabla \cdot \varphi$.

(v) Drift term with a : part involving V_n . Define

$$Y_n^0(s) := \int_{\mathbb{R}^3} V_n(y, s) \cdot \varphi(y, s) dy, \quad Y^0(s) := \int_{\mathbb{R}^3} U(y, s) \cdot \varphi(y, s) dy.$$

For each s , Cauchy–Schwarz gives

$$|Y_n^0(s)| \leq \|V_n(s)\|_{L^2(K_\varphi)} \|\varphi(s)\|_{L^2(K_\varphi)}.$$

By Assumption 5.14, there is $C_0 < \infty$ such that $\sup_n \|Y_n^0\|_{L^\infty(0, T)} \leq C_0$. Moreover,

$$Y_n^0 \rightarrow Y^0 \quad \text{in } L^2(0, T),$$

because $V_n \rightarrow U$ in $L^2(0, T; L^2(K_\varphi))$. Now

$$\int_0^T \alpha_n(s) Y_n^0(s) ds = \int_0^T (\alpha_n(s) - a_\infty) Y_n^0(s) ds + a_\infty \int_0^T Y_n^0(s) ds.$$

For the first term,

$$\left| \int_0^T (\alpha_n - a_\infty) Y_n^0 ds \right| \leq \sup_{s \in (0, T)} |Y_n^0(s)| \|\alpha_n - a_\infty\|_{L^1(0, T)} \rightarrow 0.$$

For the second term, strong convergence in $L^2(0, T)$ implies $\int_0^T Y_n^0 \rightarrow \int_0^T Y^0$, so

$$\int_0^T \alpha_n(s) \int_{\mathbb{R}^3} V_n \cdot \varphi dy ds \rightarrow a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot \varphi dy ds.$$

Thus the signed contribution in (12) satisfies

$$-\int_0^T \alpha_n(s) \int_{\mathbb{R}^3} V_n \cdot \varphi dy ds \rightarrow -a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot \varphi dy ds.$$

(vi) Drift term with a : part involving $y \cdot \nabla V_n$. Since $\varphi \in C_c^\infty$, integrating by parts in y gives

$$\int_{\mathbb{R}^3} (y \cdot \nabla V_n) \cdot \varphi = - \int_{\mathbb{R}^3} V_n \cdot (y \cdot \nabla \varphi + 3\varphi) dy.$$

Define

$$Y_n^1(s) := \int_{\mathbb{R}^3} V_n \cdot (y \cdot \nabla \varphi + 3\varphi) dy, \quad Y^1(s) := \int_{\mathbb{R}^3} U \cdot (y \cdot \nabla \varphi + 3\varphi) dy.$$

For each s , Cauchy–Schwarz gives

$$|Y_n^1(s)| \leq \|V_n(s)\|_{L^2(K_\varphi)} \|y \cdot \nabla \varphi(s) + 3\varphi(s)\|_{L^2(K_\varphi)}.$$

The $L^\infty(0, T; L^2(K_\varphi))$ bound in Assumption 5.14 gives

$$\sup_n \|Y_n^1\|_{L^\infty(0, T)} \leq C_1 < \infty.$$

Also,

$$\|Y_n^1 - Y^1\|_{L^2(0, T)} \leq \|V_n - U\|_{L^2(0, T; L^2(K_\varphi))} \|y \cdot \nabla \varphi + 3\varphi\|_{L^\infty(0, T; L^2(K_\varphi))} \rightarrow 0.$$

Therefore

$$- \int_0^T \alpha_n(s) \int_{\mathbb{R}^3} (y \cdot \nabla V_n) \cdot \varphi dy ds \rightarrow a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot (y \cdot \nabla \varphi + 3\varphi) dy ds.$$

Integrating by parts once more in the limit gives

$$a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot (y \cdot \nabla \varphi + 3\varphi) dy ds = -a_\infty \int_0^T \int_{\mathbb{R}^3} (y \cdot \nabla U) \cdot \varphi dy ds.$$

(vii) Term with b . Since $\beta_n(s)$ is independent of y ,

$$\int_{\mathbb{R}^3} (\beta_n \cdot \nabla V_n) \cdot \varphi = - \int_{\mathbb{R}^3} V_n \cdot (\beta_n \cdot \nabla \varphi) dy,$$

so

$$- \int_0^T \int_{\mathbb{R}^3} (\beta_n \cdot \nabla V_n) \cdot \varphi dy ds = \int_0^T \int_{\mathbb{R}^3} V_n \cdot (\beta_n \cdot \nabla \varphi) dy ds.$$

Let

$$Z_n(s) := \left(\int_{\mathbb{R}^3} V_n \cdot \partial_1 \varphi dy, \int_{\mathbb{R}^3} V_n \cdot \partial_2 \varphi dy, \int_{\mathbb{R}^3} V_n \cdot \partial_3 \varphi dy \right),$$

$$Z(s) := \left(\int_{\mathbb{R}^3} U \cdot \partial_1 \varphi dy, \int_{\mathbb{R}^3} U \cdot \partial_2 \varphi dy, \int_{\mathbb{R}^3} U \cdot \partial_3 \varphi dy \right).$$

Then

$$\int_{\mathbb{R}^3} V_n \cdot (\beta_n \cdot \nabla \varphi) dy = \beta_n(s) \cdot Z_n(s), \quad \int_{\mathbb{R}^3} U \cdot (b_\infty \cdot \nabla \varphi) dy = b_\infty \cdot Z(s).$$

Consequently

$$\left| \int_0^T (\beta_n - b_\infty) \cdot Z_n ds \right| \leq \|\beta_n - b_\infty\|_{L^1(0, T)} \sup_{s \in (0, T)} |Z_n(s)| \rightarrow 0,$$

because

$$|Z_n(s)| \leq \|V_n(s)\|_{L^2(K_\varphi)} \left(\sum_{j=1}^3 \|\partial_j \varphi(s)\|_{L^2(K_\varphi)}^2 \right)^{1/2}$$

and Assumption 5.14 bounds the right-hand side uniformly in n and s . Furthermore,

$$\|Z_n - Z\|_{L^2(0, T; \mathbb{R}^3)} \leq \|V_n - U\|_{L^2(0, T; L^2(K_\varphi))} \left(\sum_{j=1}^3 \|\partial_j \varphi\|_{L^\infty(0, T; L^2(K_\varphi))}^2 \right)^{1/2} \rightarrow 0.$$

Therefore

$$\int_0^T b_\infty \cdot Z_n(s) ds \rightarrow \int_0^T b_\infty \cdot Z(s) ds.$$

Hence

$$-\int_0^T \int (\beta_n \cdot \nabla V_n) \cdot \varphi dy ds \rightarrow \int_0^T \int U \cdot (b_\infty \cdot \nabla \varphi) dy ds.$$

Since b_∞ is constant in y , integration by parts yields

$$\int_0^T \int U \cdot (b_\infty \cdot \nabla \varphi) dy ds = - \int_0^T \int (b_\infty \cdot \nabla U) \cdot \varphi dy ds.$$

Step 5: assemble the limit. Passing to the limit in (12) in each term (Steps 4(i)–4(vii)) gives

$$\int_0^T \int (-U \cdot \partial_s \varphi - (U \otimes U) : \nabla \varphi + \nu \nabla U : \nabla \varphi - \Pi \nabla \cdot \varphi - a_\infty (U + y \cdot \nabla U) \cdot \varphi - (b_\infty \cdot \nabla U) \cdot \varphi) dy ds = 0.$$

Since this holds for every compactly supported vector field φ , equation (11) holds in distributions on $(0, T) \times \mathbb{R}^3$. $\square$

5.6. State-space stratification of the scale-collapse branch. The cost argument above closes the finite-cost scale-collapse branch. The remaining scale-collapse obstruction is treated by a state-space stratification, following the local estimate and autonomous-reduction route established earlier in this document. The purpose of this section is not to assert a blanket Liouville theorem for every autonomous negative-drift equation. Instead, we separate the closed branches from the precise residual defects.

Theorem 5.21 (Selected core retention). *Let a represented Type II branch be obtained from the selected-core construction developed above. Then there exist $R > 0$, a cutoff $\chi \in C_c^\infty(B_R)$, $\chi \geq 0$, a constant $\eta > 0$, and a terminal time τ_1 such that, along every selected window used below,*

$$\int_{\mathbb{R}^3} |V(y, \tau)|^2 \chi(y) dy \geq \eta \quad \text{for a.e. } \tau \geq \tau_1.$$

In particular, every translated sequence $V_n(y, s) = V(y, \tau_n + s)$ satisfies the localized nontriviality condition in Assumption 5.14, after discarding finitely many n .

Proof. The selected-core construction chooses a compact renormalized core carrying a fixed positive amount of localized kinetic mass. Since the cutoff χ is supported strictly inside that core and equals a positive spatial weight there, the selected-core lower bound gives the stated estimate. Translating $\tau = \tau_n + s$ gives the final assertion on every finite selected window. $\square$

Theorem 5.22 (Autonomous modulation selection). *Let a represented scale-collapse branch satisfy the modulation bounds obtained in the repaired gauge construction. Suppose that a sequence of windows $[\tau_n, \tau_n + T]$ is selected in the autonomous stratum. Then, after passing to a subsequence, there exist constants $a_\infty \in \mathbb{R}$ and $b_\infty \in \mathbb{R}^3$ such that*

$$a(\tau_n + \cdot) \rightarrow a_\infty \quad \text{in } L^1(0, T), \quad b(\tau_n + \cdot) \rightarrow b_\infty \quad \text{in } L^1(0, T; \mathbb{R}^3).$$

Proof. This is the autonomous-window selection conclusion of the repaired-gauge modulation analysis: the selected stratum is defined by convergence of the modulation functions to constant autonomous parameters on the fixed window length T . Passing to the subsequence supplied by that selection gives exactly the displayed L^1 -convergences. $\square$

Theorem 5.23 (Scale-collapse state-space stratification). *Let a represented Type II branch have genuine scale collapse and nonvanishing selected core. Then exactly one of the following alternatives occurs after passing to selected terminal windows:*

- (i) the branch has finite scale-collapsing cost, which is excluded by Theorem 5.7;
- (ii) the branch has finite absolute cost, which is excluded by Corollary 5.9;
- (iii) the branch admits thick autonomous windows in the sense of Definition 5.11, satisfies the selected-window estimates of Assumption 5.14, and has autonomous modulation convergence;
- (iv) the branch has a thin-drift defect: no selected fixed-length window carries a uniform positive lower bound for the averaged weighted negative drift;
- (v) the branch has an autonomous-modulation defect: thick windows exist, but no subsequence has a constant-coefficient L^1 modulation limit in the sense needed for Theorem 5.19;
- (vi) the branch has a compactness defect: the local estimates needed for Assumption 5.14 fail on the selected windows.

Proof. The scale-collapse selection theorem decomposes terminal scale-collapse windows according to the averaged negative-drift mass

$$\frac{1}{T} \int_{\tau_n}^{\tau_n+T} a_-(\tau) M_R(\tau) d\tau.$$

If the accumulated cost is finite, alternatives (i) or (ii) apply according to the cost functional under consideration. If the averaged negative-drift mass stays bounded below on a subsequence, one obtains thick windows; otherwise the branch is in the thin-drift alternative. On thick windows, either the local estimates and core retention hold, in which case Proposition 5.15 and Theorem 5.21 give Assumption 5.14, or the branch is in the compactness defect. If the estimates hold but no subsequence has the required constant-coefficient modulation limit, the branch is in the autonomous-modulation defect. The remaining case is precisely the thick autonomous alternative. $\square$

Definition 5.24 (Stationary omega-limit class). Let (U, Π) be a nonzero autonomous limit produced by Theorem 5.19. We say that it has a stationary omega-limit in the admissible class if there exists a nonzero pair

$$(W, Q), \quad W \in L^3(\mathbb{R}^3),$$

with the local energy and pressure reconstruction properties inherited from the selected windows, such that

$$(W \cdot \nabla)W + \nabla Q = \nu \Delta W + a_\infty(W + y \cdot \nabla W) + b_\infty \cdot \nabla W, \quad \nabla \cdot W = 0,$$

in $\mathcal{D}'(\mathbb{R}^3)$.

Definition 5.25 (Rigidity-covered stationary class). For a parameter pair (a_∞, b_∞) , the stationary class is rigidity-covered if every admissible stationary solution in Definition 5.24 is either zero or is not a terminal Type II branch. In the classical backward self-similar normalization this is the role of the Nečas–Růžička–Šverák rigidity theorem, after the harmless translation drift is removed. Other parameter regimes require their own stated Liouville or classification theorem.

Theorem 5.26 (Stationary-rigidity blocker). *Assume a thick autonomous branch produces a nonzero autonomous limit by Theorem 5.19. If that autonomous limit has a stationary omega-limit in the admissible class and the corresponding stationary class is rigidity-covered, then the branch is not an unresolved Type II scale-collapse branch.*

Proof. The stationary omega-limit gives a nonzero stationary profile (W, Q) in the admissible class. The rigidity-covered hypothesis says that no such nonzero profile can represent a terminal Type II branch. This contradicts unresolved membership in the scale-collapse Type II state. $\square$

Theorem 5.27 (Scale-collapse alternatives). *Let a represented Type II branch have genuine scale collapse and a nonvanishing selected core. Then one of the following mutually exclusive outcomes occurs:*

- (i) *the finite scale-collapsing cost exclusion applies;*
- (ii) *the finite absolute cost exclusion applies;*
- (iii) *the branch has a thin-drift defect;*
- (iv) *the branch has an autonomous-modulation defect;*
- (v) *the branch has a compactness defect;*
- (vi) *the branch has a nonzero autonomous reduced limit, and either a stationary-rigidity blocker applies or the remaining obstruction is the corresponding stationary/generalized self-similar Liouville problem.*

Proof. Apply Theorem 5.23. The finite-cost alternatives are exactly items (i) and (ii). The thin, modulation, and compactness defects are items (iii)–(v). In the remaining thick autonomous case, Lemma 5.13 gives $a_\infty < 0$, and Theorem 5.19 produces a nonzero autonomous reduced limit. If Definitions 5.24 and 5.25 apply, then Theorem 5.26 blocks the branch. If not, the unresolved remainder is precisely the stationary or generalized self-similar Liouville problem for the parameter pair (a_∞, b_∞) . $\square$

5.7. Conclusion. The exclusions in Sections 5.2–5.4 are proved unconditionally within the renormalized setting: no branch with finite scale-collapsing (hence finite absolute) cost can have $\lambda(\tau) \rightarrow 0$ together with a nonvanishing localized core.

Lemma 5.13 shows that any thick autonomous window admits a negative autonomous drift limit a_∞ . Lemmas 5.16 and 5.17, together with Theorem 5.19, assemble the selected-window estimates from Sections 2 and 4 into a nontrivial autonomous finite-window limit (U, Π) satisfying the full distributional autonomous equation.

Section 5.6 records the same state-space stratification as the markdown route. The autonomous reduction is a closed contradiction only in parameter regimes where a stationary omega-limit is available and the resulting stationary class is covered by a Liouville or rigidity theorem, such as the classical Nečas–Růžička–Šverák self-similar class. Outside those regimes this document records the precise thin-drift, modulation, compactness, or Liouville-defect alternative rather than asserting an unconditional closure.

6. ASSEMBLY OF THE LOCAL TYPE II EXCLUSION

Abstract. We assemble the local Type II exclusion argument for the three-dimensional Navier–Stokes equations from the local results proved in [Sections 1 to 5](#). The entry point is the local Caffarelli–Kohn–Nirenberg concentration theorem: any singular point which is not in the local Type I alternative yields a positive local Type II concentration sequence. The repaired-gauge construction places each retained nondegenerate core into a renormalized local Navier–Stokes chart. The single-core, multibubble/cascade, rough-core, scale-collapse, and virial-controlled terminal analyses then exhaust or eliminate the local state-space alternatives to which they apply. The proof of the assembly below uses only those local alternatives and does not add any stationary-profile classification theorem.

6.1. Local Entry. Let (u, p) be a suitable weak solution of the three-dimensional Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0,$$

on $\mathbb{R}^3 \times (T - \delta, T)$. For $z_0 = (x_0, T)$ and $r > 0$, set

$$Q_r(z_0) = B_r(x_0) \times (T - r^2, T),$$

$$C(z_0, r) = r^{-2} \int_{Q_r(z_0)} |u|^3 dx dt,$$

and

$$D(z_0, r) = r^{-2} \int_{T-r^2}^T \int_{B_r(x_0)} |p(x, t) - (p)_{B_r(x_0)}(t)|^{3/2} dx dt.$$

The pressure is always taken modulo functions of time, and the spatial mean normalization is the one used in the Caffarelli–Kohn–Nirenberg criterion.

Definition 6.1 (Singular set and local Type II point). Let $\Sigma(T)$ be the set of points x_0 for which u is not locally bounded in any backward cylinder $Q_r(x_0, T)$. A point $x_0 \in \Sigma(T)$ is in the local Type II alternative if it carries a positive local Caffarelli–Kohn–Nirenberg concentration sequence and no local Type I scale-invariant bound holds at (x_0, T) .

Definition 6.2 (Positive local Type II concentration sequence). A sequence $r_n \downarrow 0$ is a positive local Type II concentration sequence at (x_0, T) if there exists $\eta > 0$ such that

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n,$$

and (x_0, T) is in the local Type II alternative. The associated parabolic rescalings are

$$u_n(y, s) = r_n u(x_0 + r_n y, T + r_n^2 s), \quad p_n(y, s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Theorem 6.3 (Local concentration and Type I/Type II dichotomy). *If $\Sigma(T) \neq \emptyset$, then there are $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

At the concentration point exactly one of the following alternatives holds: there is a local Type I scale-invariant bound, or (x_0, T) is in the local Type II alternative.

Proof. This is the local concentration and blow-up dichotomy used as the entry point for [Section 6](#). □

6.2. Local Theorems Used in the Assembly. We record the local results established above in the form used below. They are quoted here as PDE statements to keep the final assembly self-contained.

Theorem 6.4 (Repaired-gauge representation). *Let a positive local Type II concentration sequence contain a nondegenerate selected core. Then, after passing to selected windows, there are local scale-center variables $\lambda(\tau) > 0$, $x_c(\tau)$, renormalized fields*

$$V(y, \tau) = \lambda(\tau)u(x_c(\tau) + \lambda(\tau)y, t(\tau)), \quad P(y, \tau) = \lambda(\tau)^2 p(x_c(\tau) + \lambda(\tau)y, t(\tau)) + \pi(\tau),$$

and modulation coefficients a, b such that

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0.$$

On each compact renormalized window one has the local suitable energy inequality, local pressure reconstruction modulo functions of time, stability under subsequences, and the compact-window bounds proved in [Section 2](#). Failure of the nondegenerate scale-center selection is part of the multibubble, cascade, or scale-rigidity alternatives.

Proof. This is the compact-window representation package of [Theorem 2.35](#). $\square$

Theorem 6.5 (Compact single-core exclusion). *No compact single-core vanishing-dissipation branch can be a local Type II sequence. More explicitly, let a represented selected-window sequence have:*

- (i) *positive local CKN concentration on a fixed compact cylinder;*
- (ii) *a uniformly bounded compact-cylinder suitable package $A + E + C + D$;*
- (iii) *nondegenerate local scale-center representation;*
- (iv) *bounded modulation coefficients;*
- (v) *local pressure reconstruction and pressure stability modulo spatial means;*
- (vi) *vanishing localized dissipation on some retained compact cylinder.*

Then the sequence is incompatible with the local Type II condition.

Proof. This is exactly [Theorem 1.18](#). $\square$

Theorem 6.6 (Multibubble and cascade exclusion). *Assume the terminal admissibility and scale-rigidity hypotheses stated in the multibubble/cascade analysis. Then no local multibubble, cascade, or gauge-degenerate Type II branch can occur in the class produced by the local active-core extraction. Equivalently, every failure of the single nondegenerate compact-core alternative is excluded by the local multibubble/cascade analysis under those stated hypotheses.*

Proof. This is the local multibubble and cascade exclusion theorem [Theorem 3.23](#), together with its removal corollary [Theorem 3.24](#). $\square$

Theorem 6.7 (Rough-core reduction). *On any compact renormalized window satisfying the local energy inequality and bounded modulation hypotheses, finite outer velocity-pressure CKN control $\mathcal{C} + \mathcal{D}$ implies finite inner windowed H^1 control. Hence failure of local windowed H^1 control on a retained inner cylinder forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ package on every relevant enclosing cylinder.*

Proof. This is the rough-core exclusion on compact windows, [Theorem 4.7](#). $\square$

Theorem 6.8 (Finite-cost scale-collapse exclusion). *No represented branch with finite scale-collapsing cost, and hence no branch with finite absolute scale cost, can have genuine scale collapse together with a nonvanishing selected core.*

Proof. This is the scale-collapse cost exclusion [Theorem 5.7](#) and absolute scale-cost exclusion [Theorem 5.9](#). $\square$

Theorem 6.9 (Scale-collapse state-space stratification). *Let a represented Type II branch have genuine scale collapse and a nonvanishing selected core. After passing to selected terminal windows, exactly one of the state-space alternatives isolated in [Section 5](#) occurs:*

- (i) *finite scale-collapsing cost;*
- (ii) *finite absolute scale cost;*
- (iii) *thin-drift defect;*
- (iv) *autonomous-modulation defect;*
- (v) *compactness defect;*
- (vi) *a nonzero autonomous reduced limit in the scale-rigid state.*

In the local Type II state space used by the multibubble/cascade and scale-collapse analyses, alternatives (iii)–(vi) are not new compact single-core mechanisms: the thin-drift, modulation, and compactness defects are exits from the compact finite-cost scale-collapse branch, and the scale-rigid state is the state handled by the scale-rigidity hypothesis in the multibubble/cascade theorem.

Proof. This is the scale-collapse state-space stratification, combined with the scale-rigidity branch used in the multibubble/cascade reduction. $\square$

Definition 6.10 (Virial-dissipative terminal state). Let $G_\nu(y) = e^{-|y|^2/(4\nu)}$. A scale-rigid terminal state is called velocity-virial dissipative if its selected terminal representative (V, P) satisfies, after the usual pressure normalization, the autonomous self-similar equation

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \nabla \cdot V = 0,$$

and the Gaussian identity below is justified by cutoff approximation on the selected renormalized windows. It is called swirl-virial dissipative if, in addition, the representative is axisymmetric, $\Gamma = RV_\theta$ is the scale-invariant circulation, and the corresponding weighted circulation identity below is justified.

Theorem 6.11 (Virial exclusion for dissipative terminal states). *Let a scale-rigid terminal state satisfy the hypotheses of [Theorem 6.10](#) and be bounded for all negative renormalized times in the sense needed for the Gaussian integrals below.*

- (a) *Suppose the velocity virial defect is absorbable: for some $\kappa > 0$ and a.e. τ ,*

$$-\frac{1}{2\nu} \int_{\mathbb{R}^3} |V|^2 (V \cdot y) G_\nu dy - \frac{1}{\nu} \int_{\mathbb{R}^3} P (V \cdot y) G_\nu dy \leq (1-\kappa) \int_{\mathbb{R}^3} |V|^2 G_\nu dy + 2\nu(1-\kappa) \int_{\mathbb{R}^3} |\nabla V|^2 G_\nu dy.$$

Then

$$\int_{\mathbb{R}^3} |V(y, \tau)|^2 G_\nu(y) dy = 0 \quad \text{for every } \tau.$$

Consequently no retained terminal core with positive Gaussian velocity mass can belong to this branch.

- (b) *Suppose the terminal state is axisymmetric and the circulation defect is absorbable: for some $\kappa > 0$ and a.e. τ ,*

$$-\frac{1}{2\nu} \int_{\mathbb{R}^3} \Gamma^2 (V \cdot y) G_\nu dy \leq (1-\kappa) \int_{\mathbb{R}^3} \Gamma^2 G_\nu dy + 2\nu(1-\kappa) \int_{\mathbb{R}^3} |\nabla \Gamma|^2 G_\nu dy.$$

Then

$$\int_{\mathbb{R}^3} \Gamma(y, \tau)^2 G_\nu(y) dy = 0 \quad \text{for every } \tau.$$

Thus every retained axisymmetric terminal core whose remaining obstruction is a positive Gaussian circulation mass is excluded by the weighted circulation identity.

Proof. For the velocity statement set

$$\mathcal{E}(\tau) = \int_{\mathbb{R}^3} |V(y, \tau)|^2 G_\nu(y) dy.$$

The Gaussian virial identity gives

$$\partial_\tau \mathcal{E} = -2\nu \int_{\mathbb{R}^3} |\nabla V|^2 G_\nu dy - \int_{\mathbb{R}^3} |V|^2 G_\nu dy - \frac{1}{2\nu} \int_{\mathbb{R}^3} |V|^2 (V \cdot y) G_\nu dy - \frac{1}{\nu} \int_{\mathbb{R}^3} P(V \cdot y) G_\nu dy.$$

The absorbability hypothesis therefore implies

$$\partial_\tau \mathcal{E} + \kappa \mathcal{E} + 2\nu \kappa \int_{\mathbb{R}^3} |\nabla V|^2 G_\nu dy \leq 0.$$

In particular $\partial_\tau \mathcal{E} + \kappa \mathcal{E} \leq 0$. Integrating from $s < \tau$ to τ gives

$$\mathcal{E}(\tau) \leq e^{-\kappa(\tau-s)} \mathcal{E}(s).$$

The ancient boundedness of the terminal state makes $\sup_{s < \tau} \mathcal{E}(s) < \infty$. Sending $s \rightarrow -\infty$ yields $\mathcal{E}(\tau) = 0$.

For the axisymmetric statement set

$$\mathcal{G}(\tau) = \int_{\mathbb{R}^3} \Gamma(y, \tau)^2 G_\nu(y) dy.$$

The weighted circulation identity is

$$\partial_\tau \mathcal{G} + \mathcal{G} + 2\nu \int_{\mathbb{R}^3} |\nabla \Gamma|^2 G_\nu dy = -\frac{1}{2\nu} \int_{\mathbb{R}^3} \Gamma^2 (V \cdot y) G_\nu dy.$$

The assumed absorption gives

$$\partial_\tau \mathcal{G} + \kappa \mathcal{G} + 2\nu \kappa \int_{\mathbb{R}^3} |\nabla \Gamma|^2 G_\nu dy \leq 0.$$

The same backward Gronwall argument gives $\mathcal{G}(\tau) = 0$ for every τ . $\square$

Remark 6.12 (Controlled swirl alone). If the axisymmetric representative satisfies $\Gamma \in L^\infty$, the same weighted circulation identity gives the a priori estimate

$$\mathcal{G}(\tau)^{1/2} \lesssim \|V\|_{L^\infty} \|\Gamma\|_{L^\infty}$$

for bounded ancient terminal states. This estimate is useful in the local state-space stratification, but by itself it is not an exclusion. The exclusion above requires the stated absorption, vanishing defect, or an equivalent condition which makes the virial inequality dissipative.

6.3. State-Space Assembly.

Theorem 6.13 (Local Type II state-space decomposition). *Every positive local Type II concentration sequence in Theorem 6.2, after passing to subsequences and selected windows in the local state-space analysis, enters one of the following alternatives:*

- (i) a compact single-core vanishing-dissipation branch;
- (ii) a multibubble, cascade, or gauge-degenerate branch;
- (iii) a rough-core branch, meaning loss of local windowed H^1 control on a retained compact cylinder;
- (iv) a finite-cost scale-collapse branch with nonvanishing selected core;
- (v) a scale-rigid state covered by the scale-rigidity hypothesis in Theorem 6.6.

In the last alternative, any terminal subbranch satisfying Theorem 6.11 is eliminated before the remaining scale-rigid state is passed to the scale-rigidity hypothesis.

Proof. This is the local state-space decomposition obtained by combining [Theorems 6.4, 6.6, 6.7](#) and [6.9](#). The representation theorem gives the nondegenerate single-core chart whenever a unique retained core is selected. Failure of that selection, splitting of positive CKN mass, compactness failure, or scale cascade is precisely the multibubble/cascade side. On a represented retained core, rough-core loss is equivalent by [Theorem 6.7](#) to failure of compact CKN control on an enclosing cylinder, hence returns to the multibubble/cascade side. Genuine scale collapse is divided by [Theorem 6.9](#) into finite-cost scale collapse or one of the scale-rigid terminal states used in [Theorem 6.6](#). These cases exhaust the local alternatives. $\square$

Proposition 6.14 (Elimination of local Type II states). *No positive local Type II concentration sequence can lie in any of the alternatives (i)–(v) of [Theorem 6.13](#).*

Proof. Alternative (i) is excluded by the compact single-core criterion, [Theorem 6.5](#).

Alternative (ii) is excluded by the local multibubble and cascade theorem, [Theorem 6.6](#), under its terminal admissibility and scale-rigidity hypotheses.

Alternative (iii) is not an independent terminal state. By [Theorem 6.7](#), rough-core loss of local H^1 control forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ package on enclosing compact cylinders. Such failure produces an active compact-cylinder concentration branch and therefore belongs to the multibubble/cascade alternative, already excluded by [Theorem 6.6](#).

Alternative (iv) is excluded by the finite-cost scale-collapse theorem, [Theorem 6.8](#).

Alternative (v) is the scale-rigid terminal state in [Theorem 6.6](#). If this terminal state is velocity-virial dissipative, [Theorem 6.11](#) forces its Gaussian velocity mass to vanish, which contradicts retention of a positive terminal core. If it is axisymmetric and swirl-virial dissipative with positive retained circulation mass, the same theorem forces $\Gamma \equiv 0$, contradicting that positive circulation mass. The remaining scale-rigid terminal states are precisely those left to the scale-rigidity hypothesis in [Theorem 6.6](#). Under that hypothesis, the state either has a nonzero bounded selected-window limit, which is incompatible with the local Type II condition, or it reduces to the compact single-core branch, which is excluded by [Theorem 6.5](#). Hence it cannot be a local Type II branch. $\square$

6.4. Assembly Theorem.

Theorem 6.15 (Local Type II exclusion). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$. Then no point of $\Sigma(T)$ lies in the local Type II alternative, within the local state-space class covered by [Theorem 6.13](#).*

Proof. Assume, for contradiction, that $x_0 \in \Sigma(T)$ lies in the local Type II alternative. By the local concentration dichotomy, [Theorem 6.3](#), there is a positive local Type II concentration sequence $r_n \downarrow 0$ at (x_0, T) . The local state-space decomposition, [Theorem 6.13](#), places this sequence, after passing to selected windows, in one of the five alternatives listed there. Each alternative is impossible by [Theorem 6.14](#). This contradiction proves that no such x_0 exists in the covered local state-space class. $\square$

Corollary 6.16 (Final local dichotomy after Type I separation). *Every singular point at time T in the covered local state-space class belongs to the local Type I alternative. Equivalently, after the Type I case is handled by the Type I part of the theory, the local Type II state space contains no remaining singular branch in that class.*

Proof. Let $x_0 \in \Sigma(T)$. By the local concentration dichotomy, [Theorem 6.3](#), a concentration point is either in the local Type I alternative or in the local Type II alternative. [Theorem 6.15](#) excludes the latter. $\square$

6.5. Conclusion. The proof uses only local PDE statements. The local concentration theorem supplies positive local CKN concentration and the Type I/Type II dichotomy. The repaired-gauge theorem supplies the local renormalized representation for nondegenerate selected cores. The single-core theorem excludes compact single-core vanishing-dissipation branches. The multibubble/cascade theorem removes splitting, cascades, gauge degeneracy, and the scale-rigid terminal state under its stated local hypotheses. The virial identities eliminate the velocity-virial dissipative terminal states and the axisymmetric swirl-virial dissipative terminal states before the remaining scale-rigid alternatives are handled by that hypothesis. The rough-core estimate sends loss of windowed H^1 control back to failure of compact CKN control. The scale-collapse theorem excludes finite-cost scale collapse and stratifies the remaining scale-collapse states locally. Thus the local state-space alternatives are exhausted without adding any further classification theorem.

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